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Mathematical Modelling of Defensive Pressure in Football

PhD Dissertation (Programa de Doctorado en Matemáticas)

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Preface and Acknowledgements

La realización de esta tesis doctoral ha supuesto una experiencia personal y académica profundamente transformadora. A los 45 años inicié el Máster Universitario en Investigación Matemática (INVESTMAT), dando comienzo una aventura con las matemáticas que se consolidó a partir de los 49 años y culmina con esta tesis en el año en que el autor alcanza los 57 años. Este trabajo se ha desarrollado de manera paralela a la actividad profesional, a lo largo de más de seis años de dedicación continuada. Este recorrido ha requerido un esfuerzo sostenido, constancia y sacrificio, pero ha sido afrontado en todo momento con entusiasmo, motivación y una profunda pasión por las matemáticas y la investigación.

El trabajo realizado en la Universitat Politècnica de València (UPV) ha sido clave en este proceso. Desde el inicio del doctorado, el autor encontró en sus supervisores —José Manuel Calabuig Rodríguez, Luis Miguel García Raffi y Enrique Alfonso Sánchez Pérez—, así como en el entorno académico del Departamento de Matemática Aplicada (DMA) y del Instituto Universitario de Matemática Pura y Aplicada (IUMPA) una acogida generosa, un acompañamiento constante y un apoyo intelectual decisivo. Más allá de la dirección académica, este camino ha permitido establecer relaciones humanas y profesionales que trascienden el ámbito estrictamente científico y que se han convertido en vínculos duraderos, fundamentales tanto en el desarrollo de la tesis como en el crecimiento personal del doctorando. También felicitar a Andrés Roger Arnau Notari, compañero de doctorando y ya Profesor Doctor, que comenzó su andadura en el DMA un poco después y colaboró con nosotros durante todos estos años.

Asimismo, deseo expresar mi agradecimiento a la entidad del Levante UD y, de manera particular, a José Manuel Pardo Gila, actual Jefe de Scouting, quien, con una mentalidad abierta y receptiva, consideró de interés nuestra propuesta de investigación. Su disposición permitió la colaboración y la formalización de un convenio de investigación con la Universitat Politècnica de València, en el marco de un proyecto basado en matemáticas, una disciplina ajena a su actividad deportiva. Confío en que las métricas defensivas desarrolladas en esta tesis puedan resultar de utilidad y aportar valor al análisis y al desarrollo del fútbol profesional.

Durante estos años, el proceso doctoral ha supuesto una auténtica inmersión en el mundo de la investigación científica. Se ha aprendido no solo a desarrollar y estructurar una investigación, sino también a formular preguntas relevantes, analizar críticamente resultados, comunicar el conocimiento de forma rigurosa y comprender el funcionamiento del ecosistema científico actual. Este aprendizaje ha incluido el descubrimiento de la Ciencia de Datos, el uso de herramientas matemáticas y computacionales de libre acceso, la participación en redes científicas internacionales a través de plataformas y redes sociales académicas, el intercambio de ideas con investigadores e investigadoras con inquietudes similares, así como la comprensión de los procesos editoriales y de publicación científica. Todo ello ha convertido el doctorado en una experiencia de aprendizaje rica, estimulante y profundamente satisfactoria, vivida con disfrute y curiosidad intelectual.

Desde una perspectiva cognitiva y formativa, el estudio profundo y continuado de las matemáticas ha tenido un impacto significativo en la manera de abordar y resolver problemas complejos. La literatura científica ha puesto de manifiesto que la actividad matemática favorece el desarrollo de capacidades cognitivas como el razonamiento lógico, la abstracción, la planificación, la creatividad estructurada y la flexibilidad mental, activando de manera intensa redes neuronales relacionadas con la comprensión, la toma de decisiones y la resolución de problemas. En este sentido, el trabajo matemático realizado durante la tesis ha contribuido no solo al desarrollo académico, sino también a una mejora sustancial en el ámbito profesional, aportando una mayor rigurosidad, precisión analítica y capacidad de estructuración en la práctica laboral cotidiana.

Aunque esta tesis se ha aplicado al estudio del fútbol, el trabajo realizado pone de relieve la transversalidad de las matemáticas y su capacidad para modelizar sistemas complejos más allá de un dominio concreto. Los modelos y métodos desarrollados ofrecen un marco abstracto y extensible para el análisis de interacciones dinámicas entre múltiples entidades, reforzando el papel de las matemáticas como herramienta integradora y como lenguaje común para abordar problemas complejos en distintos ámbitos científicos.

En conjunto, la realización de esta tesis doctoral no representa únicamente una contribución científica al análisis del rendimiento en el fútbol, sino también un proceso de crecimiento intelectual, profesional y humano. Este recorrido confirma que la investigación y el aprendizaje profundo son posibles y enriquecedores en cualquier etapa de la vida profesional, y reafirma el papel de las matemáticas como herramienta fundamental para comprender, modelizar y afrontar con rigor los problemas complejos que surgen tanto en la ciencia como en la vida cotidiana.

Por último, no puedo terminar sin agradecer de corazón a mi familia todo lo que han dado durante este proceso. A mi mujer Ana y a nuestros hijos José y Andrés, por el tiempo que les he robado, por su paciencia, su comprensión y por acompañarme incluso en esas largas conversaciones y debates filosóficos en las que las matemáticas y la ciencia ponían su acento. Sin vuestro apoyo, este camino no habría sido posible. Un beso enorme.

Resumen

La creciente disponibilidad de datos en el fútbol profesional, impulsada por los sistemas de eventos y las tecnologías de seguimiento de jugadores, ha abierto nuevas posibilidades para el análisis objetivo del rendimiento. Sin embargo, la caracterización del comportamiento defensivo ha estado históricamente limitada por la escasez de métricas capaces de describir la defensa de forma coherente, agregable entre jugadores y equipos, y con sensibilidad a la información espacial y temporal del juego. En este contexto, la presente tesis doctoral tiene como eje transversal la caracterización defensiva en el fútbol, abordada desde distintas perspectivas metodológicas y escalas de análisis, con el objetivo de desarrollar métricas interpretables, operativas y matemáticamente fundamentadas.

La tesis se articula a través de una serie de capítulos que, de manera progresiva, analizan el comportamiento defensivo utilizando datos de eventos, datos de tracking y modelos matemáticos y computacionales avanzados. El Capítulo 1 actúa de forma transversal ya que la tesis considera la aplicabilidad práctica de los modelos propuestos, incorporando aspectos computacionales y de viabilidad en tiempo real o cuasi tiempo real. Como parte fundamental de este desarrollo, se han diseñado e implementado herramientas de análisis específicas (analysis tools) que permiten operacionalizar los modelos y métricas propuestos. Estas herramientas integran los distintos enfoques desarrollados a lo largo de la tesis —datos de eventos, datos de tracking, modelos estáticos y dinámicos— y facilitan tanto el cálculo sistemático de las métricas defensivas como su visualización e interpretación.

El Capítulo 2 surge a partir de una pregunta planteada por el Levante UD, orientada a comprender cómo los equipos rivales defienden frente a dicho equipo y cómo esta información se relaciona con el éxito o fracaso de determinadas jugadas tipo. Este trabajo permite introducir técnicas de Machine Learning en el análisis del fútbol, estableciendo un marco para evaluar y caracterizar la presión defensiva de los equipos rivales a partir de datos de eventos, y sentando las bases metodológicas para los artículos posteriores.

El Capítulo 3 aborda la caracterización de la defensa desde una perspectiva estática, tanto a nivel individual como colectivo. En este trabajo se analizan estructuras defensivas basadas en la disposición espacial de los jugadores, proporcionando métricas que describen la estructura defensiva de los equipos en situaciones concretas del juego y de la totalidad del mismo, permitiendo una primera aproximación cuantitativa a la defensa colectiva y a la contribución individual de cada jugador.

El Capítulo 4 introduce una caracterización defensiva basada en la dinámica del juego, mediante un modelo matemático de presiones defensivas formulado a partir de datos de posicionamiento de los jugadores y conceptos provenientes de la física clásica. Este enfoque pone de manifiesto que el movimiento de los jugadores aporta información relevante para la comprensión del comportamiento defensivo, permitiendo describir de forma continua en el espacio y en el tiempo fenómenos como la intensidad, la acumulación y la superposición de presiones individuales y colectivas. Una aportación relevante de este enfoque dinámico es su capacidad para integrar la información defensiva individual en

métricas colectivas agregables a nivel de equipo y de partido. El carácter espacio-temporal del modelo permite capturar no solo la presencia de acciones defensivas, sino también su evolución y su influencia espacial, proporcionando una representación más fiel del comportamiento defensivo observado en el juego. Asimismo, este marco facilita la generación de visualizaciones espaciales coherentes con las métricas agregadas, permitiendo analizar las acciones dinámicas defensivas desde perspectivas colectivas e individuales.

El Capítulo 5 completa la caracterización defensiva abordando el análisis a nivel individual. A partir de las métricas desarrolladas en capítulos previos, se introduce el concepto de firma defensiva del jugador, que permite asignar a cada futbolista una representación compacta e interpretable de su comportamiento defensivo. Esta firma sintetiza patrones de agresividad y consistencia defensiva, facilitando la comparación entre jugadores y ofreciendo aplicaciones directas en contextos de evaluación del rendimiento, scouting y análisis táctico.

En conjunto, esta tesis aporta un marco coherente y progresivo para la caracterización defensiva en el fútbol profesional, contribuyendo al desarrollo de herramientas matemáticas y científicas que favorecen el análisis del rendimiento defensivo en contextos reales, como el scouting y el apoyo a la toma de decisiones.

Resum

La creixent disponibilitat de dades en el futbol professional, impulsada pels sistemes d'esdeveniments i les tecnologies de seguiment de jugadors, ha obert noves possibilitats per a l'anàlisi objectiva del rendiment. Tanmateix, la caracterització del comportament defensiu ha estat històricament limitada per l'escassetat de mètriques capaces de descriure la defensa de manera coherent, agregable entre jugadors i equips, i amb sensibilitat a la informació espacial i temporal del joc. En el context descrit, la present tesi doctoral té com a eix transversal la caracterització defensiva en el futbol, abordada des de diferents perspectives metodològiques i escales d'anàlisi, amb l'objectiu de desenvolupar mètriques interpretables, operatives i matemàticament fonamentades.

La tesi s'articula a través d'una sèrie de capítols que, de manera progressiva, analitzen el comportament defensiu utilitzant dades d'esdeveniments, dades de seguiment i models matemàtics i computacionals avançats. El Capítol 1 actua de manera transversal ja que la tesi considera l'aplicabilitat pràctica dels models proposats, en incorporar aspectes computacionals i de viabilitat en temps real o quasi temps real. Com a part fonamental del desenvolupament, s'han dissenyat i implementat eines d'anàlisi específiques (analysis tools) que permeten operacionalitzar els models i mètriques proposats. Són unes eines que integren els diferents enfocaments desenvolupats al llarg de la tesi –dades d'esdeveniments, dades de seguiment, models estàtics i dinàmics– i faciliten tant el càlcul sistemàtic de les mètriques defensives com la visualització i interpretació.

El Capítol 2 sorgeix a partir d'una pregunta plantejada pel Llevant UD, orientada a comprendre com els equips rivals defensen quan juguen contra el seu equip i com és una informació que es relaciona amb l'èxit o fracàs de determinades jugades tipus. Aquest treball permet introduir tècniques d'aprenentatge automàtic en l'anàlisi del futbol, establir un marc per a avaluar i caracteritzar la pressió defensiva dels equips rivals a partir de dades d'esdeveniments, i assentar les bases metodològiques per als articles posteriors.

El Capítol 3 aborda la caracterització de la defensa des d'una perspectiva estàtica, tant individual com col·lectiva. S'hi analitzen estructures defensives basades en la disposició espacial dels jugadors, es proporcionen mètriques que descriuen l'estructura defensiva dels equips en situacions concretes del joc i de la totalitat, la qual cosa permet una primera aproximació quantitativa a la defensa col·lectiva i a la contribució individual de cada jugador.

El Capítol 4 introdueix una caracterització defensiva basada en la dinàmica del joc, mitjançant un model matemàtic de pressions defensives formulat a partir de dades de posicionament dels jugadors i conceptes provinents de la física clàssica. Un enfocament que posa de manifest que el moviment dels jugadors aporta informació rellevant per a comprendre el comportament defensiu, i permet descriure de manera contínua en l'espai i en el temps fenòmens com ara la intensitat, l'acumulació i la superposició de pressions individuals i col·lectives. Una aportació rellevant d'aquest enfocament dinàmic és la capacitat que té d'integrar la informació defensiva individual en mètriques col·lectives agregables a l'àmbit d'equip i de partit. El caràcter espaciotemporal del model permet

capturar no només la presència d'accions defensives, sinó també l'evolució i la influència espacial que tenen, la qual cosa proporciona una representació més fidel del comportament defensiu observat en el joc. Així mateix, és un marc que facilita la generació de visualitzacions espacials coherents amb les mètriques agregades, fet que permet analitzar les accions dinàmiques defensives des de perspectives col·lectives i individuals.

El Capítol 5 completa la caracterització defensiva en abordar l'anàlisi en l'àmbit individual. A partir de les mètriques desenvolupades en els articles previs, s'introdueix el concepte de firma defensiva del jugador, que permet assignar a cada futbolista una representació compacta i interpretable del seu comportament defensiu. Una firma que sintetitza patrons d'agressivitat i consistència defensiva, i facilita la comparació entre jugadors i ofereix aplicacions directes en contextos d'avaluació del rendiment, estudi del rival i anàlisi tàctica.

En conjunt, la tesi aporta un marc coherent i progressiu per a la caracterització defensiva en el futbol professional, i contribueix al desenvolupament d'eines matemàtiques i científiques que afavoreixen l'anàlisi del rendiment defensiu en contextos reals, com l'estudi del rival i el suport a la presa de decisions.

Summary

The growing availability of data in professional football, driven by event-based systems and player-tracking technologies, has opened new possibilities for the objective analysis of performance. However, the characterisation of defensive behaviour has historically been limited by the scarcity of metrics capable of describing defence in a coherent manner, allowing aggregation across players and teams, and incorporating sensitivity to the spatial and temporal dimensions of the game. In this context, the present doctoral thesis has as its transversal axis the characterisation of defensive behaviour in football, addressed from different methodological perspectives and analytical scales, with the aim of developing interpretable, operational, and mathematically grounded metrics.

The thesis is structured around a series of chapters that progressively analyse defensive behaviour using event data, tracking data, and advanced mathematical and computational models. Chapter 1 acts transversally, as the thesis places particular emphasis on the practical applicability of the proposed models, incorporating computational considerations and feasibility in real-time or near-real-time contexts. As a fundamental component of this development, specific analysis tools have been designed and implemented to operationalise the proposed models and metrics. These tools integrate the various approaches developed throughout the thesis — event data, tracking data, static models, and dynamic models — and facilitate both the systematic computation of defensive metrics and their visualisation and interpretation.

Chapter 2 arises from a question posed by Levante UD, aimed at understanding how opposing teams defend against this team and how such information relates to the success or failure of specific play types. This article introduces Machine Learning techniques into football analysis, establishing a framework to evaluate and characterise the defensive pressure exerted by opposing teams based on event data, and laying the methodological foundations for subsequent articles.

Chapter 3 addresses the characterisation of defence from a static perspective, both at individual and collective levels. This work analyses defensive structures based on the spatial arrangement of players, providing metrics that describe the defensive organisation of teams in specific game situations as well as across entire matches, enabling an initial quantitative approach to collective defence and to the individual contribution of each player.

Chapter 4 introduces a defensive characterisation grounded in game dynamics through a mathematical model of defensive pressures formulated from player positional data and concepts derived from classical physics. This approach demonstrates that player movement provides relevant information for understanding defensive behaviour, enabling a continuous description in space and time of phenomena such as the intensity, accumulation, and superposition of individual and collective defensive pressures. A significant contribution of this dynamic approach is its ability to integrate individual defensive information into collective metrics that can be aggregated at team and match levels. The spatio-temporal nature of the model allows it to capture not only the presence of defensive actions but also their evolution and spatial influence, providing a more faithful representation of defensive

behaviour observed during play. Moreover, this framework facilitates the generation of spatial visualisations consistent with aggregated metrics, enabling the analysis of dynamic defensive actions from both collective and individual perspectives.

Chapter 5 completes the defensive characterisation by addressing the analysis at an individual level. Based on the metrics developed in the previous articles, the concept of a player's defensive signature is introduced, allowing each footballer to be assigned a compact and interpretable representation of their defensive behaviour. This signature synthesises patterns of defensive aggressiveness and consistency, facilitating player comparison and offering direct applications in performance evaluation, scouting, and tactical analysis.

Overall, this thesis provides a coherent and progressive framework for the characterisation of defensive behaviour in professional football, contributing to the development of mathematical and scientific tools that support the analysis of defensive performance in real-world contexts such as scouting and decision-making processes.

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1

Introduction

Football has undergone a profound transformation in recent years, driven by the increasing availability of data and the growing role of quantitative analysis in both research and professional practice. Advances in event-based data collection, positional tracking technologies and computational methods have opened new possibilities for describing the game beyond traditional statistics, enabling the study of football as a dynamic, spatio-temporal system.

Within this context, defensive behaviour remains one of the least formally characterised aspects of the game, despite its central importance for team performance and tactical organisation. This doctoral thesis addresses this gap by adopting a data-driven and mathematically grounded approach to the analysis of defensive behaviour in football.

This introductory chapter presents the context, objectives and structure of the thesis. It outlines the data ecosystems and analytical tools used throughout the research, defines the research objectives, and situates the subsequent chapters within a coherent methodological and conceptual framework.

This thesis adopts a data-driven and mathematically grounded approach to the study of football, with a particular focus on defensive behaviour and pressure-related interactions. Its scope is defined by the intersection of three main elements: the availability of modern football data, the methodological challenges associated with their processing and integration, and the development of interpretable quantitative models capable of capturing relevant aspects of the game.

The work presented in this thesis builds upon an extensive preliminary phase carried out during the early years of the doctoral programme, particularly between 2017 and 2018. During this period, a comprehensive review of the scientific literature on football analytics was conducted, with special attention to the state of research at that time. This review provided a detailed understanding of the dominant analytical paradigms, the types

of data commonly used, and the methodological limitations present in existing studies. Crucially, this initial phase coincided with a period in which advanced data acquisition technologies—especially automated tracking systems—were beginning to mature and to become more widely available in professional football contexts.

At that stage, much of the published research focused on offensive actions, ball possession, and attacking effectiveness, largely driven by the relative simplicity of modelling discrete on-ball events. In contrast, defensive aspects of the game, such as pressure, spatial control, and collective defensive behaviour, remained comparatively underexplored. These dimensions are inherently more complex to quantify, as they involve continuous interactions, spatial relationships and coordinated movement patterns that are not easily captured through traditional event-based representations alone. The emergence of high-resolution spatio-temporal data therefore revealed a clear research opportunity: the possibility of developing novel defensive metrics grounded in richer representations of player interactions and movement dynamics.

Motivated by this gap in the literature, the thesis progressively develops a methodological framework aimed at modelling defensive behaviour using modern football data. Rather than proposing purely descriptive indicators, the work emphasises the construction of mathematically interpretable metrics that can be linked to underlying spatial and temporal structures of the game. Throughout the thesis, particular care is taken to ensure that modelling choices remain consistent with the characteristics and limitations of the available data, as well as with the operational realities of professional football analysis.

1.1 State of the art

The increasing availability of spatio-temporal data in professional football, driven by advances in player tracking technologies and large-scale event logging systems, has profoundly transformed the way tactical behaviour is analysed. Over the last two decades, research has progressively evolved from descriptive, outcome-oriented statistics towards models that seek to capture the dynamic, interactive and spatial nature of the game. This section reviews the main scientific contributions that have shaped this evolution, with a particular focus on approaches relevant to the analysis of defensive behaviour.

1.1.1 Spatio-temporal representations of football behaviour

The formal analysis of tactical behaviour in football is fundamentally conditioned by the way the game is represented. Unlike outcome-based statistics, spatio-temporal representations aim to describe football as a continuous-time, multi-agent dynamical system, in which player trajectories, interactions and spatial configurations evolve under contextual constraints. This subsection reviews the main modelling paradigms that have been proposed to represent football behaviour in space and time, with an emphasis on their mathematical structure, descriptive power and limitations.

Early spatio-temporal pattern discovery

One of the first research directions to exploit tracking data focused on the automatic discovery of recurrent tactical patterns. These approaches typically represent the state of the game as a high-dimensional vector composed of player and ball positions, and apply unsupervised learning techniques to identify frequently occurring spatial configurations.

Grunz et al. [18] proposed a hierarchical pattern recognition framework based on self-organizing maps (SOM) and Dynamically Controlled Networks (DyCoN) to detect tactical patterns from positional data sampled at high temporal resolution. In this approach, groups of players (so-called *constellations*) were represented as spatial point sets, which were subsequently clustered to form prototype configurations. Temporal evolution was handled by sliding windows, allowing the identification of sequential tactical motifs involving attackers, defenders and the ball.

Similar clustering-based approaches were later explored using alternative representations and distance measures [28]. These methods demonstrated that tracking data contains sufficient structure to discriminate between formations, phases of play and tactical situations without relying on manually annotated events.

From a modelling perspective, these approaches implicitly assume that tactical behaviour can be decomposed into a finite set of representative states or patterns. While effective for descriptive and classification tasks, this discretisation introduces structural limitations: (i) temporal continuity is approximated by fixed-length windows, (ii) interaction intensity is not explicitly modelled, and (iii) interpretability at player level remains limited. These limitations motivated the development of representations that treat space and time as continuous variables.

Geometric and spatial control models

A complementary line of research models football behaviour through geometric constructions that quantify spatial dominance and control. In these approaches, player positions define continuous spatial partitions of the pitch, enabling the analysis of how space is occupied, shared and contested over time.

Voronoi-based models constitute one of the earliest and most widely adopted geometric representations. Fonseca et al. [14] used Voronoi diagrams to define *dominant regions* for each player and analysed their temporal evolution in futsal. Their results showed systematic differences between attacking and defending teams in terms of dominant region size, variability and regularity, highlighting that defensive behaviour is characterised by rapid spatial reconfiguration and higher temporal entropy.

Subsequent studies extended this idea to full-scale football and team-level analysis [8, 15]. These works demonstrated that geometric descriptors such as team surface area, centroid displacement and inter-player distances provide valuable information about collective organisation and compactness.

More sophisticated spatial control models incorporate not only distance, but also dynamic constraints on player movement. Spearman [39] introduced the concept of *pitch control*, defining for each location on the pitch the probability that it can be reached first by a given team, given player positions, velocities and acceleration limits. This probabilistic formulation yields a continuous control field that evolves over time and provides a mathematically grounded description of spatial dominance.

Geometric and control-based models offer an explicit, continuous representation of space and enable aggregation at multiple scales (player, sub-group, team). However, they primarily describe *where* space is controlled, rather than *how* pressure is exerted on specific opponents. In particular, interaction intensity between individual attackers and defenders is only indirectly captured.

Role alignment and structural consistency

A fundamental challenge in spatio-temporal modelling is the lack of spatial consistency caused by role switching and positional interchange. Direct comparison of player trajectories across time or matches is often meaningless unless players are aligned to functional roles.

Bialkowski et al. [7] addressed this issue by proposing a role-based alignment framework that assigns players dynamically to latent positional roles learned directly from tracking data. Each role is represented as a spatial probability distribution, and player-role assignment is solved frame-by-frame using an optimisation procedure that minimises total entropy. This methodology enables the discovery of team structures and formations in a data-driven manner and provides a stable reference frame for subsequent analysis. Extensions of this idea have been used to cluster formations across matches and seasons [28], and to compare team structures under different tactical contexts.

From a modelling standpoint, role alignment constitutes an essential preprocessing step that allows spatio-temporal metrics to be meaningfully aggregated and compared. Nevertheless, role-based representations remain descriptive in nature and do not explicitly quantify interaction processes such as defensive pressure or marking intensity.

Summary and limitations

Early spatio-temporal representations established that football behaviour can be described using continuous geometric, probabilistic and structural models derived from tracking data. Pattern discovery methods demonstrated the existence of recurrent tactical motifs, geometric models formalised spatial dominance, and role alignment provided structural consistency.

Despite these advances, several limitations persist. Most representations focus on spatial occupation rather than interaction intensity, treat defence implicitly rather than explicitly, and offer limited tools for player-level defensive evaluation. These limitations motivate the exploration of dynamic interaction-based models and form the basis for the subsequent analysis of collective coordination and defensive behaviour.

1.1.2 Collective dynamics and coordination in football

Beyond static spatial representations, football behaviour emerges from the continuous interaction of multiple agents whose movements are mutually constrained. As a result, a significant body of research has conceptualised football teams as complex adaptive systems, where collective organisation arises from local interactions rather than from centrally imposed structures. This subsection reviews the main approaches that analyse football behaviour from the perspective of collective dynamics, coordination and interaction.

Self-organising dynamics and collective behaviour

Self-organising dynamics approaches model football as a auto-organising system in which players continuously adapt their movements to teammates, opponents, the ball and task constraints. From this perspective, tactical behaviour is not prescribed but emerges through local perception–action couplings.

Several studies have analysed collective coordination by examining phase relationships, synchronisation patterns and relative motion between players or subgroups. Silva et al. [37] investigated intra-team synergies during attacking and defensive movements by analysing relative phase and time delays between players. Their results showed that defensive organisation is characterised by transient synergies that emerge and dissolve over short temporal scales, highlighting the inherently dynamic nature of defensive behaviour.

Complementary work has used entropy-based measures to quantify the regularity and predictability of collective motion. Higher entropy values have been associated with defensive phases, reflecting the need for rapid adaptation to opponent actions [14]. These findings reinforce the notion that defence is better described as a continuous adjustment process rather than a sequence of isolated events.

From a modelling standpoint, ecological dynamics provides a strong conceptual framework for understanding collective behaviour. However, the metrics commonly employed in this literature often operate at group or team level, and their abstraction makes direct interpretation at player level challenging.

Network and graph-based representations of interaction

An alternative formalisation of collective behaviour represents teams as networks, where players are nodes and interactions are edges. Initially applied to passing behaviour, network-based approaches have been widely used to study team organisation, player roles and coordination structures.

Gonçalves et al. [17] combined passing networks with positional data to analyse how network centrality measures relate to collective performance. Metrics such as closeness and betweenness centrality were shown to capture the distribution of responsibility within the team and the presence of dominant players in ball circulation. Arriaza-Ardiles et al. [3] further extended graph-based analysis by incorporating spatial constraints and analysing offensive phases as directed weighted networks. Their work demonstrated that network topology varies across matches and outcomes, reflecting tactical adjustments.

While network representations provide a compact mathematical description of interaction structure, they are fundamentally discrete and event-driven. Defensive behaviour, particularly off-ball pressure and marking, is only indirectly captured through the absence or disruption of passes. Consequently, network metrics struggle to represent continuous defensive effort and spatial interaction intensity.

Spatio-temporal interaction metrics

To overcome the limitations of purely event-based networks, several studies have proposed spatio-temporal interaction metrics derived directly from player trajectories. These approaches explicitly consider relative distances, orientations and velocities between players, enabling a more continuous description of interaction.

Research analysing inter-player distances and relative motion has shown that defensive organisation is associated with reduced spatial dispersion and tighter coupling between defenders [15]. Such metrics provide a low-level but interpretable description of collective compactness and coordination.

More recent work has explored multi-agent interaction modelling, where football is treated as a system of interacting agents governed by implicit rules and constraints [30]. These models aim to capture how local interactions propagate to collective behaviour, although they often prioritise prediction over interpretability.

Summary and implications for defensive modelling

Collective dynamics approaches have established that football behaviour emerges from continuous interaction processes and cannot be fully explained by static formations or isolated events. Self-organising dynamic highlights the transient nature of coordination, while network and spatio-temporal interaction models provide complementary mathematical formalisations of collective structure.

Nevertheless, existing approaches face two key limitations with respect to defensive analysis. First, most metrics operate at aggregate levels, making it difficult to attribute defensive contribution to individual players. Second, interaction intensity is often implicit rather than explicitly quantified. These limitations motivate the development of continuous, player-level metrics capable of capturing defensive pressure as an interaction process, which will be addressed in subsequent sections.

1.1.3 Data-driven and value-based models in football analysis

As tracking and event datasets have grown in scale and availability, a substantial body of research has focused on the development of data-driven models for prediction, classification and performance evaluation. These approaches typically formalise football matches as stochastic or statistical processes, mapping observed data to outcome-related variables such as goals, points or match results. This subsection reviews the main modelling paradigms in this area, with particular emphasis on their mathematical structure and their limitations for defensive analysis.

Event-based models for prediction and classification

Early data-driven approaches predominantly relied on event data, where a match is represented as a sequence of discrete actions (passes, shots, fouls, recoveries) annotated with time and location. From a modelling perspective, these representations naturally lend themselves to supervised learning and statistical classification tasks.

Several studies have proposed predictive models based on aggregated event counts or derived key performance indicators (KPIs), using regression, probabilistic classifiers or machine learning algorithms to estimate match outcomes [25]. These models often achieve high predictive accuracy, particularly when trained on large historical datasets, and have been widely adopted in applied performance analysis.

Within this paradigm, Chapter 2 of this thesis is situated. It explores the predictive and classificatory capacity of event-based metrics under different contextual conditions, demonstrating that aggregated indicators can capture meaningful relationships between

match events and outcomes. From a mathematical standpoint, such models can be interpreted as mappings

$$f : \mathbb{R}^P \rightarrow \mathcal{Y},$$

where $\mathbb{R}^P$ denotes a vector of event-derived features and $\mathcal{Y}$ the outcome space (e.g., win, draw, loss).

Despite their effectiveness for prediction, event-based models exhibit intrinsic limitations. The aggregation of events discards temporal ordering and spatial continuity, and defensive behaviour is only represented through sparse, outcome-dependent observations. As a result, these models provide limited insight into the underlying processes that generate defensive performance.

Efficiency and outcome-oriented performance models

A related line of work focuses on efficiency analysis, where teams or matches are evaluated in terms of how effectively inputs (e.g., shots, possession, defensive actions) are converted into outputs (e.g., goals or points). Data Envelopment Analysis (DEA) and related optimisation-based frameworks have been applied to football to estimate offensive and defensive efficiency [49].

Mathematically, these approaches model performance as a constrained optimisation problem, comparing observed input–output ratios to an estimated efficiency frontier. While such models explicitly distinguish between offensive and defensive components, they remain fundamentally outcome-oriented and rely on aggregated event variables. Consequently, they do not capture the continuous spatial and temporal structure of defensive behaviour.

Probabilistic and value-based spatio-temporal models

More recent work has sought to overcome the limitations of purely event-based approaches by integrating spatio-temporal information into probabilistic models. A prominent example is the family of expected value models, which estimate the probability or value of future outcomes conditioned on the current state of play.

Expected goals (xG) models represent one of the earliest instances of this paradigm, estimating the probability of scoring given shot location and contextual features. Extensions of xG have incorporated tracking data to account for defender positioning and movement, thereby introducing a spatially continuous perspective. Spearman [38, 39] formalised these ideas through pitch control and expected possession value (EPV) frameworks. In these models, the state of the game is represented as a continuous spatio-temporal configuration, and value is propagated through a Markovian process. Formally, the value function $V(s)$ assigns an expected reward to a game state s , and evolves according to

$$V(s) = \mathbb{E} [r(s, a) + \gamma V(s')],$$

where $r(s, a)$ denotes the immediate reward of action a , γ is a discount factor, and s' is the subsequent state.

These probabilistic models provide a mathematically principled way to combine space, time and outcome probabilities, and have been extended to quantify pass difficulty, passing

lanes and off-ball value [2, 4]. Importantly, they treat football as a continuous-time stochastic process rather than a sequence of independent events.

However, from a defensive perspective, these models primarily quantify *opportunity* and *value* rather than *interaction*. Defensive behaviour is inferred indirectly through its impact on expected outcomes, making it difficult to disentangle defensive effort, intensity and consistency at player level.

Datasets and integrated representations

The development of data-driven models has been facilitated by the release of integrated datasets combining event and tracking data [5, 31]. These datasets enable the construction of richer state representations and have accelerated research on spatio-temporal modelling, deep learning and multi-agent systems.

While such datasets significantly expand modelling capabilities, they also highlight the increasing gap between predictive performance and interpretability. As models grow in complexity, understanding how defensive behaviour contributes to outcomes becomes more challenging, reinforcing the need for explicit, interpretable defensive metrics.

Summary and positioning of the thesis

Data-driven and value-based models have substantially advanced the quantitative analysis of football, particularly in prediction and evaluation tasks. Event-based approaches provide effective but coarse representations, efficiency models formalise outcome conversion, and probabilistic frameworks integrate space, time and value within a unified mathematical structure.

Nevertheless, these approaches largely characterise defence through its effect on outcomes rather than through direct measurement of interaction processes. The limitations identified in this block motivate the transition towards explicit modelling of defensive behaviour as a continuous spatio-temporal interaction, which constitutes the focus of the subsequent analysis. In this context, the present thesis builds upon event-based prediction models (Chapter 2) and probabilistic frameworks to develop interpretable, player-level metrics of defensive pressure.

1.1.4 Defensive behaviour and pressure modelling

Defensive behaviour in football is inherently relational and dynamic. Unlike offensive actions, which are often defined by discrete events such as passes or shots, defence primarily manifests through continuous interactions aimed at constraining space, time and decision-making of opponents. This subsection reviews the main approaches that have attempted to characterise defensive behaviour, with particular emphasis on pressing and pressure, and discusses their mathematical and conceptual limitations.

Pressing and defensive constraints

One of the most extensively studied defensive concepts is pressing, typically defined as coordinated defensive actions aimed at regaining possession or disrupting the opponent's build-up. Early empirical studies characterised pressing through tactical descriptors and

observational analysis, while more recent work has sought to quantify pressing intensity using positional and event data [6, 35].

Common approaches rely on heuristic indicators such as the number of pressures applied in specific zones, time to regain possession, or event-based proxies such as PPDA (passes allowed per defensive action). While these metrics provide intuitive summaries of team-level defensive behaviour, they are fundamentally discrete and threshold-dependent. As such, they do not capture the continuous nature of defensive interactions, nor do they allow for consistent aggregation at player level.

From a modelling perspective, these approaches implicitly treat defensive pressure as a binary or categorical phenomenon, ignoring variations in intensity and temporal persistence.

Continuous representations of defensive interaction

Several studies have recognised the need to model defence as a continuous spatio-temporal process. Visual and exploratory approaches have analysed defensive pressure through proximity, relative velocity and spatial alignment between attackers and defenders [1]. These works demonstrated that defensive pressure evolves smoothly over time and often precedes observable defensive events.

More formal representations describe defensive behaviour through continuous interaction variables derived from tracking data, such as distance functions, orientation angles and closing velocities [29]. In this context, defensive pressure can be interpreted as an interaction kernel

$$P_{ij}(t) = \phi(d_{ij}(t), \dot{d}_{ij}(t), \theta_{ij}(t)),$$

where $d_{ij}(t)$ denotes the distance between defender i and attacker j , $\dot{d}_{ij}(t)$ its temporal derivative, and $\theta_{ij}(t)$ a relative orientation term.

Such formulations treat pressure as a continuous scalar field evolving in time, allowing defensive effort to be accumulated and analysed independently of discrete outcomes.

Temporal accumulation and defensive workload

Beyond instantaneous interaction, defensive behaviour is characterised by its temporal structure. Sustained pressure, recovery dynamics and repeated defensive actions contribute to overall defensive workload. Studies analysing match running performance and high-intensity actions preceding key events have shown that defensive phases are associated with distinct temporal patterns of effort [20, 21]. This formulation naturally supports aggregation across interactions and temporal windows, enabling analysis of defensive load, persistence and recovery.

Chapter 4 of this thesis builds upon this perspective by explicitly incorporating the temporal evolution of defensive pressure, showing that temporal structure is essential for understanding defensive performance beyond instantaneous measures.

Limitations of existing pressure metrics

Despite recent advances, existing pressure-related metrics suffer from several limitations. Many approaches remain tied to predefined thresholds or spatial zones, making them

sensitive to parameter choices. Others operate at team level, obscuring individual contributions. Moreover, defensive pressure is often conflated with defensive success, despite evidence that effective pressure does not always result in immediate ball recovery.

From a mathematical standpoint, the absence of a formally defined unitary pressure interaction limits the consistency and interpretability of defensive metrics. Without such a definition, aggregation across players, time and matches remains ad hoc.

Positioning of the thesis

The present thesis addresses these limitations by formalising defensive pressure as a continuous, mathematically defined interaction between players. Chapter 3 introduces a unitary pressure metric grounded in spatio-temporal proximity, enabling consistent aggregation and interpretation. Chapter 4 extends this framework by analysing the temporal dynamics of pressure, linking interaction intensity to defensive workload and persistence.

Together, these contributions provide a principled foundation for modelling defensive behaviour as a continuous spatio-temporal process, bridging the gap between descriptive pressing indicators and rigorous player-level analysis.

1.1.5 Player-level analysis and defensive characterisation

While much of the existing literature focuses on team-level organisation and collective behaviour, practical performance analysis and scouting ultimately require metrics that operate at player level. This subsection reviews approaches aimed at quantifying individual contribution and discusses their limitations when applied to defensive behaviour. It then motivates the use of distributional analysis and behavioural signatures as a principled framework for player-level defensive characterisation.

Individual performance metrics and their limitations

Traditional player evaluation in football has relied on volume-based indicators derived from event data, such as tackles, interceptions or recoveries. Although intuitive, these metrics are highly dependent on team context, game state and opponent behaviour, and often fail to distinguish between opportunity and ability.

More advanced player-level metrics have been proposed within probabilistic and value-based frameworks, attributing credit or blame to players based on changes in expected outcomes [12, 38]. Formally, these approaches assign marginal contributions

$$\Delta V_i = V(s_{t+1}) - V(s_t),$$

to player actions. While mathematically well-defined, such metrics primarily evaluate *on-ball* involvement and remain outcome-oriented.

From a defensive perspective, these approaches implicitly assume that defensive contribution is adequately captured through its effect on possession value or scoring probability. However, this assumption overlooks the continuous, off-ball nature of most defensive actions.

Distributional perspectives on player behaviour

An alternative paradigm treats player behaviour not as a set of discrete actions, but as a stochastic process generating a distribution of interaction intensities over time. Under this view, a defensive metric $P_i(t)$ is not summarised by a single aggregate value, but by its empirical distribution

$$\mathcal{D}_i = \{P_i(t) \mid t \in \mathcal{T}\},$$

defined over a relevant temporal domain $\mathcal{T}$.

Distributional descriptors such as mean, variance, skewness or quantiles provide complementary information about player behaviour. In particular, they enable the distinction between players who apply high-intensity pressure sporadically and those who apply moderate pressure consistently. This distinction is largely invisible to aggregate or event-based metrics.

Recent studies have highlighted the importance of stability and variability in performance evaluation, showing that reliable player assessment requires metrics with consistent distributional properties across matches and contexts [36].

Positioning of the thesis

Chapter 5 of this thesis addresses a gap that remains largely unexplored in the existing literature: the systematic characterisation of individual defensive behaviour beyond aggregate or outcome-oriented metrics. While prior work has primarily focused on attributing defensive value through marginal changes in possession or scoring probability, such approaches provide limited insight into the qualitative nature and consistency of defensive actions.

Building upon continuous defensive pressure metrics, Chapter 5 adopts a distributional perspective in which individual defensive behaviour is modelled as a stochastic process generating a time series of interaction intensities. Rather than collapsing this information into a single scalar, the empirical distribution of pressure values is analysed to extract descriptive properties related to intensity, variability and temporal regularity.

This approach enables the definition of player-level defensive profiles based on the shape of pressure distributions, allowing players to be differentiated in terms of aggressiveness and consistency. By focusing on distributional properties, the proposed framework provides an interpretable and context-independent characterisation of defensive behaviour, suitable for player comparison, scouting and longitudinal analysis.

In this sense, the thesis extends existing player evaluation methodologies by shifting the focus from isolated actions or marginal value attribution to the analysis of continuous interaction patterns. This constitutes a natural culmination of the progression outlined throughout this chapter, from event-based prediction models, through spatio-temporal interaction modelling, to player-level behavioural characterisation.

1.2 Data and Analysis Tools

1.2.1 Data Acquisition and Technological Evolution

Over the last decade, football has undergone a profound technological evolution in the acquisition, representation and exploitation of performance-related data. This transformation has been driven by advances in sensing technologies, computer vision, and large-scale data processing infrastructures, which have progressively enabled the systematic collection of high-resolution spatio-temporal information during matches and training sessions. From a modelling perspective, these developments have made it increasingly feasible to represent football as a time-evolving, high-dimensional multi-agent system, in which players and the ball are modelled as interacting entities whose states evolve continuously in space and time [19].

Early large-scale data acquisition in football relied primarily on manual notational analysis, producing event-based datasets that encode discrete on-ball actions such as passes, shots or duels. These datasets provide a sparse but structured representation of the game, in which actions are modelled as temporally ordered events with associated spatial coordinates. Despite their limited temporal resolution, event-based data remain widely used due to their interpretability, historical availability and standardised formats, which facilitate longitudinal and comparative analyses across competitions and seasons [26]. From a mathematical standpoint, such datasets can be interpreted as spatio-temporal point processes, forming the basis of many early quantitative models of football performance.

More recent technological advances have significantly increased the dimensionality and resolution of available data through the widespread adoption of automated tracking systems. In professional football, positional data are predominantly acquired using optical tracking technologies based on multi-camera computer vision architectures. These systems continuously estimate the two-dimensional positions of all players and the ball at sampling frequencies typically ranging between 10 and 30 Hz, yielding dense spatio-temporal trajectories suitable for fine-grained analysis of collective behaviour, physical demands and tactical organisation. The football-specific validity and reliability of such systems have been established in the literature, supporting their use in both research and applied performance analysis contexts as we have showed in previous section.

Among the leading providers of optical tracking technology, systems such as ELECTRONICS ARTS (before TRACAB Technologies then known as ChyronHego) [10] have become widely adopted across major professional leagues. TRACAB employs a distributed multi-camera architecture to reconstruct player and ball trajectories in real time, producing large volumes of positional data per match. Its deployment across competitions such as the Premier League, Bundesliga and LaLiga illustrates the degree to which automated tracking has become embedded in the operational infrastructure of modern football. Importantly, these systems typically generate multiple data streams corresponding to different stages of processing, ranging from low-latency real-time outputs to fully post-processed datasets, each characterised by different levels of accuracy and suitability for analytical purposes [23].

In parallel with optical systems, wearable technologies such as GPS and inertial measurement units are widely used to capture physical performance metrics, particularly in training environments. While these systems offer complementary information related to player load and movement dynamics, their use in official matches is subject to regulatory constraints. As a result, optical tracking systems remain the primary source of positional data in competitive match settings, reinforcing their central role in contemporary football analytics.

The coexistence of event-based data, optical tracking data and physical performance measurements has substantially expanded the analytical landscape but has also introduced significant methodological challenges. These heterogeneous data sources are often collected by independent systems, characterised by different temporal resolutions, coordinate definitions and annotation protocols. Consequently, the integration and synchronisation of multiple data streams has emerged as a critical prerequisite for advanced modelling approaches.

In response to these challenges, a growing body of research has focused on the development of methodological frameworks for data validation, harmonisation and synchronisation. Approaches based on ball dynamics, spatio-temporal constraints and optimisation criteria have been proposed to improve the temporal and spatial accuracy of integrated datasets, thereby reducing uncertainty in the localisation of key events and enhancing the robustness of downstream analytical models [9, 48].

Beyond technical and methodological considerations, the widespread adoption of tracking technologies reflects a broader institutional and strategic commitment to data-driven innovation in football. Data acquisition systems such as TRACAB are certified under the FIFA Quality Programme for Electronic Performance and Tracking Systems, providing an additional layer of standardisation and quality assurance. Moreover, FIFA's Strategic Objectives for the 2023–2027 period explicitly emphasise increased investment in digital technology and artificial intelligence, including initiatives aimed at enhancing fan engagement and supporting innovation for future generations [13]. These strategic priorities further underline the central role of advanced data acquisition technologies within the global football ecosystem.

Taken together, these developments indicate that the technological evolution of football data acquisition over the last five to ten years has not only increased the volume and resolution of available data, but has also reshaped the methodological standards of the field. A detailed understanding of the underlying acquisition systems, their providers, and their integration within professional practice is therefore essential for the quantitative modelling approaches developed throughout this thesis.

The relevance of these technological and methodological advances extends beyond academic research and is clearly reflected in professional practice. Data acquisition and analytics have become integral components of performance departments in modern football, a trend further evidenced by the emergence of specialised postgraduate programmes

and innovation platforms promoted by leading football organisations and educational institutions. Initiatives such as LaLiga Business School, Sports Data Campus, FC Barcelona's Barça Innovation Hub, the Real Madrid–Universidad Europea master's programme in football analytics, and Unisport's programmes in big data and artificial intelligence applied to sport illustrate the consolidation of data-driven methodologies as a core component of contemporary football analysis [11, 22, 40, 46, 47].

In summary, the technological evolution of football data acquisition over the last five to ten years has fundamentally reshaped the way the game can be represented, analysed and modelled. Understanding the mathematical structure, limitations and integration challenges of modern football data is therefore a necessary foundation for the quantitative modelling approaches developed throughout this thesis.

1.2.2 Data Availability and Nature

This section focuses on the availability and nature of football data, reviewing the main types of datasets used in professional and academic contexts and discussing their respective strengths and limitations. and addresses the analysis tools, processing and structuring of football data, detailing the methodological decisions required to ensure data quality, consistency and suitability for quantitative modelling.

The data-driven analysis of football relies on a heterogeneous ecosystem of data sources that differ substantially in terms of availability, structure, resolution and accessibility. These data sources are generated by a combination of commercial providers, technological platforms, governing bodies and research-oriented initiatives, each operating under distinct objectives and constraints. As a result, football data cannot be regarded as a uniform or homogeneous resource, but rather as a collection of complementary representations of the game, each capturing different aspects of performance.

One of the most established categories of football data is event-based data, which describe discrete actions occurring during a match, such as passes, shots, duels or set pieces. These datasets are typically produced by specialised commercial providers and are generated through a combination of manual annotation, computer vision techniques and, increasingly, machine learning models. Event data are widely used in both professional and academic contexts due to their interpretability, structured nature and long historical availability. However, access to high-quality event datasets is generally restricted by commercial agreements, and their use in academic research often depends on institutional collaborations with clubs or data providers. In addition to proprietary event data feeds, some providers have released openly accessible event datasets for research purposes, such as the StatsBomb Open Data repository [45] or Metrica Sports [27].

In addition to event data, modern football analysis increasingly relies on positional tracking data, which provide continuous spatio-temporal information on the movement of players and the ball. Tracking data can be obtained through different Electronic Performance and Tracking Systems (EPTS), including optical tracking systems based on multi-camera computer vision architectures and wearable sensor-based solutions. Among

these, optical tracking systems are the predominant source of positional data in official competitions, as their use is compatible with regulatory frameworks and does not require players to wear additional hardware during matches. Due to their technological complexity, infrastructure requirements and validation procedures, tracking data represent one of the most challenging and costly data sources to obtain in professional football.

The availability of tracking data is further shaped by institutional and regulatory factors. In recent years, FIFA has established quality and certification standards for EPTS technologies, defining validation protocols, accuracy requirements and data specifications. These initiatives aim to ensure consistency, reliability and interoperability across different tracking systems and providers, and they play a central role in shaping the landscape of positional data available for performance analysis and research.

Beyond proprietary and commercially restricted datasets, a growing number of initiatives have contributed to the release of openly available football data for research purposes. Some technology providers and research-oriented organisations have made sample datasets publicly accessible, including synchronised event data, positional tracking data and metadata. These datasets are typically accompanied by detailed documentation and are released in standardised formats, such as flat tabular representations or structured formats aligned with FIFA data transfer specifications. Open datasets play a crucial role in supporting methodological development, benchmarking and reproducibility within the research community, despite their more limited scope when compared to full-season professional datasets.

Football data sources may also include additional streams of information that complement on-field performance data. These include match metadata (such as team line-ups, substitutions and match context), physical performance indicators derived from wearable technologies, as well as economic, contractual and media-related data. While these sources are not always available at the same level of granularity or reliability as event and tracking data, they contribute valuable contextual information and support broader analytical objectives.

Overall, the availability and nature of football data are the result of a complex interplay between technological capabilities, commercial interests, regulatory frameworks and research practices. Understanding the origin, accessibility and limitations of the different data sources is essential for defining appropriate analytical questions and for selecting modelling approaches that are consistent with the characteristics of the underlying data. This overview provides the foundation for the more detailed examination of specific data types presented in the following sections of this section.

Event-based data

Event-based data constitute one of the most established and widely used data modalities in football analytics. They describe a match as a temporally ordered sequence of discrete actions, such as passes, shots, duels, fouls or restarts of play, each occurring at a specific instant in time. In contrast to continuous representations of the game, event-based data provide a sparse but semantically rich description focused on moments of direct interaction

with the ball.

From a modelling perspective, event-based data abstract the continuous dynamics of football into a set of discrete observations, each associated with categorical and numerical attributes. Mathematically, these datasets can be interpreted as marked spatio-temporal point processes, where each event corresponds to a labelled point in a spatial domain and at a specific time. This abstraction has historically facilitated the application of statistical analysis and machine learning techniques, particularly in contexts where off-ball behaviour and continuous spatial interactions were not explicitly required.

Despite their interpretability and widespread use, event-based representations exhibit inherent limitations. By construction, they capture only selected moments of on-ball activity and omit large portions of the match in which relevant tactical and defensive behaviour unfolds without direct ball interaction. As a result, spatial organisation, coordinated movement and defensive pressure are only indirectly observable through event data, a limitation that is particularly relevant for the analysis of defensive behaviour.

Generation of commercial event data. Modern commercial event datasets are generated through hybrid acquisition pipelines that combine manual annotation, computer vision techniques and machine learning models. Providers such as StatsPerform [44](formerly Opta Sports) operate large-scale infrastructures capable of producing event data in near real time while maintaining consistency across competitions and seasons. These pipelines yield highly structured datasets that follow stable ontologies of event types and qualifiers, and are designed to be integrated with complementary data feeds such as positional tracking or metadata.

Access to such datasets is typically restricted by commercial agreements. In the context of this thesis, the empirical study used in Chapter 2 relies on confidential event data provided by a professional football club under a collaboration agreement. These data were derived from a commercial provider and encoded using a proprietary event format, reflecting a common applied research scenario in professional football analytics.

Opta F24: XML-based representation of event data. A representative example of commercial event data is provided by the Opta F24 feed, which encodes match events in a hierarchical XML format. Unlike flat tabular representations, the F24 format explicitly models semantic relationships between competitions, matches, teams and events through nested XML elements.

An Opta F24 document is structured around the following main components:

- `<OptaDocument>`: root element,
- `<Competition>`: competition metadata,
- `<MatchData>`: match-level information,
- `<TeamData>`: team and player identifiers,

- **<EventData>**: collection of match events.

Each individual event is represented by an **<Event>** node that encapsulates both core attributes and contextual qualifiers. The core attributes include the event type, temporal location, spatial coordinates and the player and team involved. Table 1.1 summarises the most relevant attributes for analytical purposes.

Table 1.1: Core attributes of an Opta F24 event

Attribute	Description
type_id	Encoded event type (pass, shot, goal kick, etc.)
period_id	Match period identifier
min, sec	Discrete temporal location of the event
team_id	Team responsible for the event
player_id	Player associated with the event
x, y	Normalised pitch coordinates
outcome	Binary or categorical event outcome

From a mathematical standpoint, these attributes define each event as a labelled point in a spatio-temporal domain, with time discretised at the second level and space represented through normalised coordinates.

Qualifiers and semantic enrichment. A distinctive feature of Opta F24 data is the use of qualifiers, which provide additional contextual information that refines the interpretation of an event. Qualifiers encode aspects such as body part used, pass height or direction, pressure context or set-piece situations. This mechanism allows a relatively compact event ontology to express a wide range of football actions through combinations of event types and qualifiers.

While qualifiers significantly enrich event semantics, their interpretation depends on external provider-specific documentation. As a result, F24 data are not self-describing and require access to proprietary ontologies in order to be fully understood, introducing a dependency between raw data and external semantic definitions.

Open and FIFA-aligned event data: Metrica Sports. In contrast to proprietary commercial feeds, Metrica Sports [27] provides several openly available football datasets specifically designed for research and educational purposes. These datasets include synchronised event data, positional tracking data and metadata, and are released in formats aligned with FIFA data transfer specifications.

A key characteristic of Metrica event data is the availability of an explicit and publicly documented event taxonomy. Events are organised according to their functional role within the game, including ball actions, duels, recoveries and pressure-related defensive actions. In the Metrica terminology, pressure is defined as an intentional defensive action aimed at constraining an opponent in possession, regardless of whether direct contact with the ball occurs. This explicit representation of defensive intent is particularly relevant for

the modelling objectives pursued in this thesis.

Metrica event data are provided both in FIFA-aligned XML format and in a structured JSON representation optimised for analytical use. While both formats encode the same underlying events, the XML representation prioritises interoperability and standard compliance, whereas the JSON format facilitates direct integration into data analysis pipelines. Importantly, the JSON representation supports interval-based temporal descriptions for certain defensive actions, providing a more natural bridge between discrete event data and continuous positional tracking.

Context-enriched event data: StatsBomb. StatsBomb adopts a different design philosophy for event-based football data, emphasising semantic richness and contextual information. Through its Open Data initiative, StatsBomb has released fully open event datasets encoded in a self-describing JSON format. Each event includes explicit semantic labels and event-specific structures, resulting in heterogeneous but highly expressive representations.

In addition to standard event descriptors, StatsBomb data include contextual information such as binary pressure indicators and freeze-frame snapshots of nearby player positions at the moment of an event. Although freeze-frame data do not provide continuous tracking, they offer partial off-ball context that bridges the gap between classical event logs and full positional datasets. These features make StatsBomb data particularly relevant for the study of local spatial context and decision-making under pressure.

Comparison of event data paradigms and role within this thesis. The comparison between Opta F24, Metrica Sports and StatsBomb highlights three complementary paradigms of event-based football data. Opta F24 represents a traditional commercial approach focused on comprehensive logging of discrete actions. StatsBomb prioritises contextual enrichment within a purely event-driven framework. Metrica Sports used in this thesis occupy an intermediate position, combining explicit event semantics, FIFA-aligned standardisation and native integration with positional tracking data.

Within this thesis, event-based data play a foundational but evolving role. The study carried out in Chapter 1 relies exclusively on commercial event data to analyse specific game situations using discrete action representations. Subsequent studies progressively incorporate event data that are explicitly documented and integrated with tracking data, enabling the modelling of defensive behaviour through spatial and temporal relationships. This progression motivates the transition to positional tracking data in the following section.

Positional tracking data

Positional tracking data provide a continuous spatio-temporal representation of football matches by recording the locations of all players and the ball at regular time intervals throughout the game. Unlike event-based data, which capture discrete actions, tracking data describe the underlying movement dynamics of the match, offering a dense and high-

resolution view of player behaviour and collective organisation over time.

From a mathematical perspective, positional tracking data can be interpreted as a set of time-indexed trajectories in a two-dimensional spatial domain. Each player and the ball are represented as moving points whose positions evolve continuously according to the kinematics of the game. This representation is particularly well suited for the analysis of spatial structure, interaction patterns and collective behaviour, which are only indirectly observable through event-based representations.

Data acquisition technologies and EPTS. In professional football, positional tracking data are mainly obtained through Electronic Performance and Tracking Systems (EPTS). These systems include optical tracking technologies based on multi-camera computer-vision architectures, as well as wearable sensor-based solutions used primarily in training environments. In official competitions, optical tracking systems constitute the predominant source of positional data, as they do not require players to wear additional hardware during matches.

Modern optical tracking systems rely on computer-vision-based tracking pipelines that reconstruct player and ball trajectories from synchronised multi-camera video feeds installed around the stadium, enabling the capture of spatio-temporal data at high sampling frequencies suitable for both analytical and broadcast applications [16, 24, 43]. These systems operate at sampling frequencies typically ranging from 10 to 25 Hz and are capable of producing large volumes of spatio-temporal data suitable for both analytical and broadcast applications. Due to their technological complexity and infrastructure requirements, access to tracking data remains more limited than access to event-based data and is often governed by commercial agreements or institutional frameworks.

Structure of tracking data. Tracking data are commonly provided as time-indexed records in which each time step corresponds to a frame containing the spatial coordinates of all players and the ball. At each frame, the state of the match is represented by a set of two-dimensional positions, expressed in a provider-specific coordinate system and sampling resolution.

In the datasets used in this thesis, tracking files contain raw positional measurements without embedded football-specific semantics. Variables such as team possession, phase of play or defensive responsibility are not explicitly encoded. Instead, tracking data describe only the kinematic evolution of players and the ball over time. This separation between movement data and semantic interpretation is a defining characteristic of positional tracking datasets and has important methodological implications for subsequent analysis.

Role of event data in the interpretation of tracking data. Although positional tracking data provide detailed information on player and ball movement, they do not, by themselves, encode key contextual aspects of the game. In particular, concepts such as ball possession, attacking and defending phases, or ball-in-play status are not directly observable from

positional data alone.

In this thesis, these contextual variables are inferred through the integration of tracking data with event-based information and metadata. Event data provide discrete indicators of ball interactions and game state changes, which can be temporally aligned with the continuous tracking timeline. For example, possession is inferred by analysing sequences of passes, recoveries and turnovers, while the distinction between active play and stoppages is obtained from the availability of ball position measurements in the tracking data. This integration is essential for associating spatial behaviour with meaningful football concepts and for defining defensive metrics based on pressure and proximity.

Metadata and reference information. Positional tracking datasets are accompanied by metadata files that define the structural and contextual properties of the match. These metadata include information on pitch dimensions, coordinate system orientation, team identities, player identifiers, substitutions and match periods. Metadata establish the reference framework required to interpret raw positional measurements consistently across time and between teams.

In particular, metadata specify the spatial reference system in which tracking coordinates are expressed and provide the information necessary to maintain player identity consistency in the presence of substitutions. Without this contextual information, tracking data cannot be reliably analysed or compared across matches. While metadata are introduced here as a fundamental component of tracking datasets, their operational use—such as coordinate transformations, handling of substitutions and temporal alignment—is addressed in detail in the following subsection on data processing and structuring.

Advantages, limitations and role within this thesis. Positional tracking data offer significant advantages for the quantitative analysis of football. They enable the study of collective spatial organisation, player interactions and continuous defensive behaviour that are not directly observable through event-based representations. These properties make tracking data particularly suitable for modelling defensive pressure, spatial control and team compactness.

At the same time, tracking data present important limitations. Their interpretation depends on derived variables and modelling assumptions, such as possession inference or role assignment, which are not directly measured. Moreover, tracking systems are subject to measurement noise, occlusions and occasional data loss, particularly in high-density situations around the ball.

Within this thesis, positional tracking data constitute the primary source of information for the development of defensive metrics based on spatial pressure and player interactions. In Chapter 3 and Chapter 4, tracking data are combined with event data and metadata to construct mathematically interpretable representations of defensive behaviour at both individual and collective levels. This use of tracking data represents a methodological progression from discrete, action-based analysis towards a continuous and spatially grounded

representation of the game, motivating the data processing and structuring strategies described in the next subsection.

1.2.3 Data Processing and Structuring

The availability of rich football datasets does not, by itself, guarantee meaningful quantitative analysis. Raw data obtained from event logs, positional tracking systems and metadata files are heterogeneous in structure, temporal resolution and semantic content, and cannot be directly used for mathematical modelling without appropriate processing. Data processing and structuring therefore constitute a critical methodological stage that bridges the gap between data acquisition and analytical modelling.

In this thesis, data processing is understood not merely as a set of technical preprocessing operations, but as the definition of explicit and reproducible rules that transform raw measurements into structured analytical representations of the game. These rules encode assumptions about temporal continuity, spatial organisation and game context, and must be made explicit in order to ensure interpretability and reproducibility.

Conceptual overview of the processing pipeline. The processing framework developed in this thesis integrates three complementary data modalities: event-based data, positional tracking data and metadata. Each modality contributes distinct information and operates at different levels of abstraction. Event data provide discrete indicators of ball-related actions and game state changes, tracking data describe the continuous movement of players and the ball, and metadata define the contextual and structural properties of the match.

The objective of the processing pipeline is to transform these heterogeneous inputs into a unified analytical representation in which all data streams are temporally aligned, spatially consistent and context-aware. This representation serves as the foundation for the defensive metrics and models developed in the subsequent chapters.

Temporal alignment and synchronisation. Temporal alignment is a prerequisite for integrating event-based and tracking data. Event data encode actions using discrete timestamps, while tracking data consist of regularly sampled frames acquired at fixed frequencies. Metadata introduce additional temporal structures, such as match periods and stoppages, which must be consistently reconciled with both timelines.

In this thesis, tracking data are treated as the primary temporal backbone due to their regular sampling and continuous coverage of the match. Event timestamps are mapped onto this backbone by associating each event with the closest corresponding tracking frame or with a short temporal window of frames surrounding the event time. This strategy preserves the high temporal resolution of the tracking data while embedding discrete actions within their continuous spatio-temporal context.

Temporal alignment is inherently approximate. Small offsets may arise due to differences in recording systems or annotation delays, and alignment must therefore balance precision with robustness. These approximations are treated as part of the integration

noise rather than as modelling errors, and their impact is mitigated through temporal aggregation strategies introduced later in this section.

Spatial reference systems and coordinate normalisation. Raw positional tracking data are provided using provider-specific spatial reference systems that are optimised for data acquisition rather than for analytical modelling. To enable consistent spatial analysis, all tracking data used in this thesis are transformed into a common Cartesian reference frame centred at the geometric centre of the pitch and expressed in metric units.

This transformation involves scaling normalised coordinates to physical dimensions, translating the origin to the centre of the pitch and applying axial symmetries to ensure a consistent orientation of the coordinate axes. An additional symmetry is applied to second-half data to maintain a constant attacking direction across the entire match. The explicit formulation of these transformations ensures that spatial relationships are invariant to pitch orientation and match half, facilitating aggregation and comparison across time.

Positional tracking data provided by Metrica Sports are expressed in a normalised two-dimensional coordinate system. Let

$$P = (x, y) \in [0, 1]^2 \quad (1.1)$$

denote the position of a player or the ball in this native reference frame. The origin (0, 0) is located at the top-left corner of the pitch, the horizontal axis increases from left to right, and the vertical axis increases from top to bottom.

The transform process for coordinate normalisation used in this thesis is given by:

$$\text{WidthFieldSize} = 105 \text{ m}, \quad \text{HeightFieldSize} = 68 \text{ m}. \quad (1.2)$$

The first transformation consists of scaling the normalised coordinates to metric units. This yields

$$P'(x', y') = (x \cdot \text{WidthFieldSize}, y \cdot \text{HeightFieldSize}), \quad (1.3)$$

where $P'(x', y')$ represents the position in metres within the original Metrica reference frame.

Although this transformation expresses positions in physical units, the resulting coordinate system remains asymmetric and non-Cartesian from an analytical standpoint, as the origin is located at a corner of the pitch and the vertical axis is inverted.

For the dynamic analysis of player interactions and defensive pressure developed in this thesis, a Cartesian inertial reference frame is defined with its origin located at the geometric centre of the pitch. Spatial coordinates are expressed in metres, following the International System of Units, and the axes are oriented according to standard mathematical conventions.

The transformation from the native Metrica coordinates to the centred reference frame is obtained through a translation and an axial symmetry. Formally, let $P''(x'', y'')$ be the position in metres after scaling. The transformation is defined as

$$P''(x'', y'') = \left(x' - \frac{\text{WidthFieldSize}}{2}, \frac{\text{HeightFieldSize}}{2} - y' \right). \quad (1.4)$$

This operation simultaneously:

- relocates the origin of the coordinate system to the centre of the pitch;
- restores a conventional Cartesian orientation, with the vertical axis increasing upwards.

The resulting reference frame is symmetric with respect to both spatial axes and provides a natural setting for the computation of distances, spatial gradients and interaction-based metrics.

To ensure directional consistency, an additional transformation is applied to all positional data recorded during the second half of the match. Specifically, a point symmetry with respect to the origin of the centred reference frame is applied:

$$P'''(x''', y''') = (-x'', -y''). \quad (1.5)$$

This transformation maps the second-half positions onto the same attacking orientation as the first half. As a result, the attacking direction of each team remains consistent throughout the entire match, facilitating the construction of spatial heat maps, density estimates and aggregated defensive metrics.

Handling metadata and match context. Metadata play a central role in the processing pipeline by defining the contextual framework required for consistent interpretation of event and tracking data. Metadata specify pitch dimensions, coordinate conventions, team and player identifiers, substitutions and match periods. This information is essential for maintaining identity consistency across data streams and for correctly interpreting spatial and temporal measurements.

In addition to formal match structure, contextual information about the effective continuity of play is obtained by combining metadata with tracking data. In the datasets used in this thesis, the ball-in-play status is inferred from the availability of ball position measurements in the tracking data, with missing ball coordinates indicating stoppages. This information is used to restrict analyses to meaningful phases of play and to avoid artefacts introduced by dead-ball situations.

In the datasets used in this thesis, the ball-in-play status is not provided as an explicit variable. Instead, it is inferred directly from the tracking data of the ball. Specifically, the ball position variables (`ball_x` and `ball_y`) are recorded as valid spatial coordinates when the ball is in play, and take missing values (NaN) when the ball is not active.

Inference of possession and game phases. Ball possession is not directly encoded in the tracking data and must be inferred by integrating event-based information with positional data. Possession is modelled as a piecewise-constant state that changes only when specific possession-related events occur. Each event is interpreted according to an explicit set of rules that determine whether possession is gained, maintained or lost, based on its type,

sub-type and outcome.

The inference of possession follows the operational definition adopted in the Metrica Sports documentation. According to this definition, a player is regarded as being in possession of the ball as soon as they touch the ball or make a conscious decision to delay collecting it when they have the opportunity to do so, with the intention of taking control of the ball at a different position.

Once a player has gained possession, possession is considered to persist until one of the following events occurs:

- another player takes over possession of the ball;
- a shot is deflected by the goal frame or by another player;
- the ball is significantly deflected by the referee;
- the ball goes out of play.

Importantly, possession is not considered lost when an opposing player makes contact with the ball but does not significantly alter the intended outcome of the action, for example when a pass still reaches its target or the original player retains control.

The Metrica event taxonomy classifies possession-related actions into several event groups and sub-types. Table 1.2 summarises the event taxonomy used for possession inference, while Table 1.3 describes the interpretation rules applied to each event type.

Table 1.2: Event types and sub-types relevant for possession inference (Metrica)

Event type	Sub-types
SET PIECE	Set piece, FK attempt, Retaken
RECOVERY	Interference 1, Interference 2
PASS	Body part, Attempt, Deflection, Offside
BALL LOST	Body part, Attempt, Interference 1, Intervention, Deflection, Offside
BALL OUT	Body part, Attempt, Intervention, Deflection, Offside, Own goal
SHOT	Body part, Deflection, Shot direction, Shot result, Offside
FAULT RECEIVED	–
CHALLENGE	Challenge, Fault, Challenge result

The combination of event types and sub-types reflects the intentional and contextual nature of ball possession, requiring possession changes to be inferred from both the semantic classification of events and their temporal ordering. Each event is associated with a player and a team, allowing possession to be assigned at both the individual and team levels. In ambiguous situations, such as challenges or deflections, possession is resolved according to the event outcome and the subsequent sequence of actions.

Table 1.3: Interpretation of event types for ball possession inference

Event type	Interpretation for possession
SET PIECE	Ball is brought back into play by the acting team
RECOVERY	Possession gained through significant interference or restart
PASS	Possession maintained or transferred within the same team
BALL LOST	Possession lost to the opposing team or referee interference
BALL OUT	Possession terminated; ball leaves the field of play
SHOT	Possession ends following a shot attempt
FAULT RECEIVED	Possession retained; free kick awarded
CHALLENGE	Possession contested; resolved by subsequent event

Table 1.4 reports the empirical distribution of event types and sub-types used to infer ball possession in Chapter 4 of this thesis. The dataset contains a total of 4.059 event records distributed over 2.513 tracking frames, indicating that multiple events may occur within the same temporal frame.

As possession is modelled as a piecewise-constant state, only a single event per frame can be retained to define the possession status until the occurrence of the next possession-changing event. Consequently, when multiple events are recorded within the same frame, a prioritisation rule based on the possession criteria described above is applied in order to select the event that governs the possession state.

Once possession is assigned at the time of an event, it is propagated forward in time until the occurrence of a subsequent possession-changing event. This propagation allows possession to be associated with each tracking frame, enabling the classification of match states into attacking and defending phases. This inferred possession signal constitutes a fundamental input for all defensive metrics developed in the thesis.

In the study described in Chapter 4 of this thesis, event-based temporal windows are used to analyse defensive behaviour immediately following a loss of ball possession. The objective is to characterise how defensive pressure exerted by a team evolves in the immediate aftermath of a ball loss or ball recovery by the opponent. To this end, all match events corresponding to interceptions, ball thefts or lost challenges are identified from the event dataset. Specifically, the following event categories are considered as initiating events for a ball-loss window:

- BALL LOST -- INTERCEPTION (126 events),
- BALL LOST -- THEFT (26 events),
- BALL LOST -- CROSS (10 events),
- CHALLENGE -- LOST (158 events).

These events yield a total of 320 candidate ball-loss situations extracted from the event dataset.

Once the initial frame corresponding to each ball-loss event is identified, the subsequent trajectory of the team that has gained possession is analysed until one of the following termination conditions is satisfied:

- the team in possession loses the ball again;
- the ball leaves the field of play, regardless of whether possession is formally retained;
- play is interrupted due to a stoppage.

From the initial set of 320 events, 8 cases are excluded from the analysis: one due to the ball not being in play at the event time, and the remaining cases due to originating from free-kick situations. The final dataset therefore consists of 312 ball-loss sequences suitable for dynamic defensive analysis.

Construction of analytical representations. The processed data are further transformed into analytical representations that encode the match state in a mathematically interpretable form. At any given time, the match state is represented by the spatial configuration of all players and the ball, together with contextual variables such as possession and phase of play.

At any given time t , the state of a football match can be described by the spatial configuration of all players and the ball, together with contextual variables such as possession and phase of play. Formally, the match state is represented as:

$$S(t) = \{P_1(t), \dots, P_N(t), B(t), \Pi(t)\},$$

where $P_i(t) \in \mathbb{R}^2$ denotes the position of player i , $B(t)$ denotes the ball position when available or Nan when the game is Dead, and $\Pi(t)$ denotes the inferred team in possession at time t .

Depending on the analytical objective, different representations are constructed, including player-centric, team-centric and mixed player-team views. In addition, temporal aggregation and event-driven temporal windows are employed to focus the analysis on specific game situations, such as defensive behaviour following a loss of possession. These representations form the structured inputs to the defensive models and metrics presented in the subsequent chapters.

Data quality, uncertainty and robustness. The integration of heterogeneous football data introduces unavoidable sources of uncertainty related to measurement noise, annotation ambiguity and temporal or spatial misalignment. Rather than assuming idealised data, the processing framework adopted in this thesis explicitly acknowledges these limitations and incorporates strategies to mitigate their impact.

Temporal aggregation, neighbourhood-based representations and rule-based inference of derived variables reduce sensitivity to frame-level noise and short-lived inconsistencies. As a result, the analytical representations prioritise robustness and interpretability over excessive sensitivity to measurement precision.

Reproducibility and generalisation. All processing steps and assumptions are explicitly documented, enabling reproducibility and facilitating adaptation to alternative data providers or competitions. The processing framework is designed to be modular and largely provider-independent, allowing individual components to be substituted without altering the core analytical logic.

Where possible, the methodology is demonstrated using open-access datasets, ensuring that key aspects of the analysis can be independently verified. At the same time, the framework is compatible with proprietary industrial datasets, reflecting the realities of applied football analytics.

A key aspect of reproducibility is the explicit documentation of all processing steps and assumptions. In this thesis, each transformation applied to the raw data—temporal alignment, spatial normalisation, possession inference and construction of analytical representations—is described in detail and justified in relation to the analytical objectives.

By clearly separating raw data from derived variables and analytical constructs, the processing pipeline makes explicit which aspects of the analysis are directly observed and which are inferred. This transparency allows other researchers to replicate the methodology using different datasets or to modify specific components while preserving the overall analytical structure.

Although the empirical analyses in this thesis rely on specific data providers, the processing framework is designed to be largely provider-independent. Concepts such as temporal synchronisation, centred spatial reference systems, event-driven possession inference and neighbourhood-based representations are not tied to a particular data format or vendor.

This modular design enables the substitution of individual components of the pipeline, such as event taxonomies or coordinate systems, without altering the core analytical logic. As a result, the framework can be generalised to datasets obtained from alternative providers, provided that equivalent information is available.

Operational considerations and near-real-time feasibility. Beyond reproducibility in the academic sense, an additional practical requirement in modern football analytics is deployability under strict time constraints. In professional environments, metrics and models are often expected to operate in real time or near real time in order to support two major use cases: (i) broadcast and media production, where live visualisations enrich the narrative of the match as it unfolds, and (ii) coaching and performance departments, where timely indicators can inform tactical instructions, substitutions, or risk management decisions during the game.

This operational requirement is increasingly aligned with the wider industry trend towards live data capture and delivery. Commercial providers deliver match event feeds in real time [41, 42], for media and professional clients, and optical tracking systems can capture the locations of players and the ball at high sampling rates (e.g. 25 Hz [24]),

enabling the production of live spatio-temporal analytics and on-screen graphics during broadcasts.

Within this thesis, the computational feasibility of the proposed defensive metrics was assessed through the implementation of the dynamic pressure model developed in the Chapter 4. Using the Metrica dataset, the full computation of the defensive metrics was obtained in approximately three minutes. The implementation was executed in R using RStudio [32] (version 2024.12.0 Build 467) on a laptop equipped with an Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz and 16 GB RAM running Windows x64. While this execution time does not constitute real-time performance in the strictest sense, it demonstrates near-real-time feasibility in a standard portable computing environment, and suggests that further optimisation (e.g. compiled implementations, parallelisation, incremental updates, or streaming computation) could reduce latency to levels compatible with live operational workflows.

From a methodological perspective, near-real-time constraints have important implications for model design. First, representations and metrics should ideally rely on processing steps that can be updated incrementally as new frames arrive, rather than requiring repeated full-match recomputation. Second, the inference of derived variables (e.g. possession, ball-in-play status, or phase segmentation) must be robust to short-lived integration noise and timestamp offsets. Finally, the choice of temporal windows and aggregation strategies involves a trade-off: longer windows improve robustness to noise, while shorter windows increase responsiveness for live decision support.

Overall, incorporating near-real-time deployability as an explicit design consideration strengthens the practical relevance of football analytics research. It ensures that the resulting metrics are not only scientifically interpretable and reproducible, but also compatible with the operational needs of broadcasters and coaching staff in real match conditions.

1.2.4 Analysis Tools

All data processing, analysis and visualisation tasks carried out in this thesis were implemented using open-access software environments. The core analytical workflows were developed in the R programming language, with **RStudio** [32] as the main integrated development environment. Rather than relying on pre-existing scripts, closed-source platforms or standardised toolkits, the entire analytical codebase was developed specifically for this thesis and evolved progressively alongside the formulation of the proposed models.

This approach reflects a deliberate methodological choice: analytical tools were not treated as external utilities, but as an integral part of the modelling process. As new questions emerged during the research, custom scripts were designed and refined to address specific challenges related to data ingestion, temporal alignment, spatial transformations, possession inference and the computation of defensive metrics. This tight coupling between methodological development and software implementation ensured full control over the analytical pipeline and allowed modelling assumptions to be explicitly encoded and tested.

All scripts and source code used in this thesis were created specifically for this research. The implementation was carried out using **RStudio 2024.12.0+467** (“Kousa Dogwood” release, cf37a3e5488c937207f992226d255be71f5e3f41, 2024-12-11) running on Windows x64, together with **R version 4.5.1 (64-bit)**.

Interactive analysis and visualisation with Shiny

In order to support exploratory analysis, result validation and interpretation of the proposed defensive metrics, interactive visualisation tools were developed using the **Shiny** framework [34]. A dedicated dashboard was created and used extensively throughout this thesis, enabling dynamic inspection of defensive pressure metrics at multiple temporal and spatial scales.

The application can be accessed at:

- <https://ccatalan.shinyapps.io/KPIsFootball/>
- <https://indicame.upv.es/KPIsFootball/>

This dashboard allows the user to analyse defensive metrics computed over the entire match or within selected temporal windows. It supports multiple analytical perspectives, including individual player–player interactions, player–team interactions and fully aggregated team-level representations. The ability to interactively select temporal windows proved particularly valuable for identifying and analysing specific match situations triggered by event data and associated with precise tracking frames.

Figures 1.1 and 1.2 illustrate two views of the Shiny application. The first tab presents dynamic visualisations of defensive pressure metrics—such as Momentum, Force and Defensive Work—developed in Chapters 3 and 4, with full control over temporal filtering. The second tab displays the matrix of defensive pressures and interactions between players from both teams, corresponding to the selected temporal interval.

Reproducible methodological documentation with Quarto

In addition to interactive dashboards, the **Quarto** publishing system [33] (version 1.6.39) was used to document and communicate the methodological development of the defensive metrics proposed in this thesis. Quarto is an open-source scientific and technical publishing framework that enables the creation of reproducible, interactive documents combining narrative text, executable code and dynamic visualisations.

Using Quarto, an interactive HTML document was produced to present the full methodological pipeline underlying Chapter 4 “*From Static to Dynamic: A New Framework for Measuring Defensive Pressure in Football*” which includes structured explanations of the data processing steps, event handling, tracking integration and metric formulation,

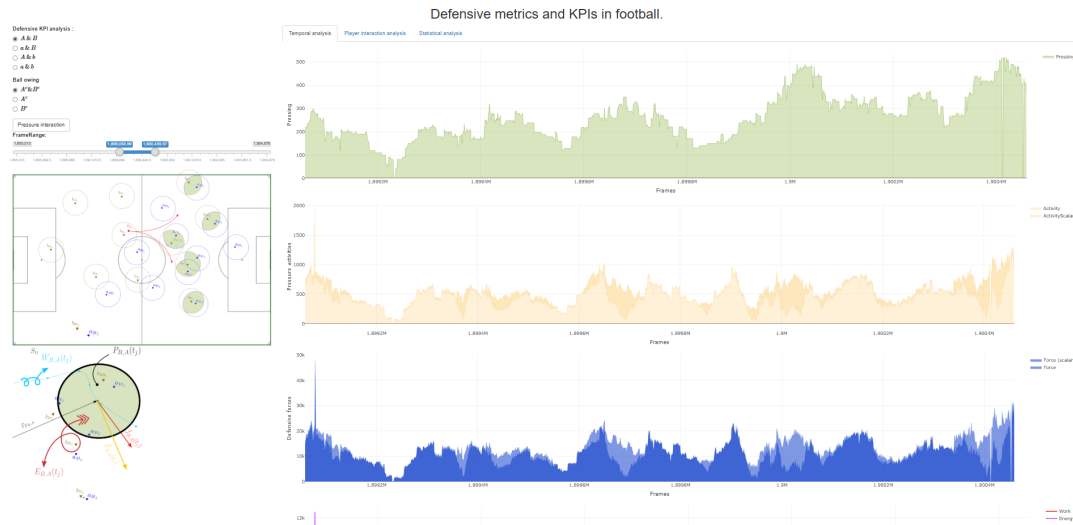


Figure 1.1: The figure displays the main interface of the Shiny application developed for the analysis of defensive pressure metrics. A fixed control panel on the left allows the user to select the analytical scale: team–team interactions ($A \& B$), player–team interactions ($a \& B$, $b \& A$), or individual player interactions ($a \& b$). Within the same panel, it is possible to visualise ball possession for one team or both teams, and, most importantly, to define and restrict specific temporal windows in order to analyse individual actions or complete phases of play. The main area of the interface presents, in its first tab (*temporal analysis*), dynamic visualisations of the defensive pressure metrics—Momentum, Force and Defensive Work—introduced and developed in Chapter 4

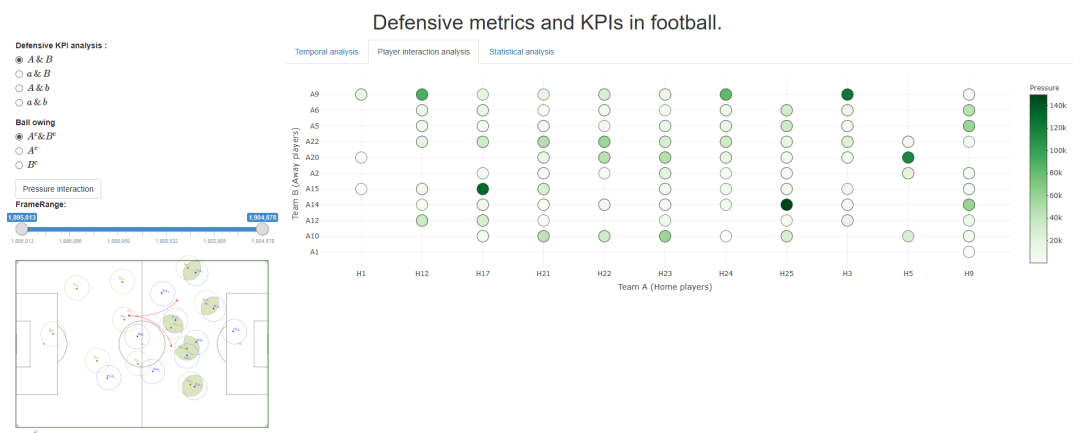


Figure 1.2: The figure shows the second tab of the Shiny application, where the matrix of defensive pressures and player interactions is displayed. The matrix is computed for the temporal window selected in the lateral control panel and provides a compact representation of defensive interactions between players from both teams during the selected phase of the match.

together with the corresponding source code.

The interactive document can be accessed at:

- <https://cccapaccioni.github.io/ModellingFootball/Modelling/KPIs.html>
- <https://indicame.upv.es/KPIs.html>

Figure 1.3 shows the landing page of the Quarto-generated HTML document. Its interactive structure, navigable index and embedded code blocks facilitate a transparent understanding of the modelling process and significantly lower the barrier to reproducibility and critical inspection.

The screenshot displays the landing page of a Quarto-generated HTML document. The page features a title, author information, publication date, abstract, and keywords. A sidebar on the right contains a 'Contents' menu with a scrollable index. The main content area shows the beginning of the document, including a section for '1 ENVIRONMENT' and '1.1 Packages', which lists necessary packages and their installation commands in R.

A mathematical model to study defensive metrics in football: individual, collective and game pressures
Dynamical System of Pressures.

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ABSTRACT
We present a complete model for the calculation of defensive dynamic pressures that is able to evaluate and analyse the movement of defensive pressures from a spatio-temporal point of view in football, associating it with the equivalence to a Dynamic System of Variable Mass Particles.

KEYWORDS
football, soccer, modelling, pressure, dynamical system

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1 ENVIRONMENT

1.1 Packages

Necessary packages to use in R.

```

1 # R version: R-4.4.2 (OK)
2 # R version: R-4.5.1 (Necessary for ggstream package)
3 # RStudio version: 2024.12.0 Build 467
4 # Quarto version:
5
6 if(!require(here))install.packages("here");library("here")
7 if(!require(psych))install.packages("psych");library("psych")
8 if(!require(tidyverse))install.packages("tidyverse");library("tidyverse")
9 if(!require(reshape2))install.packages("reshape2");library("reshape2")
10 if(!require(readr))install.packages("readr");library("readr") #para usar la funci?n read.f
11 if(!require(readtext))install.packages("readtext");library("readtext") #para poder leer e
12 if(!require(XML))install.packages("XML");library("XML") #para leer archivos .xml y ejecu
13 if(!require(xml2))install.packages("xml2");library("xml2") #para leer archivos .xml
14 if(!require(dplyr))install.packages("dplyr");library("dplyr") #para ordenar archivos y man
15 if(!require(purrr))install.packages("purrr");library("purrr") #para manipular datos
16 if(!require(e1071))install.packages("e1071");library("e1071") #para calcular Kurtosis, Fis
  
```

Figure 1.3: .The figure shows the landing page of the interactive HTML document created with **Quarto**. This document provides a structured and reproducible presentation of the methodology and software pipeline used to compute the defensive metrics developed in this thesis, including the organisation of the codebase and the main processing stages.

Simulation and dynamic visual validation

To further support the interpretation and validation of the proposed defensive models, a dedicated **simulator** was developed to represent player kinematics and the temporal evolution of defensive pressure metrics. This simulator was used to generate images and videos illustrating representative defensive situations, enabling a direct visual link between the mathematical formulation of the metrics and the underlying football behaviour.

Beyond its illustrative role, the simulator became an essential analytical tool for corroborating specific defensive actions, inspecting short spatio-temporal windows and diagnosing edge cases in the behaviour of the dynamic pressure model. Its use strengthened confidence in the interpretability and football relevance of the proposed metrics.

A key practical advantage of this application is that it provides a visual validation mechanism in scenarios where match video footage is not available. In several of the datasets analysed in this thesis, access to broadcast video was not possible; under these conditions, the simulator played a crucial role in verifying defensive actions, interpreting metric behaviour and ensuring the football relevance of the analytical results.

In the interest of transparency and reproducibility, the simulator, together with the Quarto HTML document, source code, generated images and videos, has been deposited in the public repository: <https://github.com/cccapaccioni/ModellingFootball>. This repository also contains several of the scripts used to generate visualisations and analytical outputs throughout the thesis.

Methodological implications

The toolchain developed in this thesis reflects a tightly integrated approach to data processing, modelling and visualisation. By designing all analytical tools in-house and using open-access technologies, the research ensures full transparency of methodological choices and enables the replication and extension of the proposed framework.

More broadly, this approach highlights the role of software development as a first-class methodological component in modern football analytics. The analytical tools presented here are not merely auxiliary instruments, but constitute an essential bridge between raw data, mathematical modelling and football-specific interpretation.

1.3 Objectives

The primary objective of this thesis is to **characterise defensive behaviour in football at individual and collective levels by developing quantitative metrics and models** that integrate spatial, temporal and contextual information from modern football data sources.

The specific objectives are:

- **To design, implement and validate a tailored data analysis framework for football analytics, based on custom-developed scripts and open-access software**

tools, specifically conceived to support the modelling and analysis of defensive behaviour.

This objective emphasises the development of original analytical workflows, visualisation tools and computational environments created throughout the thesis using open-access platforms such as *R*, *RStudio*, *Shiny* and *Quarto*. Rather than relying on pre-existing toolkits, the framework is progressively constructed to address the specific requirements of the proposed models, ensuring transparency, flexibility and reproducibility.

- **To develop mathematically interpretable metrics for the analysis of defensive behaviour, grounded in spatial and temporal relationships between players, and applicable at multiple levels of granularity, from individual actions to collective team organisation.**

This objective focuses on the formalisation of defensive metrics that capture pressure, interaction and spatial influence, avoiding purely outcome-based indicators and enabling interpretation at player, team and mixed player–team levels.

- **To investigate defensive behaviour in specific game situations, such as ball-loss events and transitions between attacking and defending phases, by analysing short-term spatio-temporal dynamics triggered by discrete match events.**

This objective supports the study of defensive responses as dynamic processes unfolding over time, rather than as static snapshots, and motivates the use of event-driven temporal windows combined with continuous positional tracking data.

- **To assess the robustness, generalisability and practical feasibility of the proposed framework and metrics under real-world constraints, including data quality and computational requirements, using both proprietary and open-access datasets.**

Particular attention is given to the applicability of the proposed methods across different data providers and competition contexts, as well as to their potential deployment in applied environments such as performance analysis, coaching support and near-real-time or broadcast-oriented applications.

These objectives guide the organisation of the thesis and are addressed progressively across its chapters.

The methodological objective related to data integration and representation is developed in the introductory Chapter 1, while the remaining objectives are addressed through the empirical studies presented in Chapters 2 to 5. The final chapters synthesise the results, discuss their implications and evaluate the extent to which the stated objectives have been achieved.

1.4 Thesis structure

This doctoral thesis is structured into a series of chapters as follows:

Chapter 1 establishes the conceptual, methodological and technical foundations of the thesis. It introduces the scientific motivation for the quantitative study of defensive behaviour in football and presents the data ecosystems, analytical tools and bespoke computational workflows developed throughout the research. Particular emphasis is placed on the integration of event-based data, positional tracking data and metadata, as well as on the custom scripts and visualisation tools implemented using open-access environments to support the proposed models and metrics. This chapter also defines the research objectives and outlines the overall structure of the thesis.

Chapters 2, 3, 4, 5 contain the core scientific contributions of the thesis and are structured as a compendium of studies. There are primarily composed of four research articles, which together address the modelling and quantification of football performance from complementary perspectives. The common goal is to advance towards a scientific framework capable of **describing defensive game** in quantitative terms, bridging mathematical theory, positional modelling, and applied decision-making in real contexts.

Chapter 2, “*Using Neural Networks and Hierarchical Cluster Analysis to Study Goal Kicks in Football*”, is based on event data and explores the application of machine learning techniques as a tool for the classification of defensive behaviours observed during goal-kick situations, serving as an introductory and methodological foundation for the thesis. The study is conducted in the specific context of the matches played by Levante UD and focuses on the defensive characterisation and classification of opposing teams prior to the application of learning algorithms.

The principal objective of this chapter is to establish a **defensive taxonomy** of opponent teams based on **hierarchical cluster analysis** with descriptors extracted from game events. These defensive team characterisation, formally introduced and defined within the chapter, provide a structured representation of team behaviour that is subsequently used as input for **neural networks**. In this sense, machine learning is employed as a classification mechanism built upon a prior mathematical characterisation, rather than as a purely data-driven or black-box approach.

Beyond its specific findings, this work had a fundamental value within the development of the thesis. It enabled a deep understanding of the structure, processing, and management of event data, while introducing **machine learning methodologies** into the analytical framework of this research in a controlled and interpretable manner. As such, this chapter establishes the conceptual and computational basis for the subsequent development of formal mathematical models aimed at the quantitative analysis of defensive pressure.

Catalán, C., Roger, A., Calabuig, J. M., García Raffi, L. M., Pardo Gila, J. M., Pons, R. and Sánchez Pérez, E. A.

“Using neural networks and hierarchical cluster analysis to study goal kicks in football”.

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Chapter 3, “*A Mathematical Model to Study Defensive Metrics in Football: Individual, Collective and Game Pressures*”, introduces a mathematical framework for the analysis of defensive pressure, proposing new metrics at the individual, collective, and team levels. It develops a comprehensive **mathematical model of defensive pressure**, combining positional and relational information to quantify defensive pressure at the individual, collective, and team levels. This work establishes the basis for a **new generation of defensive metrics**, capable of describing the tactical load and structural organisation of defensive systems in a quantitative and reproducible manner.

Catalán, C., Calabuig, J. M., García Raffi, L. M. and Sánchez Pérez, E. A.

“A Mathematical Model to Study Defensive Metrics in Football:
Individual, Collective and Game Pressures”.

Mathematics 2024, 12, 3854 (pages 1-18). DOI: 10.3390/math12233854 .

Chapter 4, “*A Dynamic Framework for Defensive Pressure Assessment in Football*”, extends the previous models by developing a dynamic and spatio-temporal representation of defensive pressure, with a focus on real-world applicability in near real-time contexts. It constitutes the natural evolution of the previous models, advancing towards a **dynamic and continuous representation of defensive behaviour**. Unlike traditional static metrics, this framework considers pressure as a spatio-temporal and emergent process, dependent on motion, synchronisation between players, and player interactions. Its development aims to deliver **innovative defensive indicators** applicable in **near real time**, with potential use for coaches, performance analysts, analytics platforms, and media outlets, thereby facilitating the tactical interpretation of defensive behaviour in professional contexts.

Catalán, C., Calabuig, J. M., García Raffi, L. M. and Sánchez Pérez, E. A.

“A Dynamic Framework for Defensive Pressure Assessment in Football”.
preprint (ResearchGate) (2026).

DOI: 10.13140/RG.2.2.30460.19841

Finally in Chapter 5, the thesis also includes an additional study entitled “*The Defensive Signature of the Player: Characterising Aggressiveness and Consistency in Defensive*”.

Pressure”, which applies the proposed defensive metrics to the individual player level, illustrating their potential use for player profiling, performance evaluation, and scouting purposes.

This work builds upon the defensive metrics introduced in the previous chapters and focuses on their application at the **individual player level**. The study proposes a framework to characterise players according to their defensive behaviour, introducing the concept of a **defensive signature** that captures both aggressiveness and consistency across matches and game contexts. By combining descriptive statistical analysis with graphical representations, this article demonstrates how defensive metrics can be used to differentiate player profiles, support comparative analysis, and enhance player evaluation processes. From the perspective of the thesis, this chapter extends the applicability of the proposed models beyond team-level analysis, highlighting their potential relevance for **scouting, performance assessment, and decision-making** in professional football environments.

Catalán, C., Calabuig, J. M., García Raffi, L. M. and Sánchez Pérez, E. A.
 “The Defensive Signature of the Player:
 Characterising Aggressiveness and Consistency in Defensive Pressure”.
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Together, these studies reveal a **methodological and conceptual progression**: from the exploration of event data and the introduction of artificial intelligence techniques, to the design of mathematical and dynamic models capable of quantifying and characterising the defensive game, supporting its interpretation in real time. Therefore, consolidate the main scientific contributions of this doctoral thesis, providing a **solid bridge between data analytics, mathematical theory, and the tactical practice** of modern football.

Chapters 6 and 7 refer to discussion and conclusions.

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Table 1.4: Distribution of event types and sub-types used for possession inference (Chapter 4)

Event type	Sub-type	Count
BALL LOST	CLEARANCE	11
BALL LOST	CROSS	10
BALL LOST	DEEP BALL	3
BALL LOST	END HALF	2
BALL LOST	FORCED	7
BALL LOST	GOAL KICK	3
BALL LOST	HAND BALL	2
BALL LOST	HEAD	19
BALL LOST	INTERCEPTION	126
BALL LOST	OFFSIDE	2
BALL LOST	REFEREE HIT	1
BALL LOST	THEFT	26
BALL LOST	THROUGH BALL	2
BALL LOST	NA	98
BALL OUT	CLEARANCE	11
BALL OUT	CROSS	1
BALL OUT	DEEP BALL	2
BALL OUT	HEAD	5
BALL OUT	THROUGH BALL	1
BALL OUT	NA	47
CARD	YELLOW	1
CARRY	NA	1395
CHALLENGE	ADVANTAGE	4
CHALLENGE	AERIAL	72
CHALLENGE	FAULT	38
CHALLENGE	GROUND	187
CHALLENGE	LOST	158
CHALLENGE	TACKLE	76
CHALLENGE	WON	157
FAULT RECEIVED	NA	12
PASS	CROSS	2
PASS	DEEP BALL	12
PASS	GOAL KICK	8
PASS	HEAD	31
PASS	OFFSIDE	2
PASS	THROUGH BALL	6
PASS	NA	1067
RECOVERY	BLOCKED	3
RECOVERY	GOAL	1
RECOVERY	INTERCEPTION	127
RECOVERY	SAVED	6
RECOVERY	THEFT	26
RECOVERY	NA	131
SET PIECE	CORNER KICK	8
SET PIECE	DIRECT	1
SET PIECE	FREE KICK	28
SET PIECE	INDIRECT	27
SET PIECE	KICK OFF	4
SET PIECE	RETAKEN	2
SET PIECE	THROW IN	50
SHOT	BLOCKED	3
SHOT	GOAL	2
SHOT	HEAD	1
SHOT	OFF TARGET	8
SHOT	ON TARGET	9
SHOT	OUT	6
SHOT	SAVED	7
SHOT	WOODWORK	2

2

Introduction

This chapter constitutes the starting point of the doctoral research and represents the initial step towards a quantitative and computational analysis of football, within the broader objective of modelling and characterising defensive behaviour. It focuses on the study of goal kicks as a key phase of the game, using **event data** collected from professional competitions to identify and classify tactical patterns through the application of **machine learning** and **statistical clustering techniques**.

The analysis is conducted in the specific context of the matches played by Levante UD, with the aim of classifying the teams they faced according to their behavioural patterns during goal-kick situations. Although goal kicks are not defensive actions per se, they represent a structured game phase that strongly conditions subsequent defensive organisation and pressure-related behaviours. By analysing how opposing teams approach this phase, this chapter provides an indirect but informative perspective on their defensive tendencies and strategic intent.

This chapter explores the use of **neural networks** and **hierarchical cluster analysis** to detect structures and regularities in how teams initiate possession from goal kicks and how these behaviours differ across opponents. Through this classification, this study illustrates how data-driven approaches can be used to group teams according to their tactical behaviour, offering a first step towards the systematic modelling of opponent strategies.

From the perspective of this thesis, this chapter played a crucial dual role. On the one hand, it enabled a comprehensive understanding of the **structure, processing, and manipulation of event data**, deepening the practical knowledge of the data pipelines commonly available in professional football analytics. On the other hand, it marked the introduction of **neural networks and machine learning methods** into the analytical framework of this research, providing exploratory tools that would later support the development of formal mathematical models.

Therefore, this first publication served as both a **technical and conceptual foundation** for the thesis. While its focus lies on classification rather than explicit pressure modelling, it establishes the methodological basis and analytical mindset from which the later chapters evolve towards a rigorous and quantitative description of defensive pressure using positional and temporal information.

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Using Neural Networks and Hierarchical Cluster Analysis to study goal kicks in football

Abstract

The objective of this study is to build a mathematical model that predicts the success of a goal kick in soccer. The model is based on an ensemble of Neural Networks whose inputs are 5 features extracted directly from the goal kick and one more that depends on the opposing team. This new variable is calculated using a Hierarchical Cluster Analysis and divides the possible opponents by their defensive strategy. It is shown that it has a relevant importance in the result of a goal kick, which validates the idea of using the presented methodology that takes into account the opponent's tactics when analyzing specific plays of a soccer match.

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Keywords: Football, neural networks, clustering, match analysis, soccer, mathematical model.

2.1 Introduction

As in other disciplines, in recent years data science is increasingly being used in the world of football. From a technical point of view analysts use two type of data: *tracking data* and *eventing data*. The use of cameras and image analysis techniques allow the monitoring of a complete football match on very small time scales. This sequence constitutes the so called *tracking data*. Usually tracking data consists of a set of two-dimensional coordinates for all the players and also the ball sampled at $25Hz$ —that is, 25 frames per second—. In some cases the third coordinate of the ball—the height—is also provided. This is, for instance, the case of [24]. Although it is a very rich dataset, the large amount of data and the difficulty in obtaining and handling it can sometimes be a barrier to its use. On the other hand *event data* is a file collecting all the events occurring in the match—goals, goal kicks, free kicks...—. Event data started to be stored by *Statsperform*—previously Opta Sports—in 1996 (see [23]). Unlike with tracking data, the process of collecting event data is not automatic. It has also some problems as, for instance, we do not know the context of an event—some providers, as [22], has started including this advanced information—. Recently [25] propose an automatic method to obtain event data from tracking data by using a computational framework.

Historically, researchers, coaches and data scientists mainly used statistical techniques for match analysis. However, with the increase in the amount of data to be analysed, artificial intelligence has become an emerging and fruitfully tool. Following John McCarthy, one of the fathers of Artificial Intelligence, Artificial Intelligence is “*The science and engineering of making intelligent machines that have the ability to achieve goals like humans do*”. In contrast to the intelligence of human beings, we speak of Artificial Intelligence to refer to the intelligence displayed by a machine. Artificial Intelligence is developed in different fields in which humans can have similarities with machines. Some of these fields (or branches) are Voice Recognition, Pattern Recognition, Image Processing or Natural Language Processing. One of the branches of Artificial Intelligence is known as Machine Learning.

Machine Learning basically focuses on the creation of systems capable of learning automatically and has three paradigms: Supervised Learning, Unsupervised Learning and Reinforcement Learning. In Supervised Learning the system is provided with a set of questions—or characteristics—as well as the answers—or labels—to these questions from a set of observations. From this data the system has to be able to predict the answers to other questions that are not in the given observations. In the case of Unsupervised Learning only the questions/characteristics are provided. There is no base of answers from observations against which the system can compare to do the learning. Finally, for the Reinforcement Learning, learning is done by recognizing an environment that provides rewards which the system must maximise.

Of course, as in other research areas, Machine Learning is widely used in football analysis to obtain information from data. It has been used for the last fifteen years both using event data—see for instance [17] and [18]—and tracking data—as the reader may consult in [10] or [21]—. In terms of the different types of learning, there do not seem to be any applications of Reinforcement Learning to football—the reader can consult [4] for an application to the economy field—. However, the applications of Supervised and Unsupervised Learning in football are wide-ranging. This is, for instance, the case

of [3] that use event data and Supervised Learning for prediction and classification—via k -Nearest Neighbours—or [26] that use tracking data and Supervised Learning for clustering—they use in this case k -means—. Recently, [9] have critically reviewed the application of Machine Learning in football in relation to attacking play.

In this paper we are interested in the answer of the following question of Levante UD's interest in the way a match is restarted: “*How successful is a goal kick going to be?*”. As it can be useful to adapt the strategy to be used in a match to each opponent, we add: “*And how does the opposing team influence it?*”. In order to address this question we have used the event data corresponding to the complete 2020/2021 season of Levante UD from Spanish LaLiga. We have created a mathematical model that allows us to predict the degree of success of a goal kick from six variables that, in a way, identify it. For this purpose, we begin by grouping the Spanish first division teams according to their defensive characteristics, since in the situations we are interested, Levante UD is the team that has the possession of the ball. That grouping is obtained by using an Unsupervised Learning clustering algorithm, and represents one variable of each goal kick: the *Rival Type*. Clustering algorithms have already been used for instance in [2]—the authors use k -means—or [27]—they use k -means and Decision Trees (and also Principal Component Analysis, PCA)—. The other five variables used are “easily” obtained from the data. This is, for instance, the case of the *Zone variable*—whose possible values are Right, Center or Left—. This variable indicates the area of the field of play that the ball reaches when the goalkeeper kicks the ball. Or also the *duration* of each goal kick. This is explained in the second section of this article.

Third section is devoted to the methods and results. From the six variables defining our model—and presented in the second section—, the model should be able to determine the degree of success of the goal kick. To this end, the possible values of the function are: 1 (failure), 2 (potential success) and 3 (total success). In short, the function defining our mathematical model is:

$$f : \mathbb{R}^6 \rightarrow \{1, 2, 3\}.$$

Neural Networks is the Supervised Learning technique we use in order to answer our question. This technique have been used, for instance, in [14] and [1]. In this section we start by explaining the architecture of a Neural Network, and, after this, we present the obtained results.

The fourth and last section is dedicated to the conclusions and future works.

2.2 The mathematical algorithm

We start this section by explaining the preparation of the data. As we said before our initial data consists in a collection of different events occurred in a match. In our case they are prepared by Levante UD and corresponds to the season 2020/2021. The dataset is created from data given by providers and it is usually made in real-time by a combination of manual annotations, Computer Vision and some Artificial Intelligence models. The reader may consult <https://www.statsperform.com/opta/> of StatsPerform (one of the data providers from Levante UD in 2020/2021 season). For this work we have extracted from the eventing data the information of goal kicks. For this purpose, we consider that a *goal kick play* is a sequence of events that starts when the goalkeeper—o some outfield

player—re-starts the game with a goal kick and ends when Levante UD loss the possession of the ball or the game stops. In addition to storing all events between the start and end of the goal kick play we have also collected the next event. This event will be helpful to determine the reason why the play has stopped. In Figure 2.1 we can see an example of a whole goal kick play, composed by 17 events on 9 seconds.

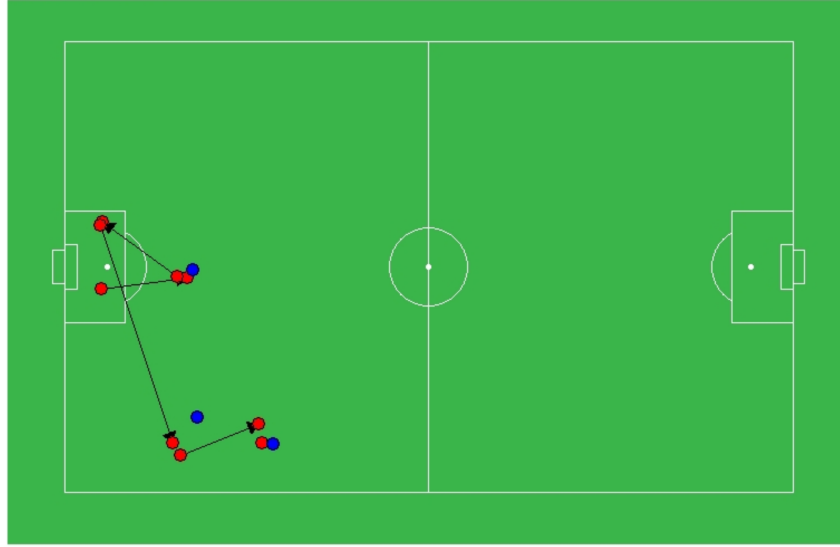


Figure 2.1: Example of a goal kick play. In red Levante UD players and in blue the opponent ones. The arrows represent passes between Levante UD players.

Let us explain this example in more detail. Figure 2.2 represents the first four events of the example. First (Event 1) the goalkeeper of Levante UD (which is colored in red) passes the ball to player number 9. This corresponds to one row in our dataset and the value of the *type.name* column is *Pass*. This is shown on the left side of Figure 2.2. In the center of the Figure player 9 carries the ball. In this case we have two events (Events 2 and 3) of the goal kick play. These two events are labeled in our data as *Control* and *Carries* (again in the *type.name* column) corresponding to two arrows in the database. Finally on the right part of the figure a new event (Event 4) appears: the number 9 of the Rival Team (color blue) is pressing player 9 of Levante UD. This is again a new row of the data. In this case the value of the *type.name* column is now *Pressure*.



Figure 2.2: Events of the first part of the example of goal kick play (in red the players of Levante UD). On the left side Event 1 is a pass (from player 1 to 9). On the center part we show two events: Event 2 is a control and Event 3 a conduction. Finally on the right side Event 4 is a pressure of player 9 (in blue) of the opponent team.

Next events of the example are displayed in Figure 2.3. On the left side a new event (Event 5) is showed: player 9 passes the ball to player 3. On the right part we see three events (Events 6 to 8): player 3 *controls* the ball and *carries* it and then he *passes* the ball to player 2.



Figure 2.3: Events of the second part of the example of goal kick play. On the left side Event 5 is a pass from player 9 to player 3. On the right side we show three events: control of player 3 (Event 6), a small conduction (Event 7) and a pass (Event 8) to player 2.

The final part of the goal kick is shown in Figure 2.4. On the left side we show four events (Events 9 to 12). Event 9 is a pressure of player 16 (blue dot), Event 10 a control and Event 11 a ball conduction (from player 2). Event 12 is a ball pass to player 12. Finally on the right side we show five events (Events 13 to 17). First a control and a conduction of the ball. Then player 6 (blue dot) appears having a duel (Event 16). Last Event 17 of this goal kick example is the intercept of the ball of player 6.

In summary, from the original data we first have a new database that groups sequences of events to identify a goal kick play. With all this we obtain a new database with 282 goal kicks. Figure 2.5 shows the distribution of the 38 matches of the season and the average (which is 7.42).

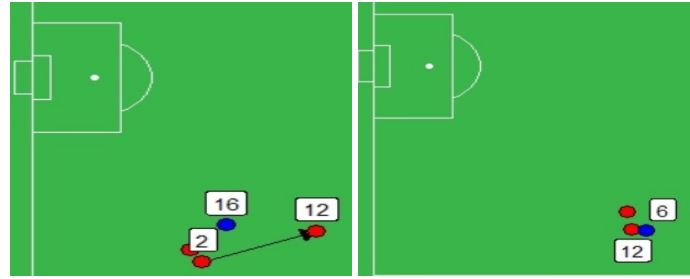


Figure 2.4: Events of the final part of the example of goal kick play. On the left side we show four events: Event 9 is a pressure of player 16 (in blue). A control (Event 10), a conduction (Event 11) of player 2 and a ball pass (Event 12) from player 2 to player 12 are the next events. On the right side we show five events: a control (Event 13) and a conduction (Event 14) by player 12, a duel against player 6 of the opposing team (Event 15) and the loss (Event 16) and intercept (Event 17) ending the play.

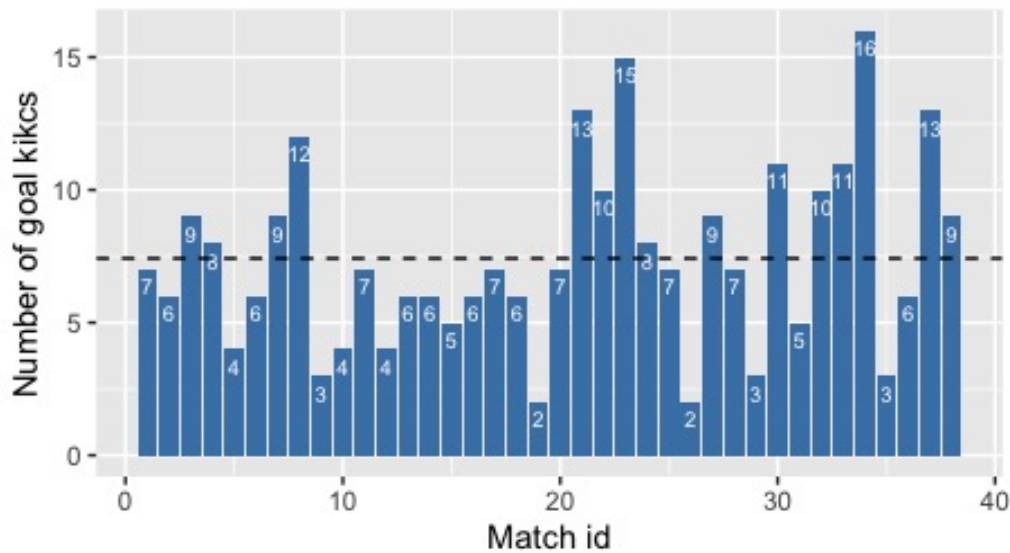


Figure 2.5: Distribution of the number of goal kicks per match and average in the 2020/2021 season.

As can be seen in Figure 2.6 a clear trend towards the short goal kick is observed.

2.2.1 Clustering of LaLiga teams by their defensive behavior

First of all, we are interested in correlating the opponent team with the success of a goal kick. But the data we have is of only one season, which makes no sense to study each possible opponent separately (as only about 15 goal kicks per season per opponent are available). We solve this problem studying the possible opponents that follow a similar strategy as one group of teams. To do so, we study only the relevant characteristics of that team when defending an opponent's goal kick. At the beginning the only three selected features are the number of

- a) recoveries,
- b) pressures, and

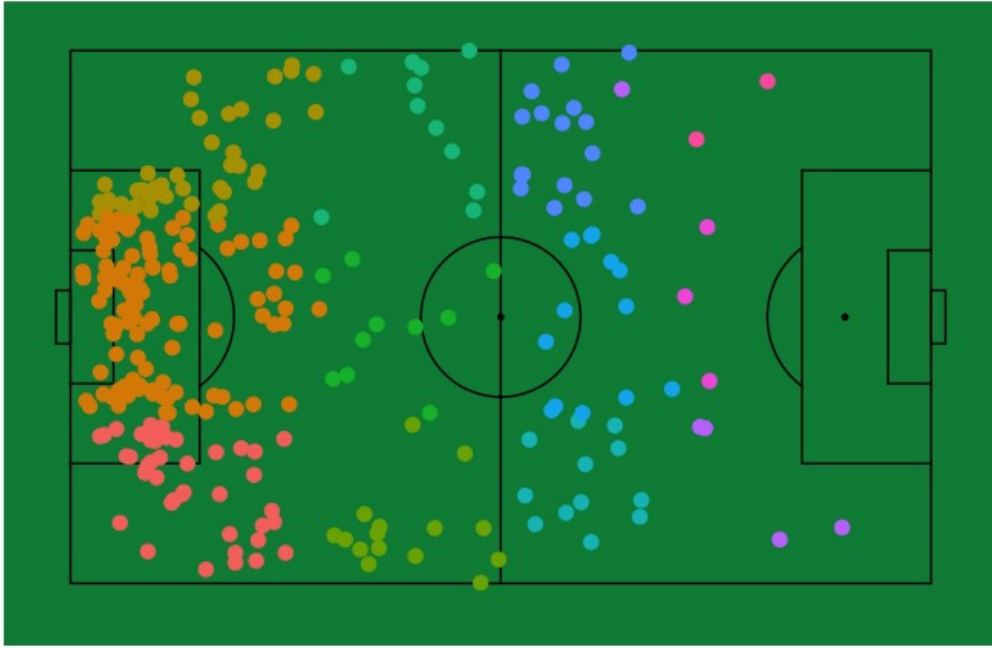


Figure 2.6: Distribution of goal kicks in the 2020/2021 season. Each dot represents the end of a goal kick that Levante UD made in this season. The colors refers only to the different zones of the pitch (see Figure 2.11).

c) fouls,

that LaLiga teams made in that competition during the season under study. These three variables are directly obtained from the raw data. Indeed each event of the dataset is labelled by using a “type name” (for instance, Pass, Block or Dribble). In our case we count, for each opponent team, the number of events of each of the three labels: Ball Recovery, Pressure and Foul Won.

The idea of a clustering algorithm is to obtain a classification of our data—the rival teams in our case—into groups—the clusters—so that distances within the groups are small but distances between groups are large. Clustering have been used in Football in recent papers as [7]—using k -means with Tracking data—or [19]—which also use k -means but with both Tracking and Event data—.

In our case we use the *Hierarchical Cluster Analysis* (HCA), that is an Unsupervised Learning technique that involves the creation of clusters that have a predominant top-down ordering. The Hierarchical Cluster Analysis is divided in two categories: Agglomerative and Divisive. For our work we use the Agglomerative case. This is the more common type and works as follows: in the first step, each point—a team in our case—constitutes an individual cluster. Afterwards, in each iteration we merge clusters together. We finish the iterations when there is only one cluster left.

In order to merge clusters there are several methods and distances that can be used. First, from the distances point of view we can use Euclidean distance—classical d_2 —, Manhattan distance— d_1 —or Maximum distance— d_∞ —. In our case we have used the

so-called *Canberra Distance* which is given by the expression:

$$d_C(x_1, x_2) = \sum_{j=1}^p \frac{|x_{1j} - x_{2j}|}{|x_{1j}| + |x_{2j}|}.$$

We use this distance because is usually used for non-negative values—as, for instance, counts—. Note that these distances are also the distances between the clusters in the first step—recall that here the clusters are the individual teams—. For the next steps several methods can then be used to calculate the distance between clusters. For instance, we can define the distance between two clusters as the distance between the two most similar objects from these clusters—that is the minimum—or between the most dissimilar—which would be the maximum—. In our case we use the *Ward's Method* of minimum variance. Due to Joe H. Ward, this method tries to minimise the total within-cluster variance. This is a generally popular and effective method producing more compact clusters. Actually minimizing within cluster variance can be done by using the sum of errors—ward.D—or the sum of squared errors—ward.D2—. In this article we use ward.D.

After applying the Hierarchical Cluster Analysis with the Canberra Distance and the Ward.D method the corresponding dendrogram is shown in Figure 2.7. By descending on the three, we can determine the number of clusters to choose. We have selected four clusters, as also shown in the same figure. A representation of the defensive features studied by clusters are shown on Figure 2.8.

The three characteristics appear on the figure, where a *Principal Component Analysis* (PCA) has been applied to the data only with a visual purpose. The PCA is a technique that recomputes the original variables in new ones, where the new variables are a linear combination of the previous. This is done in a way that the first ones are the ones with more information (the ones with highest variation) of the data, see [11] for a visual explanation of the method.

Analyzing the results, we can see that the number of teams on each cluster is not equally distributed and the clusters are not very metrically disjoint. In fact, clusters 3 (blue) and 4 (green) intersects on the representation of Figure 2.8, but joining those clusters (selecting only three groups) does not seem to improve the clustering, because clusters would be more unbalanced and teams with very different playing styles would belong to the same cluster.

For example, in the Type 4 (green) teams we can find FC Barcelona and Cádiz CF, two teams with very different defensive behavior. For instance, FC Barcelona conceded 38 goals, with an average of 1 goal per match (being the fourth better team). However Cádiz CF received 58, with 1.53 of average (being the second worse team). FC Barcelona was the team with the highest ball possession in the season studied and Cádiz CF the one with significantly less possession (almost half that of FC Barcelona). Both are grouped in the same cluster, since the three variables studied are similar for the two teams, FC Barcelona because is attacking most of the time and Cádiz CF, on the other hand, prefers to wait defending until the opponent loses the ball.

To solve those problems, as each team has not been defending the same time, we modify this three features and computed all per hour without the possession of the ball (as a percentage). It results in the following formula (and equivalent ones for the Pressure/h

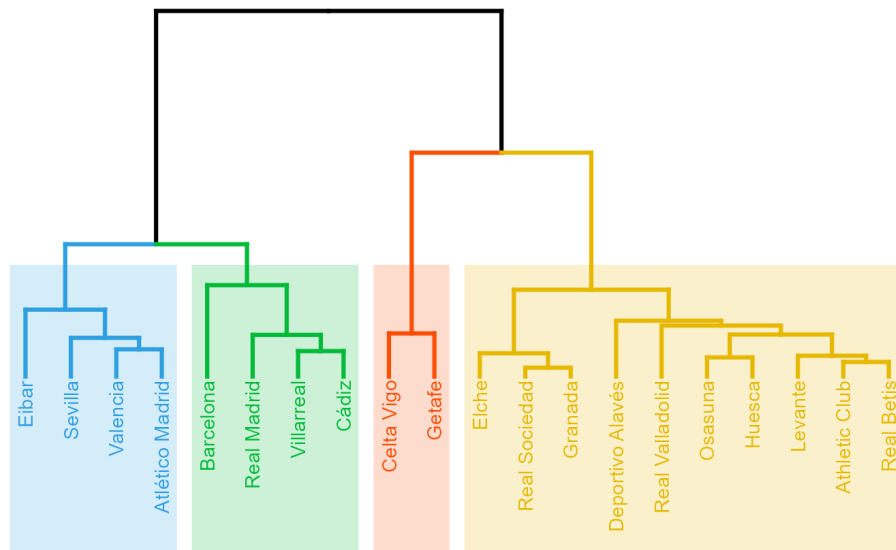


Figure 2.7: Cluster dendrogram of team defensive Type. The numbers assigned to each group are 1 (red), 2 (yellow), 3 (blue) and 4 (green).

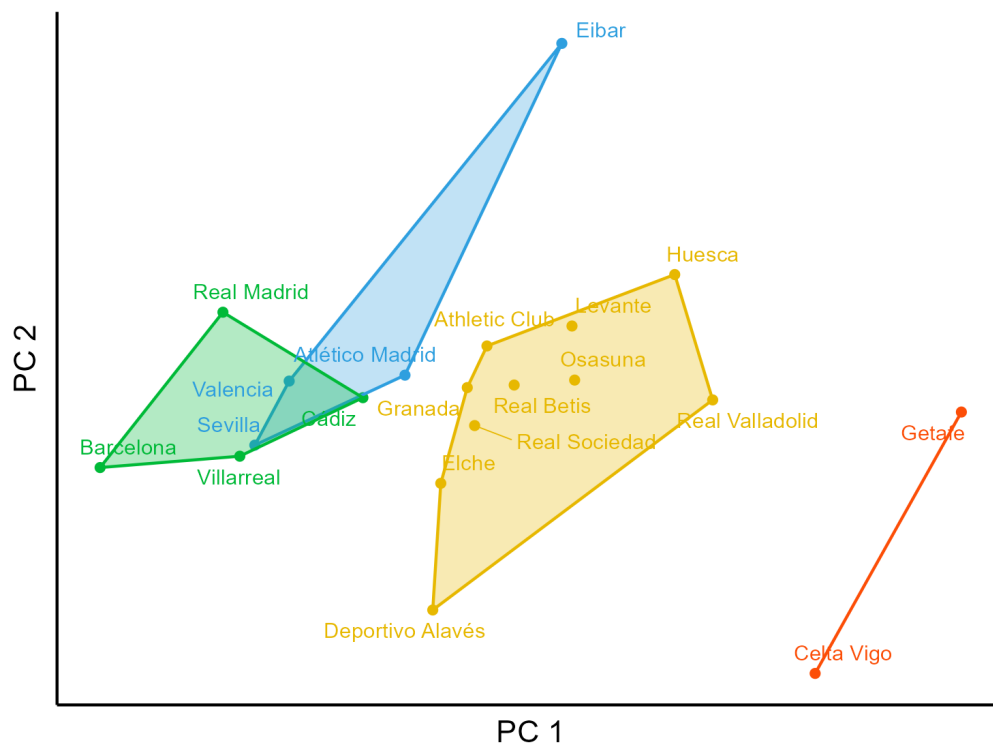


Figure 2.8: Two principal components of the PCA applied to the defensive data of the teams, which both represents more than 88% of the variance.

and Fouls/h):

$$\text{Recoveries/h} = \frac{\text{Number of recoveries}}{1 - \text{Possesion mean}} \cdot \frac{60}{90 \cdot 38},$$

since each team played 38 matches on one LaLiga season.

We apply the same method with this three new variables:

- recoveries/h,
- pressures/h, and
- fouls/h.

The resultant tree can be seen in Figure 2.9.

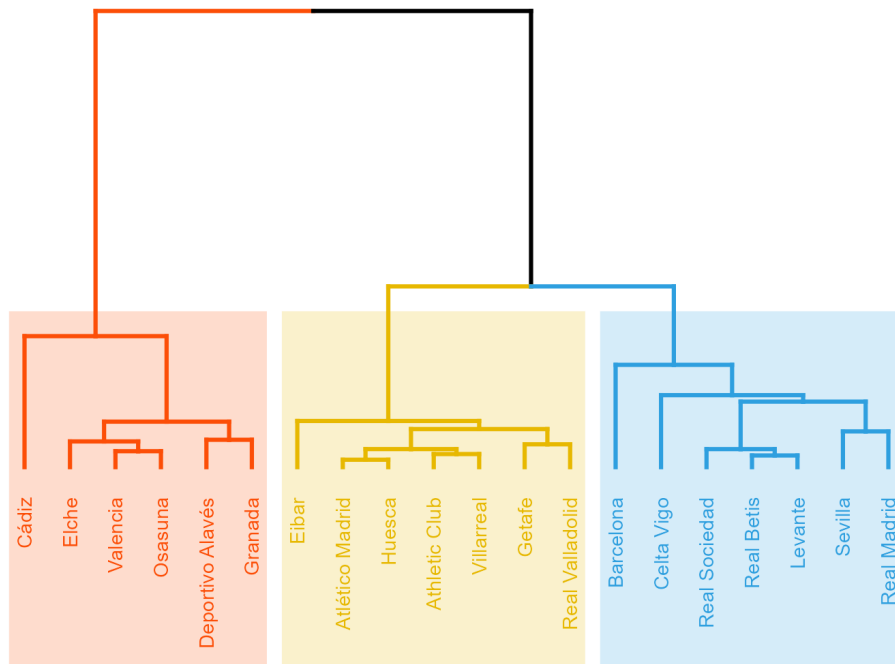


Figure 2.9: Cluster dendrogram for obtaining *Rival Type* variable. The numbers assigned to each group are, from left to right, 1 (red), 2 (yellow) and 3 (blue).

In this case, we determined three clusters, since adding one more cluster would make Cádiz CF form a singleton cluster. The studied characteristics of each group can be seen in Table 2.1.

Analysing the results, we obtain a more reasonable grouping of teams, with more metrically separate clusters, see Figure 2.10:

- Type 1 teams—which we will call *defensives*—are characterised by a lesser predisposition to recover the ball and more time defending, waiting for the opponent's action.
- Type 3 teams—*recoverers*—show a more offensive behavior and put more pressure on the opponent to win the ball quickly when defending.
- Finally, Type 2 teams—*neutrals*—represent a middle ground between the two previous clusters.

On Table 2.1 the average defensive features of those clusters are presented. Observe that Levante UD belongs to the *recoverers* teams.

Team type	Recoveries/h		Pressures/h		Fouls/h	
	mean	sd	mean	sd	mean	sd
1. DEFENSIVES	50.39	3.95	159.70	11.46	17.60	2.22
2. NEUTRALS	60.44	3.37	186.48	6.43	19.86	0.96
3. RECOVERERS	67.40	5.50	213.96	21.95	22.81	2.67

Table 2.1: Average defensive characteristics per hour without possession of each group obtained using the HCA algorithm.

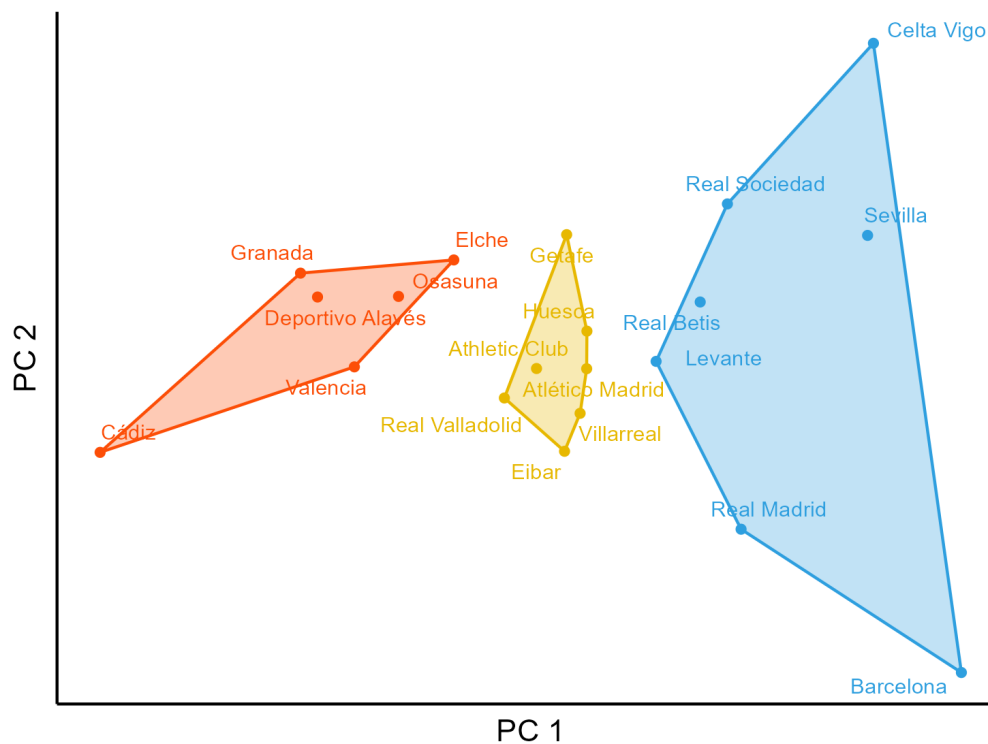


Figure 2.10: Two principal components of the PCA applied to the new defensive features of the teams. Both components represents more than 95% of the variance.

2.2.2 Characteristics of a goal kick play

Given a goal kick play, we select the most relevant variables to generate the database that will be used for the predictive model. Our final dataset will be composed by the 282 goal kicks/rows each one with the following six variables.

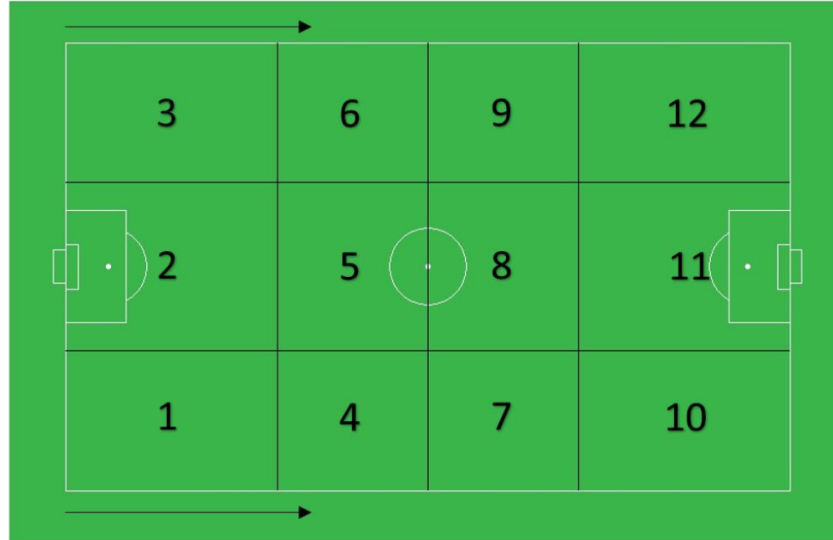


Figure 2.11: The soccer field is divided into 12 zones, assigning a number to each one. The vertical lines dividing the field are placed at 35, 60 and 85 meters; and horizontal ones at 25 and 55 meters.

- Serve Zone.* Using the field area where the kick is directed—see Figure 2.11—, we assign the values 1 for right kick (zones 1, 4, 7 and 10), 2 for center kicks (2, 5, 8 and 11) and 3 for left ones (3, 6, 9 and 12). Although the dimension of the pitch can change between stadiums, the size is rescaled to 120×80 meters, with Levante UD goal always situated at $x = 0$.
- Duration.* Time difference—in seconds—between first and last event.
- Last Player.* Denotes the playing position of the last player in possession—observe that this is always a Levante UD player—. It is an integer between 1 (goalkeeper) and 24 (left forward) which denotes how offensive or defensive is the position.
- First Rival Player.* Position of the next player in possession in the first event after Levante UD loses the ball or play is stopped. Observe that it can be a player from the opposing team—for which we use the same classification to the previous one—, a Levante UD player or none player. If it is a Levante UD player, a value of 30 is assigned to this variable and if the game ends on that action, a value of 32 is assigned.
- Time.* Time difference—also in seconds—between the last event before Levante UD loses the ball or the game stops and the next one.

As we said in the introduction, these five variables are almost automatically obtained from the original database. Indeed, for the Serve Zone variable we just use the three coordinates (x, y, z) of the first player taking the ball after the goalkeeper. The Duration is just the time difference between the last and the first event of a goal kick. For the Last Player we use the coordinates of the position of the last event of each goal kick. However

in football we cannot ignore the opposing team when trying to analyse events in the game. For this reason, we have introduced the variable resulting of the above clustering.

- f) *Rival Type*. This new variable describes the defensive behavior of the opposing team.

2.2.3 Classification of the result of a goal kick game

As we want to use a Supervised Learning technique to predict the result of a goal kick, we need to label our data. To do so, we add to our database a new column, called *Result*, that represents the degree of success of each goal kick. Although the objective on a football match is score more goals than the opposing team, we can not classify only goal kicks that end on a goal, since it is a very uncommon situation in football matches —less than 2.5 goals are scored on average per game in the studied LaLiga season—.

For us a goal kick has been successful if possession is maintained with some control of play after the kick has been taken, or in the case of a lost kick if there is no danger of a counter-attack or conceded goal. On this premise we have labeled each goal kick of our event data following next ideas:

- a) A goal kick that is delivered directly to the opponent will be a *failure*—labelled as 1—. It is also a miss when the goal kick movements do not proliferate sufficiently to go beyond the first third of the field or the ball is lost without going beyond the middle of the field.
- b) A *potentially successful* play—labelled as 2—when the ball is lost beyond the middle of the field without reaching the offensive zones, or when a foul is received in the own half of the field.
- c) Finally, it would be *total successful*—labelled as 3—in plays that reach the last third of the field, end in a goal, or after a foul in the opponent's half.

The labels are distributed as follows: 90 failures, 94 potential successes and 98 total successes. The distribution is presented in Figure 2.12.

Figure 2.13 shows the distribution of goal kick success against each opponent class. Note that Levante UD belongs to the *recoverers* cluster (blue), and the result of the goal kicks versus other teams of the same cluster is equally distributed on the 3 classes. In goal kicks versus *defensives* teams, a high degree of success is obtained. Finally, the result against *neutral* teams has a lower proportion of success than the *recoverers*. That surprising fact can be explained by the fact that Levante UD belongs to the latter group.

If we divide the classification of the *failures* into *total failures*—the goal kicks that end with loss of possession in the areas closest to one's own goal—and *potential failures*—the others—, the previous fact becomes clear (Figure 2.14). We can observe a worse result against *neutral* teams than *recoverers* in goal kicks that ends up in a dangerous defensive situation for Levante UD.

Once the six variables identifying a goal kick are defined and the labeling is done, we return to our original question: “*How successful is a goal kick going to be?*”. To answer this question from the numerical value of the six variables the system has to be able to establish how successful the goal kick is. In summary our mathematical model is a function

$$\mathbb{R}^6 \rightarrow \{1, 2, 3\},$$

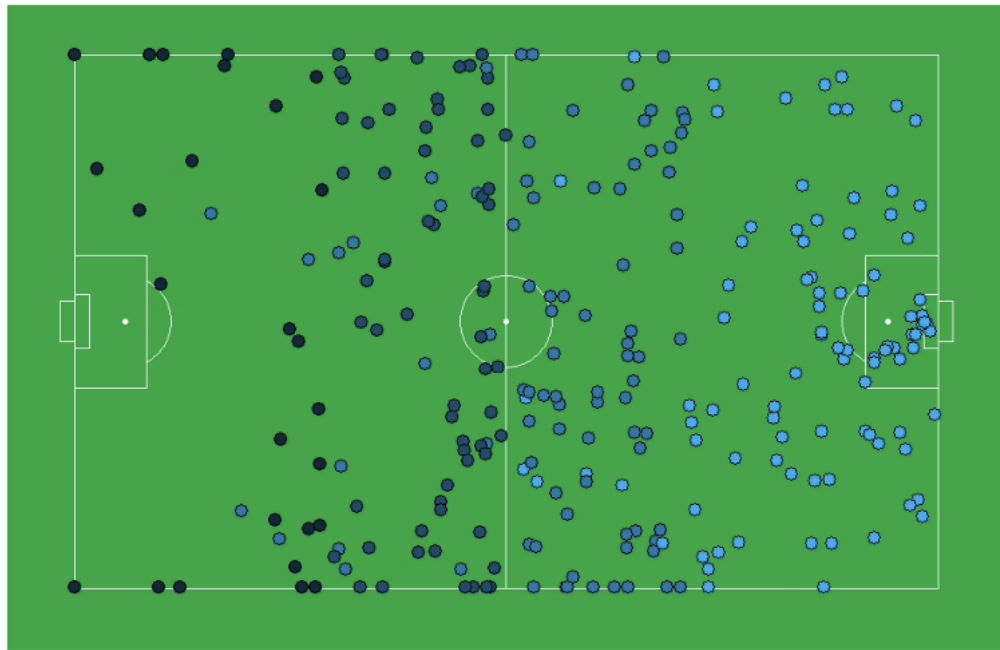


Figure 2.12: Field distribution of the labels in our database: failure, potential success and total success. The goal kick is from the left-hand goal.

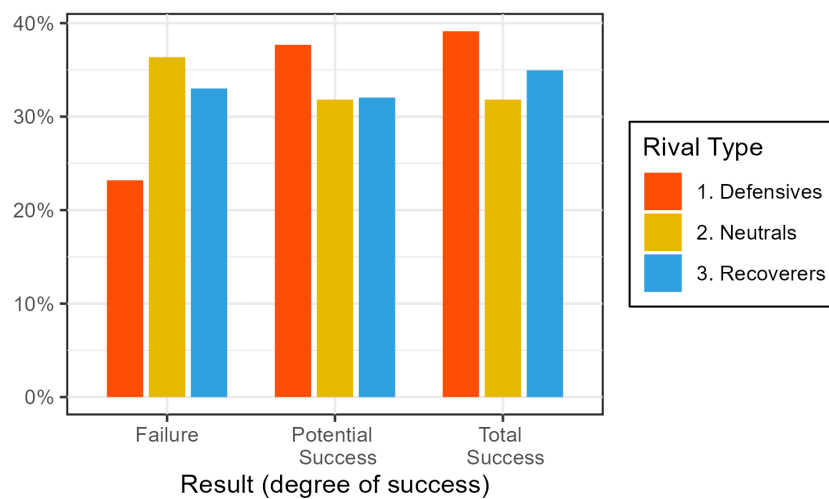


Figure 2.13: Distribution of goal kicks versus each rival type on the three possible results (failure, potential success and total success).

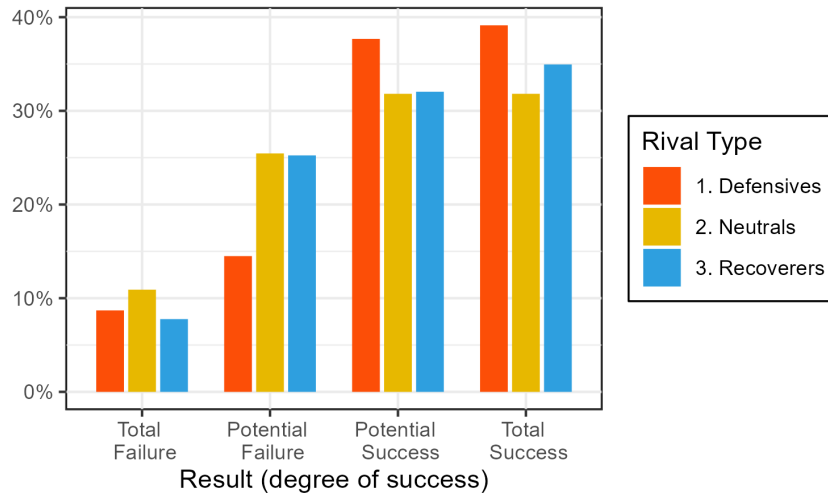


Figure 2.14: Distribution of goal kicks according to possible outcomes (*failure* is divided into *total failure* and *potential failure*). This classification is for comparative purposes only and is not used in the following section.

which takes a goal kick, identified with a vector align $v = (\text{Serve Zone}, \text{Duration}, \text{First Rival Player}, \text{Last Player}, \text{Time}, \text{Rival Type}) \in \mathbb{R}^6$, align and must give the goal kick's probability of being a *failure*, a *potential success* and a *total success*. The prediction of the model will be the most probable class, so the final model will be a function

$$f : \mathbb{R}^6 \rightarrow \{1, 2, 3\},$$

that assigns to v —and so, the goal kick—a predicted result.

2.3 Methods and results

To calculate the likely success of a goal kick we have used a *Neural Network*. Neural Networks have recently been used—for event data in football in [15] or [20] and also for tracking and event data in [12], in this case by using Convolutional Neural Networks—. In our case, in order to use Neural Networks as a Supervised Learning technique to classify the goal kick—into the three categories of success—we use the labels above computed.

The first step in using Neural Networks is to determine their hyperparameters. In order to find the best possible ones we have sampled 250 different architectures of networks between 1 and 5 hidden layer with a number of neurons each one between 4 and 40. Between each two layers of neurons, a linear transformation is applied with weights to be trained, composed with an activation function. The activation functions for the intermediate layers are the *Rectified Linear Unit* function (ReLU), given by:

$$\text{ReLU}(x) = \max(0, x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

for $x \in \mathbb{R}$, applied independently on each component. On the last layer we have used the *SoftMax* function—commonly used in the output layer of a classification network—with

3 values defined on each component by:

$$\text{SoftMax}(\bar{x})_i = \frac{e^{x_i}}{\sum_{j=1}^3 e^{x_j}},$$

for $\bar{x} = (x_1, x_2, x_3) \in \mathbb{R}$ and $i = 1, 2, 3$. The learning rate is also randomly sampled between 10^{-1} and 10^{-3} . The loss function used to compute the model is *sparse categorical crossentropy*. The *accuracy* metric is used to choose when to stop network training. See [8] for more information and the explicit formulae of those metrics.

Then, the model is a composition of functions as the following.

$$\text{SoftMax}(b_n + W_n \cdot \text{ReLU}(b_{n-1} + W_{n-1} \cdots \text{ReLU}(b_1 + W_1 \cdot X))),$$

where $X \in \mathbb{R}^6$ represents the input. The matrices W_i and vectors b_i for $i = 1, 2, \dots, n$ are the parameters of the model, that will be optimised when the hyperparamters be fixed. Thus, the model output is not only a prediction of the level of success of a goal kick, but also an estimate of the possibility of being in each class of success. The 5 first variables used are also scaled to be in the interval $[0, 1]$. The variable *Rival Type* is encoded—using the so called one hot encoding—into 3 variables (one per each type of opponent), so an opponent of type 1 take the value $(1, 0, 0)$, one of type 2, $(0, 1, 0)$ and one of type 3, $(0, 0, 1)$. So the input vector, v , is transformed to a new one, X with 8 components between 0 and 1.

As the dataset used in this study is too small, a cross-validation ensemble is implemented—see [13] for more information about this technique—. For that purpose, the 282 goal kick plays are divided in 6 groups of 47 and model will be trained on 6 different distributions of the training, validation and test sets. Each experiment gives a result with the final result of the ensemble being the mean—see Figure 2.15—.

For each sample of hyperparameters, 6 independent Neural Networks are trained—each one with a different distribution of training, validation and test sets—then the loss and accuracy are computed. To train all the networks, the *Keras* R package—[5]—based on *Tensorflow* is used—[16]—. As one might expect, with the small number of data we have used—only 282 goal kicks— big networks with many layers, neurons and trainable parameters might not reveal an improve on the accuracy. The results are shown in Figure 2.16 and the best ones are presented in Table 2.2. Those results validate that adding more complexity to the network is not justified, and a network with only one layer has the best performance.

Once the best hyperparameters are chosen: one hidden layers of 16 neurons with a learning rate of 0.1, that architecture is fixed. We can see the resulting model on Figure 2.17.

To analyse the performance of the model, the Neural Network is trained individually on the 6 samples of the cross validation, as explained before. For all of those samples—with training, validation and test sets each one—, 20 independent networks are trained, and the final output—one number between 0 and 1 for each success class—is computed as the mean of the 20 networks, see the diagram in Figure 2.18.

This doubled ensemble technique is used to reduce two of the main sources of error, in particular on neural networks with a small data set:

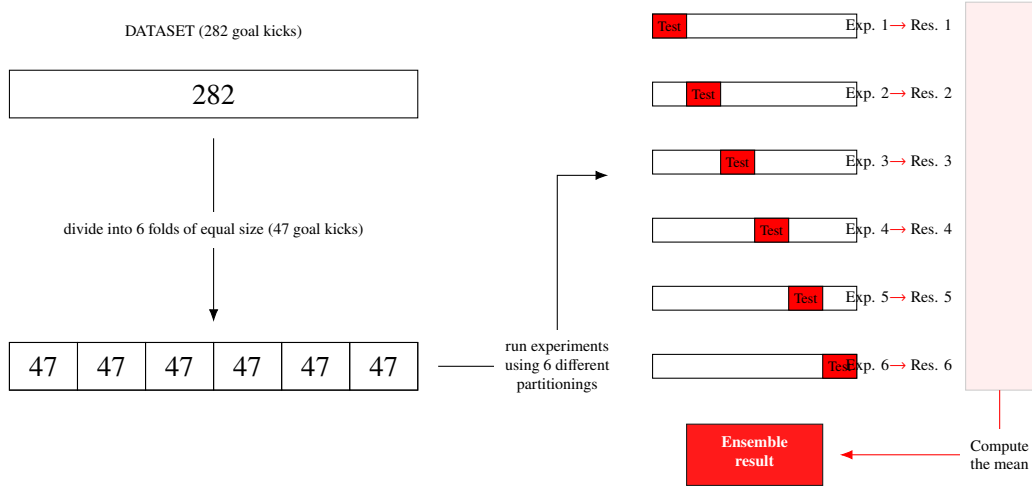


Figure 2.15: Cross-validation ensemble: the complete dataset is divided into 6 folds. We run 6 experiments with different partitions (of test, training and validation). Each experiment gives a result. The final result of the ensemble is the mean of the obtained results of the experiments.

layers	neurons	lr	param	loss	accuracy
1	16	0.10	195	1.017	0.5656
1	15	0.10	183	0.998	0.5585
1	31	0.10	375	1.005	0.5585
1	39	0.10	471	1.043	0.5532
1	33	0.10	399	1.028	0.5514
2	14, 16	0.10	417	1.081	0.5496
1	32	0.10	387	1.012	0.5461
1	38	0.10	459	1.015	0.5461
1	22	0.10	267	1.025	0.5425
2	38, 38	0.02	1941	1.007	0.5390

Table 2.2: Architectures of the Neural Networks with best score. *lr* represents the learning rate and *param* the number of trainable parameters. The loss and accuracy on the validations set are also given.

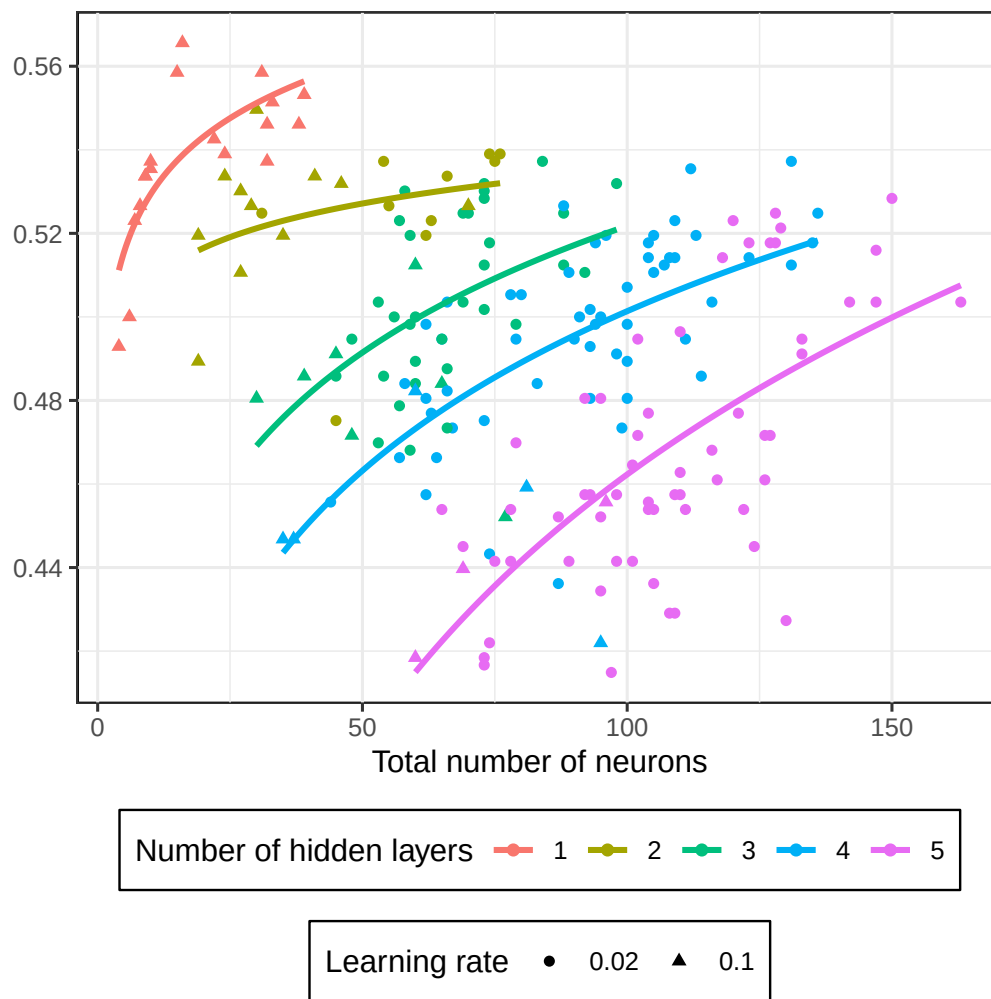


Figure 2.16: Accuracy obtained in the sample hyperparameters of the network. Only the best learning rate result for each architecture is shown.

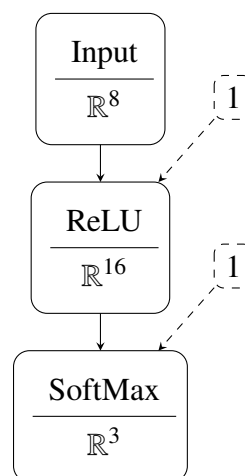


Figure 2.17: Diagram of the resultant Neural Network. Each arrow represents a fully connected layer and dashed squares represents the bias of each layer.

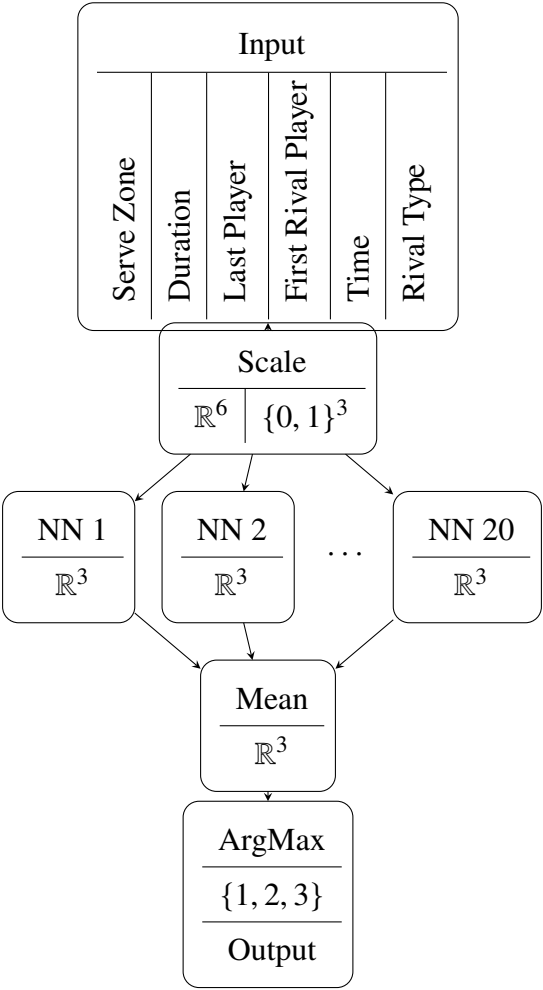


Figure 2.18: Diagram of the ensemble model. The architecture of each Neural Network (NN) is shown in Figure 2.17.

- a) The selection of the test and training sets, which can cause that the error computed on the test set is not representative of the overall goal kicks. Therefore, a cross-validation ensemble is implemented, see Figure 2.15.
- b) The random sample of starting weights on the neural networks influences on the outcome after training, resulting in a huge variance when training independent networks on the same set. In each cross-validation experiment, an ensemble of 20 neural networks is trained, each with different initial training parameters (the network weights) to reduce model variance, see Figure 2.18. We have checked the result using the average of the 20 predictions and all metrics are better than the single predictions given by the best of the 20 neural networks launched. This is shown in Table 2.3.

Average/Individual		
accuracy	precision	recall
0.543 /0.525	0.531 /0.526	0.537 /0.522
f1score	MCC	Kappa
0.524 / 0.517	0.313 /0.286	0.307 /0.282

Table 2.3: Comparative table of the results taking into account the average of the predictions or the individual results for the neural network. Best result occurs with the average of the 20 predictions.

The performance of the final model can be seen in Table 2.4. All of the metrics presented take values in the interval $[0, 1]$ —where 0 represents a totally wrong classification and 1 a perfect one—. An accuracy of 54% is obtained, and the model predicts especially well the *failure* and *total success* actions, while it is more hesitant in the *potential success* goal kick plays.

accuracy	precision	recall	F1-Score
0.543	0.531	0.537	0.524
Success class	1	2	3
F1-Score	0.559	0.394	0.618

Table 2.4: Method performance results. Each value is calculated as the mean of the model on each 6 samples of the cross validation. The first values are computed over the whole test set and those in the second table, separated by each success class.

To analyse the sensitivity of the last variable introduced, the *Rival Type*, we evaluate the test set on the model modifying that variable. We obtain than, in 26.2% of the cases, the result of the model changes. In the case where the variable *Rival Type* is modified, the accuracy of the model decreases to 27.1%, half of our model.

Studying the results, the model performance is not sufficiently high to classify perfectly the goal kicks using the studied variables. This is not surprising, since in football—as in many other sports—predicting the result of a move is very difficult by the number

of variables that influence. Even so, the model shows the importance of classifying the opponent before training those models.

2.4 Conclusions

In this paper we present a methodology to deal with a problem of predicting the success of a soccer play. The model we present is an ensemble of Neural Networks with inputs such as *Kick Zone*, *Duration* and other variables depending on how the game is restarted. A new variable which defines the opponent's defensive style, *Rival Type*, is added. The output of the model is a probability that a kickoff play will be a *failure*, a *potentially success*, or a *success*.

The study conducted to classify the teams defensively is obtained through a clustering algorithm: Hierarchical Cluster Analysis. Using the variables *recoveries*, *pressures*, *fouls* and *possession*, it seems to group the teams reasonably and well separated. The 20 teams of LaLiga are grouped in three clusters which we call *defenders*, *neutrals* and *recoverers*. In the future, this clustering could be refined using other statistics commonly collected by data providers, such as the distance from their own goal to the zone they are defending. Other relevant variables may be those that study what the team usually does after recovering the ball, as it may influence on the tactics used when attacking these teams.

Once the clustering is done, the new variable *Rival Type* is added to the database. Then, a predictor of the success of a goal kick based on neural networks is implemented. To establish the network architecture, 250 possible networks are trained. The best hyperparameters obtained are 1 hidden layer with 16 neurons and *ReLU* as activation function; and the output layer with the 3 outputs and the *SoftMax* function. The final model is an ensemble of 20 neural networks using the previous architecture. The performance obtained is that 54% of the test goal kicks—those which have not been seen by the model—are well classified.

Our models can be a good tool for soccer clubs when studying plays from a tactical-strategic point of view. Moreover, since we have international football data providers for our study (coming from the usual data sources used by clubs), our model and methods can also be applied to other leagues (such as the Premier League, Bundesliga...). But they need to be trained with a large amount of data, especially because the most important plays (those that end in a goal or result in a dangerous enemy counterattack, for example) are the least common. As a consequence, the accuracy we obtain in our study is not very high. However, we consider that the incorporation of the *Rival Type* variable seems to have influence, since the model shows an important sensitivity when this variable is modified for the same plays. In fact, the accuracy of the model drops to half of the original model when the opposing team is voluntarily dropped out.

On the other hand, it may be that the differences displayed in Figure 2.14 are not entirely due to the type of team faced and that other factors may play a role. Obviously, this study has not considered the individual metrics corresponding to each of the players of the football match. In future work, individual metrics of the players participating in the match could be included in the model.

Finally, other machine learning techniques can be used. For future work we can explore the possibility of using nonlinear models such as autoencoders. These methods

have recently been used in other fields to obtain better results taking into account the nonlinearity of the problems (see [6]).

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3

Introduction

This chapter represents a significant methodological and conceptual evolution within the framework of this doctoral thesis. Building on the foundations established in Chapter 2—where event data and neural networks were employed to identify tactical patterns—this work moves towards the development of a **mathematical model** designed to quantify and interpret **defensive pressure** in football.

Unlike traditional descriptive or event-based analyses, this study relies on **tracking data** and the spatial–temporal interaction between players to construct a rigorous mathematical representation of defensive behaviour. The model proposes a hierarchy of three analytical levels: individual, collective, and global (or game) pressure. Each level captures a different dimension of the defensive process, from the immediate action of the individual player to the coordinated behaviour of the defensive block and the overall systemic response of the team.

This framework introduces the notion of defensive pressure as a **continuous and measurable quantity**, rooted in spatial geometry, positioning, and relative player distances. Through this approach, defensive behaviour can be objectively characterised, offering an alternative to the traditional event-based metrics commonly used in football analytics.

From the broader perspective of the thesis, this paper constitutes a key step in the transition from data exploration to mathematical formalisation. Up to our knowledge it represents the first complete attempt to define and validate **quantitative defensive metrics** derived from theoretical principles rather than heuristic observation. These metrics not only enhance the scientific understanding of defensive organisation, but also provide a foundation for models capable of producing tactical insights in near real time.

In addition, the work highlights the potential application of these metrics for **coaches, analysts, and performance platforms**, enabling the monitoring of defensive behaviour,

the assessment of defensive performance, and the communication of key insights through data-driven media narratives. In this sense, it bridges the gap between mathematical modelling and practical football analysis, illustrating how abstract models can inform decision-making processes in high-performance contexts.

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A Mathematical Model to Study Defensive Metrics in Football: Individual, Collective and Game Pressures

Abstract

Performance analysis, utilizing video technology and recent technological advancements in soccer stadiums, provides a wealth of data, including player trajectories and real-time game statistics, which are crucial for tactical evaluation and decision-making by coaches and players. This data allows for the definition of metrics that not only enriches the experience for soccer fans through enhanced visual displays, but also empowers coaching staff and managers to make informed, real-time decisions that directly impact match outcomes. Ultimately, it serves as a pivotal tool for improving team strategy based on comprehensive post-match data analysis. In this article, we present a mathematical model to study the concept of pressure between players and, subsequently, between teams. We first explore the concept in a fixed frame of a match, determining what we call *influence areas* between players. We introduce the unit pressure function and analyse the total number of pressure interactions. Then, we apply these concepts to football matches, considering various factors such as players and the radius of the area of influence, examining pressure efficiency through mean unitary pressure. Lastly, a real case study is presented, showcasing visualizations like a heatmap matrix displaying individual and collective pressure, as well as team pressure balance.

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3.1 Introduction

Top-level soccer is an increasingly competitive sport and, as a consequence, performance analysis is becoming an important tool for supporting coaches and players in the decision-making process, providing relevant information about the play. For decades, the use of video has been a key element in the tactical analysis of teams before and after a match [9]). The technological advances developed in recent years in soccer stadiums now make it possible to obtain a large amount of data related to the game, both individually and collectively. These data, which contain the trajectories of the players, the ball or the movement of the refereeing team in real time, allow a very precise mathematical analysis of the parameters that characterise the course of the matches. In addition, broadcasters have realised that coverage of sporting events can be enhanced by providing analysis of the action, often by specialised professionals who use visual tools and dashboards to describe key events and sequences during the game. There is also an apparent consumer appetite for statistics and attractive visual data displays, and there are many websites dedicated to providing this type of information. Therefore, there is a general interest in improving the visualizations and filling them with technical content that can provide more information to soccer fans. This is of interest not only to the fans, but also to the coaching staff and managers, who can analyse the data, even in real time, and make relevant decisions that will affect the outcome of the match. It is thus intended to serve as a tool for improving the strategy of teams, based on real data, after the complete data describing the match are processed.

The basic ways to obtain the real trajectories and positions of objects and players are mainly achieved via electronic performance and tracking systems (EPTSs). Technologies based on optical tracking system (OTS) capture and can analyse 25 frames per second, which would already add up to a total of 135.000 sets of values for 90 min of play per game, and if we consider 22 players, a game ball and the referee team constitute big-data of the order of $3 \cdot 10^6$ data vectors in $\mathbb{R}^3$ Euclidean space to be stored, which are not very feasible to analyse without an adequate modeling and treatment to obtain relevant and useful information for potential customers in use.

There are widely used indicators such as shots on goal, number of passes, tackle rates, team ball possession and distances covered, although their use is a matter of question [6]. Then, it is a task for data science to derive indicators from raw data that describe relevant situations of the game properly [2]. In particular, a lot of studies have analysed tactical indicators related to goal scoring in soccer (see, for example [11]). In some of them, one of the concepts that recursively appears when we talk about football is the concept of pressure. Several attempts to define pressure can be found in the literature. In 1996, ref. [12] introduced the *minimum moving time pattern (MMT)* for a player at any one moment as a pattern in which each point has the minimum time to move from that player's current position to that point; and also defined the *dominant region* of a player at any one moment as a region where the player can arrive at earlier than all the other players.

The concept of playing style is discussed in the reference [4], with special emphasis on what variables can be measured to determine it. In particular, in attacking or offensive phases of play, players move into positions towards the zones of the field in order to receive the ball or score a goal, and it is in this phase of play when pressure, con-

sidered as limiting the passing options and limiting space available, becomes fundamental.

One of the most characteristic and essential elements of soccer defense is pressing but it has not yet received much attention from researchers in football analytics. In [13], the authors analysed the automatic extraction of football events using tracking data that implies among others fact the definition of the possession of the ball to determine in-game events. For this purpose, they developed a decision tree-based algorithm, and in the conclusions, they established as an improvement of their algorithm *extending the possession zone's definition to encompass a variable radius/shape based on pitch location, proximity of opponents and player velocity*. The concept of pressure (which is directly related to that of possession) proposed in this article is based on the concept of this zone around the player and addresses, among other things, the determination of this radius.

In [1], the authors developed a computational method to analyse and quantify the dynamics of pressure and pressing behavior during football matches. To study team tactics in various scenarios, they designed an interactive visual tool called time mask, which enables the selection of multiple disjointed time intervals based on specified conditions.

In 2019, [5] proposed a method to quantify attacking performance in football. They introduced a parametric variable, referred to as *dangerousness*, which estimates the probability of scoring a goal at any given moment when a player is in possession of the ball. This metric is influenced by four variables, including one termed *pressure*, which measures a defender's ability to prevent the ball carrier from completing an action. Pressure was defined as the proximity of the defender to the player with the ball, assuming a threshold distance beyond which pressure diminishes. They modeled pressure using four sub-areas with varying radii, shaped by the angle between the player and the goal, resulting in a complex formula.

In [7], the authors introduced the *Pressure Passing Efficiency Index (PPEI)* as a tactical performance indicator. This index evaluates high-quality through-balls by emphasizing passes that outplay multiple opponents, factoring in the pressure on both the passer and the receiver.

In [10] author introduced a novel metric to quantify its effectiveness in different contexts. He assessed pressure-calculating probabilities of recovering the ball and conceding a goal-scoring opportunity in the near future. They XGBoost was stacked with a logistic regression model to estimate probabilities.

In [3] authors explored the area of defensive performance in football by analyzing successful defensive phases using space and time characteristics of defensive pressure. They used data from German Bundesliga games. The study distinguishes pressure on the ball-carrier, the group of attackers closest to the ball, and the entire team, finding that pressure is higher in areas closer to the ball and as the defensive play approaches its end.

In this paper, we apply our definition of pressure to the analysis of data from a real football match. These data have been obtained from *Metrica company* ([8]). These data cover two games and were anonymised, meaning that there are no references to the names of players, teams or competitions. The dimensions of the field are the same for both games: 105×68 m. Additionally, tracking and event data were synchronised. Detailed information is given about the definition and explanation of all events types and subtypes.

The article is organised as follows. After this introductory section, in the second section, we first study the notions of pressure on a fixed frame of the match. To do that, we start by introducing the concept of influence area ($\mathcal{A}$) between two players. This function depends on the distance between the players and also on the ball radius. An approximation of this function (using a secant line) yields the definition of pressure ($\mathcal{P}$) between the players. This notion allows us to define the pressure between one team over a player and finally between the two teams. Since the cumulative pressure values can be very large then we introduce the unit pressure function (p) which it is computed by dividing the pressures by its maximum. Then, we study the total number of pressure interactions. In the final part of this section we apply the previous formulas to study the pressure on a complete match of football. Now the pressure depends not only on the players and on the ball radius, but also on the frame. We divide this section in three parts. In the first one, we determine a ball radius in order to have a significant (but not too many) pressure interactions. Once this value ($r = 5\text{ m}$) is fixed, we study the pressure along a set of frames ($\mathcal{F}$): the key point here is that we sum the pressures of all players defending the ball. Since, in a match, not all the players play the same time in the third part of this section, we study the efficiency of the pressure by introducing the mean unitary pressure ($\widehat{p}$). This provides a number in $[0, 1]$ easier to use. In the third section we present a real case study providing different visualizations using our formulas. Hence, we start by showing a heatmap matrix that allows us to show (in the same table) the individual and the collective pressure (both exerted and received). In the second part, a butterfly chart (also called tornado chart) is presented. This visualization allows us to study the balance of pressures of the teams in a match. In the last section, we discuss the results and arrive at some conclusions.

3.2 Materials and methods

3.2.1 Pressure functions on a frame of a match

Let A and B represent the two teams on the pitch. In this section, we first introduce the notion of pressure between the players $a \in A$ and $b \in B$ and also the corresponding pressure between one team and a player and also between the two teams. In order to simplify the computations, an approximation of these functions—using a secant line—will be also used. Next, we introduce the unitary concepts of pressure, and we finish this section by providing the total number of pressing interactions.

Throughout this section, we set a frame so that there is no time dependence, and the definitions depend only on the radius of the ball. In the next section, we use our results to analyse a complete match—with the corresponding time/frame dependence—and we also look at how to determine the radius to compare the pressure between players and teams. We start by defining the notion of influence area between two players.

Definition 3.1. A real number, $r > 0$ is fixed. We define the influence area between the players $a \in A$ and $b \in B$, as the truncated area of $B(a, r) \cap B(b, r)$. That is,

$$\mathcal{A}_{a,b}^r = \mathcal{A}^r(d_{a,b}) := \begin{cases} 2r^2 \arccos\left(\frac{d_{a,b}}{2r}\right) - \frac{d_{a,b}}{2} \sqrt{4r^2 - d_{a,b}^2}, & 0 \leq d_{a,b} \leq r, \\ 0, & d_{a,b} > r, \end{cases} \quad (3.1)$$

where $d_{a,b}$ is the Euclidean distance, $d_{a,b} = \|a - b\|_2$, and a and b stands for the coordinates of each player—in $\mathbb{R}^2$.

Note that the objective of making the influence area function equal zero when $d_{a,b} > r$ is to take into account that when two players are far away enough, this area cannot be considered. Figure 4.1 shows the truncated intersection areas between the balls $B(a, r)$ and $B(b, r)$ —that is, the influence area between the players a and b —.

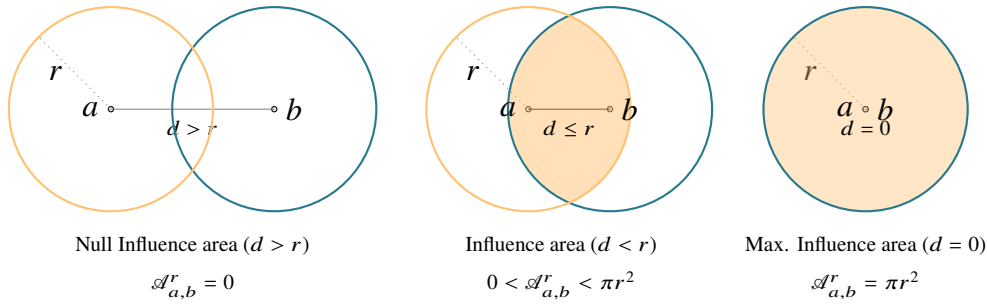


Figure 3.1: Influence area between players $a \in A$ and $b \in B$ depending on the radius r .

Thinking on the real-time computations, we can approximate the influence area function $\mathcal{A}^r$ on the interval $[0, r]$ by the secant line from $(0, \mathcal{A}^r(0))$ to $(r, \mathcal{A}^r(r))$. This provides our notion of pressure.

Definition 3.2. We define, for $r > 0$, the pressure exerted by player $a \in A$ over player $b \in B$ as

$$\mathcal{P} = \mathcal{P}^r(d_{a,b}) := \begin{cases} \pi r^2 - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right) r d_{a,b}, & 0 \leq d_{a,b} \leq r, \\ 0, & d_{a,b} > r. \end{cases} \quad (3.2)$$

The linear approximation provided in this case gives accurate results. In fact, it allows us to approximate the influence area function with small errors. Figure 3.2 shows a visualization of how the influence area and the pressure functions behaves for different values of r . With all this approximations we will speed up the calculations so that we can display the results dynamically in near-real time.

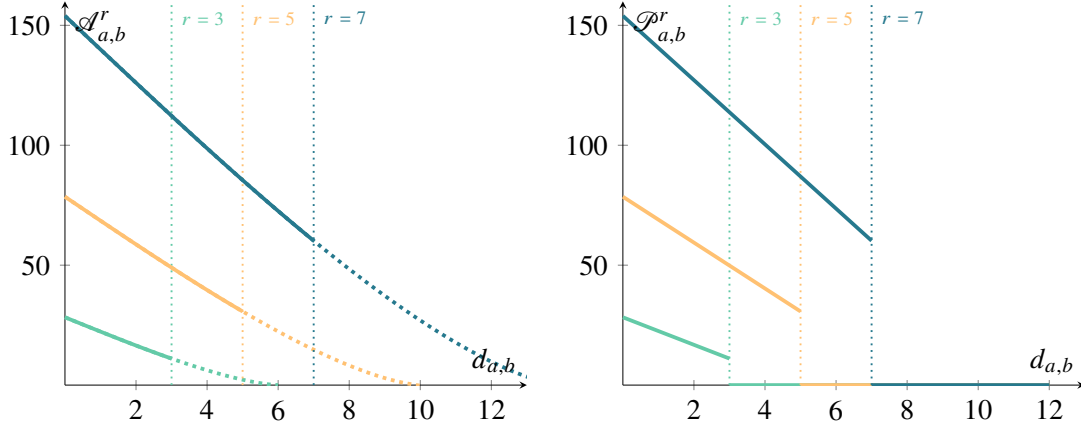


Figure 3.2: Representation of the influence area function—left—and its approximations—right—for different ball sizes, $r=3, 5$ and $7m$ —as a function of the distance $d_{a,b}$.

In a similar way, we can define the pressure between a team and a player and between two teams.

Definition 3.3. Given $r > 0$, the **pressure exerted by team A over the player, $b \in B$** , is defined as

$$\mathcal{P}_{A,b}^r = \sum_{a \in A} \mathcal{P}_{a,b}^r. \quad (3.3)$$

Definition 3.4. With a fixed $r > 0$, the **pressure exerted by team A over team B** is defined by

$$\mathcal{P}_{A,B}^r = \sum_{b \in B} \mathcal{P}_{A,b}^r = \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}^r. \quad (3.4)$$

Note that some (actually many) players in $a \in A$ will exert zero pressure over player $b \in B$. In this case, $\mathcal{P}_{a,b}^r = 0$, and then (many) terms of the sum will be null.

Sometimes, in order to show the results and make them more understandable—note that the cumulative values can be, in general, very large—it is interesting to use the unit pressure, which is denoted by the small letter $p_{a,b}^r$. To define it, recall the values of the pressure for the three singular situations given by $d_{a,b} = 0, r$ and $2r$ are

$$\begin{cases} \mathcal{P}_{a,b}^r(0) &= \pi r^2 =: f_{\max}, \\ \mathcal{P}_{a,b}^r(r) &= \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)r^2 \approx 0.391 f_{\max}, \\ \mathcal{P}_{a,b}^r(2r) &= 0 =: f_{\min}. \end{cases} \quad (3.5)$$

To define the unit pressure, it is enough to divide the pressure function by its maximum value, $f_{\max}$ —that is, the area of the circumference of radius r :

$$p_{a,b}^r = \frac{\mathcal{P}}{f_{\max}} = \frac{1}{\pi r^2} \mathcal{P}. \quad (3.6)$$

We finish this section by counting the number of people of a team exerting pressure—in the sense that the pressure is different to zero—.

Definition 3.5. Let $b \in B$. We will write $n_{A,b}^r$, for the number of players on team A who exert (non null) pressure over the player $b \in B$. In other words it is defined by the following:

$$n_{A,b}^r = \text{card}(\{a \in A : \mathcal{P}_{a,b}^r \neq 0\}). \quad (3.7)$$

We define the **total number of pressure interactions** on the pitch as

$$N^r = \sum_{b \in B} n_{A,b}^r. \quad (3.8)$$

Finally, just observe that the total number of interactions computed from each side—the pressure that all the members of B receive from the players of A , and the pressure that the members of A exert over the team B —coincide. This can be understood—and proved—by applying graph theory just thinking on two disjoint sets representing a graph consisting of $n_A^r + n_B^r$ vertices and N^r edges—see Figure 3.3.

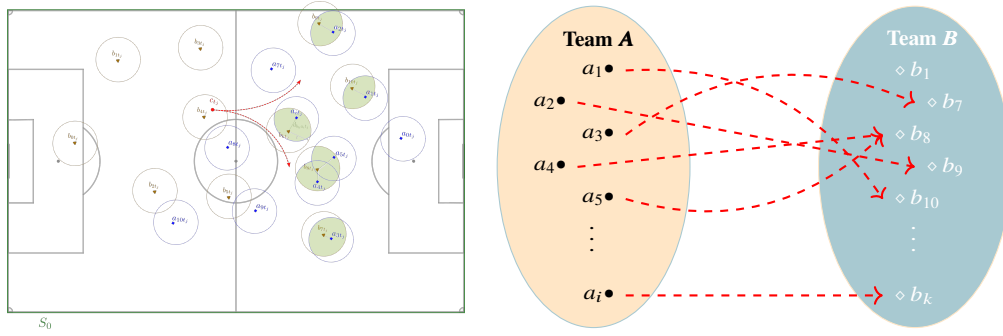


Figure 3.3: Example of total number of pressure interactions between the two teams on a frame.

3.2.2 Modelling the Pressure in a Complete Football Match

Now, we will use the formulas introduced in the previous section in order to analyse the pressure in a complete match. As a starting point, we take a set of frames $\mathcal{F}$. Note that, now, our pressure definitions depend on the frame $f \in \mathcal{F}$ —since the distance between a and b depends actually of $f \in \mathcal{F}$ —. To simplify our notation, we write

$$\mathcal{P}(f) := \mathcal{P}^r(d_{a,b}(f)). \quad (3.9)$$

Of course, if any of the players are not on the pitch in the frame $f \in \mathcal{F}$, then $\mathcal{P}(f)$ is defined as zero.

Step 1. Determining the radius. In order to obtain a useful tool to analyse a football match, first, we want to determine which values of r could be optimal to obtain a metric.

Let us take the first part of the game, which is composed of a total of 70.578 frames. The origin of these data was commented in Section 3.1. Figure 3.4 shows the positions on the pitch—in a particular frame $f \in \mathcal{F}$ —for four different radii— $r = 1$, $r = 4$, $r = 7$ and $r = 10$ —and the corresponding total number of pressure interactions.

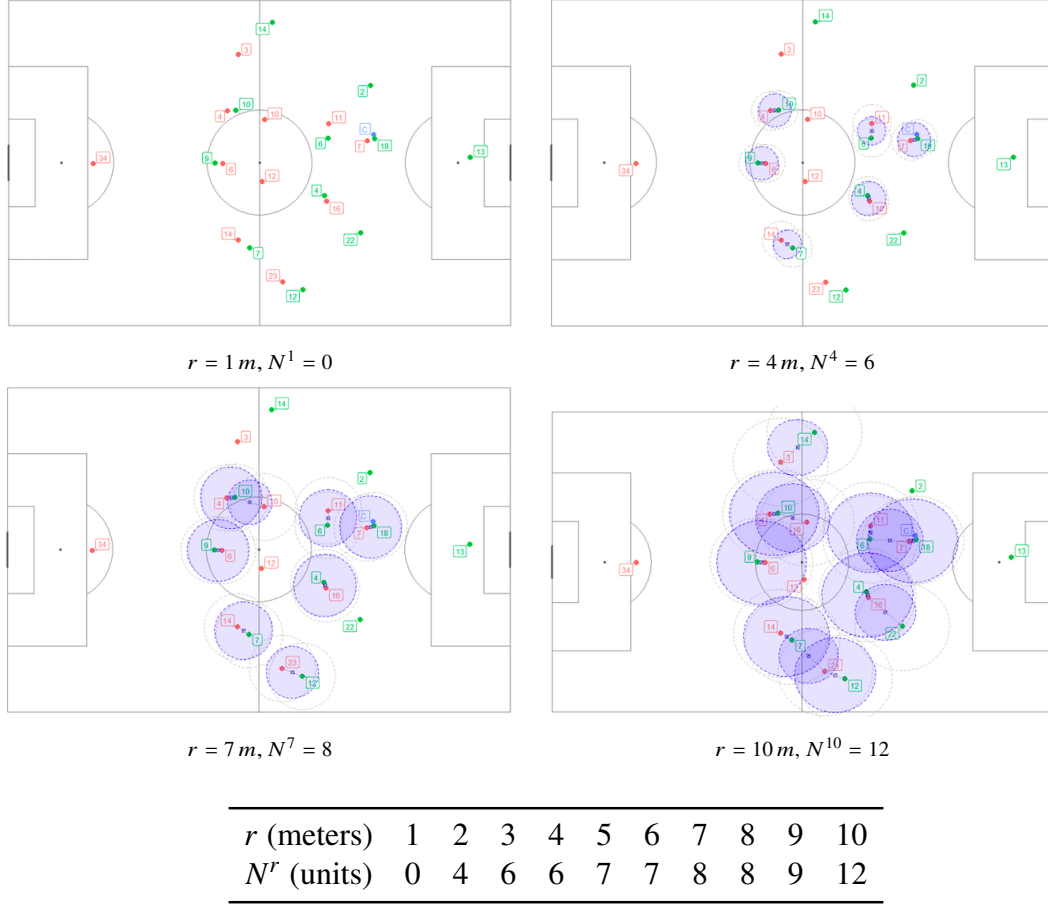
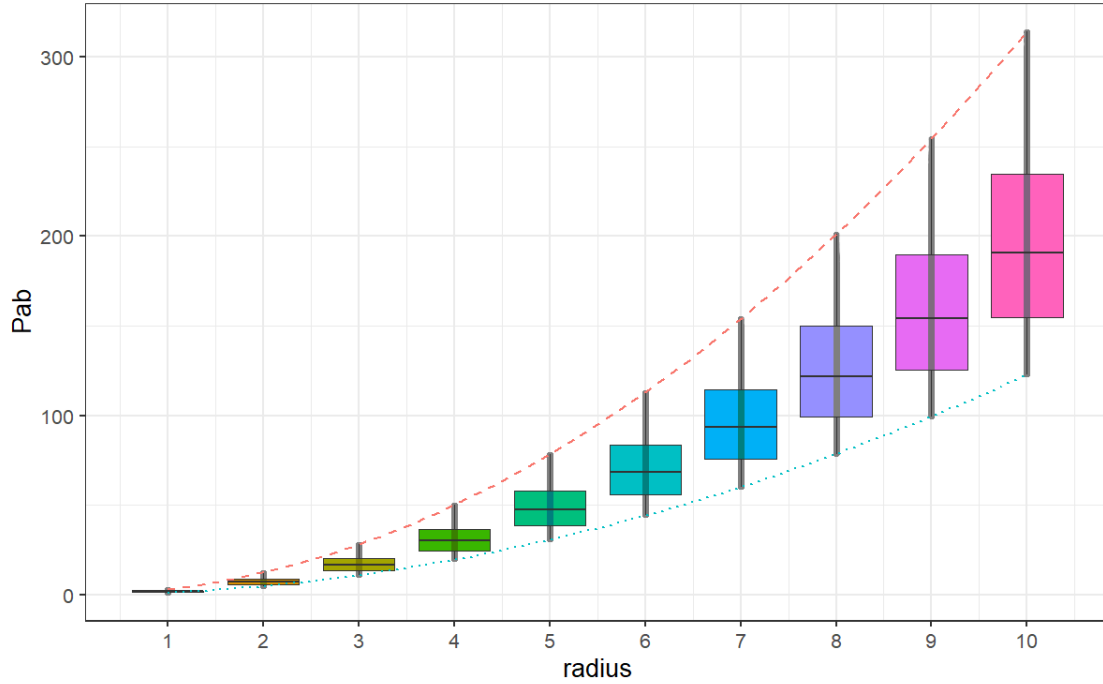


Figure 3.4: $\mathcal{P}$ sensibility to the ball radius variation in a frame $f \in \mathcal{F}$.

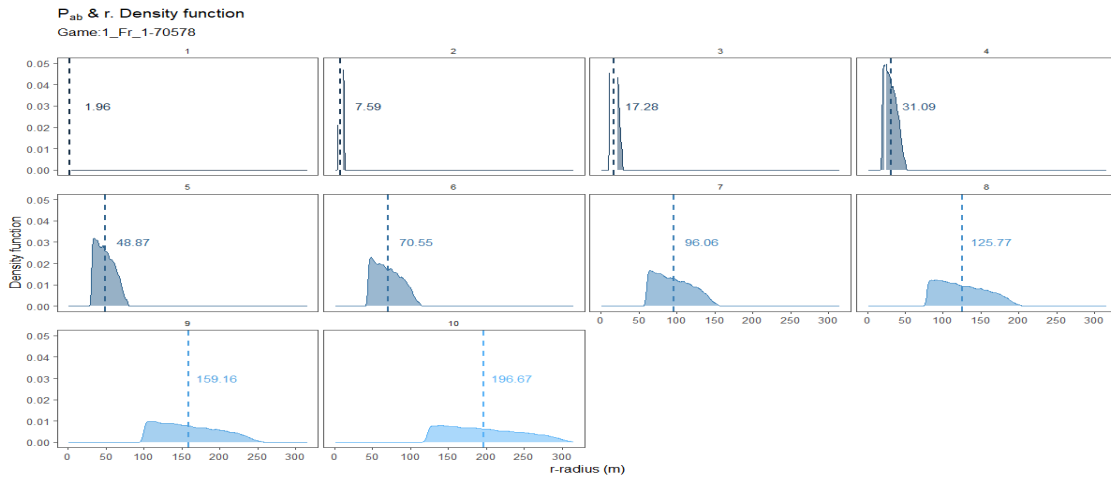
In the general case, we want a ball radius $r > 0$, which provides a large number of interactions $N^r(\mathcal{F})$ —of course, depending of the set of frames $\mathcal{F}$ —, but avoiding the possibility that the associated pressures have large oscillations. The decision on what is the optimal value of the ball radius have to be the result of a compromise between (a) a significant number of interactions, N^r , to allow accurate estimates, and (b) avoiding considering distant players as participants in the pressure. The pressure depends on the number of interactions, as explained above. The data obtained shows quite a few outliers, with higher values: the number of high outliers is due to the frames belonging to corners and throw-ins where players tend to group together more.

Figure 3.5a shows a Box and Whisker plot for the pressures, depending on the value of r and also on the frame set $\mathcal{F}$. The dotted lines show the $\mathcal{P}^r(0) = f_{\max}$ of the pressure

and the $\mathcal{P}_{a,b}^r(r) = 0.391 f_{max}$ value. It can be seen that the inter-quartile range increases considerably as the radius, r , increases. Moreover, if we look at Figure 3.5b, where we plot the density functions obtained via the sampling of the corresponding values of $\mathcal{P}$, we observe that the distributions flatten considerably as r increases. In principle, excessive flattening is not in our interest to obtain a good model, since too much dispersion would lead to a lack of precision in the results.



(a)



(b)

Figure 3.5: Pressure distributions for different radii along the set of frames $\mathcal{F}$. (a) Boxplot of pressures; (b) Density of pressures.

In order to reach a balance between high values of N^r —now along the set of frames $\mathcal{F}$ —and concentrated distributions for the pressure, we chose to use the shape factors of

the distributions, namely the kurtosis coefficient. In Figure 3.6, we show the plot of the normalised mean of the interactions for each value of r and the normalised kurtosis values for each of these radii. As we see, we reach the desired equilibrium point for the model for a radius on the interval $[4, 6]$ —we can take, for instance, the middle point $r = 5\text{ m}$ —.

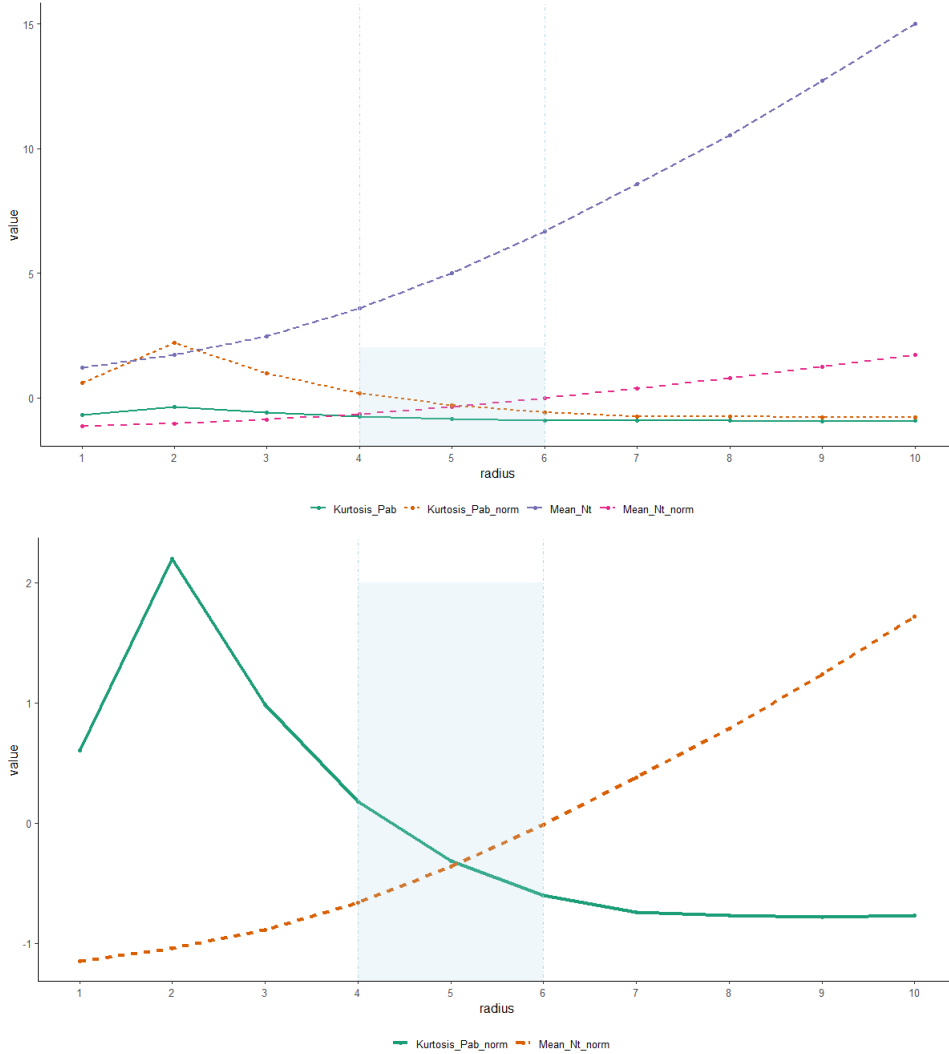


Figure 3.6: Optimal ball radius using the normalised mean of the interactions and the normalised kurtosis of the number of interactions between players. We set the radius as $r = 5$ meters.

Step 2. Studying the pressure. Once the parameters of the model and the corresponding approximation has been determined, we want to use our pressure formulas in order to study the pressure between players and teams on the complete match. First, we fix a set of frames $\mathcal{F}$. **Let us assume A is the defending team**—and, therefore, B has possession of the ball—. Then, we write that the players of A exert pressure over the players of B , or equivalently, the players of B receive pressure over the players of A . Hence, we make a disjoint partition, $\mathcal{F} = \mathcal{F}_A \cup \mathcal{F}_B$ where $\mathcal{F}_A$ are the frames where A defends—i.e. does not have the possession of the ball—and $\mathcal{F}_B$ the corresponding situation where team B is defending.

Definition 3.6. Let $r > 0$ and $\mathcal{F}$ denote a set of frames. We define the **pressure exerted by the player $a \in A$ over the player $b \in B$ along the set $\mathcal{F}$** as

$$\mathcal{P}(\mathcal{F}) = \sum_{f \in \mathcal{F}_A} \mathcal{P}(f),$$

where $\mathcal{P}(f)$ stands for the pressure on the frame f —see (3.9).

Example 3.7. Suppose we take a set, $\mathcal{F} = \{f_1, f_2, f_3\}$, where in f_1 and f_2 are defended by team A—so B has the possession of the ball, and in f_3 is the team B , which defends—see Figure 3.7.

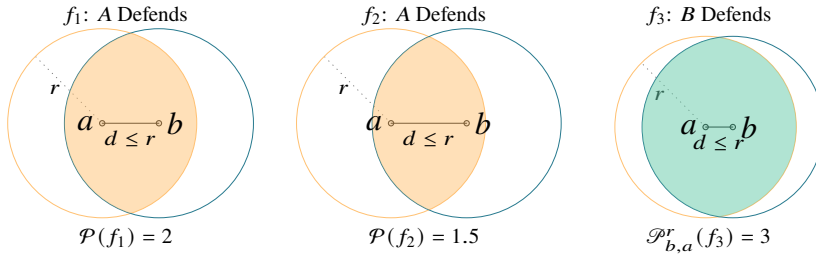


Figure 3.7: Example of pressure with a set of three frames.

Then, we have that

$$\mathcal{P}(\mathcal{F}) = \mathcal{P}(f_1) + \mathcal{P}(f_2) = 3.5 \quad \text{and} \quad \mathcal{P}_{b,a}^r(\mathcal{F}) = \mathcal{P}_{b,a}^r(f_3) = 3.$$

In a similar way, we define the **pressure exerted by the team A over player $b \in B$ along the set $\mathcal{F}$** as

$$\mathcal{P}_{A,b}^r(\mathcal{F}) = \sum_{a \in A} \mathcal{P}_{a,b}^r(\mathcal{F}) = \sum_{a \in A} \sum_{f \in \mathcal{F}_A} \mathcal{P}_{a,b}^r(f).$$

Note that $\mathcal{P}_{A,b}^r(\mathcal{F})$ can also be seen as the **pressure received by $b \in B$ along the set $\mathcal{F}$** . Also, the **pressure exerted by team A over team B along the set $\mathcal{F}$** (or the received by team B along the set $\mathcal{F}$) is given by

$$\mathcal{P}_{A,B}^r(\mathcal{F}) = \sum_{b \in B} \mathcal{P}_{A,b}^r(\mathcal{F}) = \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}^r(\mathcal{F}) = \sum_{b \in B} \sum_{a \in A} \sum_{f \in \mathcal{F}_A} \mathcal{P}_{a,b}^r(f).$$

Step 3. Efficiency of the pressure. Our starting point here is that not all the players play the same time—there are players that are sent off or substituted—. To deal with this problem, we study the **mean unitary pressure** notion. In a sense, this attempts to explain the effectiveness of pressure. Since the cumulative values of the individual exerted pressures during the whole match can be very high, one can use instead the unit pressures, $p_{a,b}^r$, as defined in (3.6). More precisely:

$$p_{a,b}^r(\mathcal{F}) = \frac{\mathcal{P}(\mathcal{F})}{\pi r^2} = \frac{1}{\pi r^2} \sum_{f \in \mathcal{F}_A} \mathcal{P}(f). \quad (3.10)$$

Recall that we are assuming that A is the defending team—so the ball possession corresponds to B —, and $\mathcal{F}_A$ are the set of frames where A defends. Hence, let us write $\mathcal{F}_A^a$, for the set of frames where A is defending and player $a \in A$ is on the pitch. Note that the cardinal of this set corresponds to the time where a is on the pitch and team A defends. Actually, since, in our database, each *second* has 25 frames, then the time is $t_A^a = 25\text{card}(\mathcal{F}_A^a)$. Therefore, the following applies:

Definition 3.8. Let $r > 0$ and $\mathcal{F}$ a set of frames. We define the **mean unitary pressure exerted by the player $a \in A$ over player $b \in B$ along the set $\mathcal{F}$** as

$$\widehat{p}_{a,b}^r(\mathcal{F}) = \frac{p_{a,b}^r(\mathcal{F})}{\text{card}(\mathcal{F}_A^a)} = \frac{1}{\text{card}(\mathcal{F}_A^a)} \frac{1}{\pi r^2} \sum_{f \in \mathcal{F}_A} \mathcal{P}(f). \quad (3.11)$$

Note that $\widehat{p}_{a,b}^r$ is a number on the interval $[0, 1]$.

3.3 Results: A real case study. Visualizations

In this section we study the pressure exerted by the teams in one entire match of the anonymised game (Sample Game 1) offered by ([8]). We propose different visualizations to represent our notions. Visualization is of primary importance to make our results useful. As we said before, since the cumulative pressures are very large, we prefer to use the corresponding unitary pressure values—that is the functions $p_{a,b}^r$.

The first visualization proposed is based on the construction of heatmap matrices. In Figure 3.8, we present the two matrices: in the figure above, team A defends, and in the figure below, is team B the defender team. The interpretation of the tables is as follows:

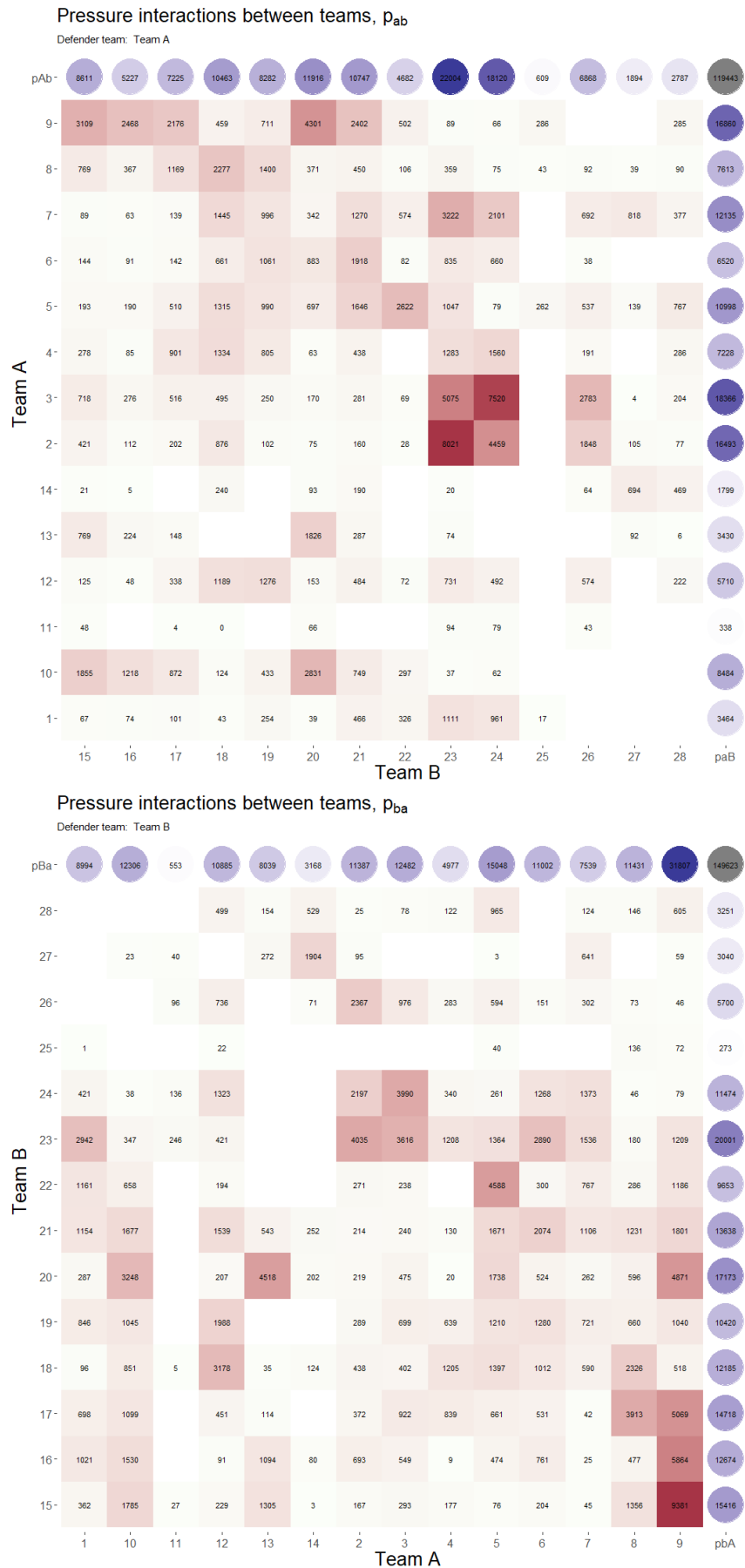


Figure 3.8: Individual and collective pressure unitary metrics. Above: exerted by team A and received by team B. Below: exerted by team B and received by team B.

1. **Individual exerted unitary pressure.** The i row of the matrix gives the individual pressure exerted by the corresponding labeled player—marked in red colors. For instance, the three players who have exerted the most pressure when team A defends—the figure above—are the ones labeled with the numbers 2 (8.021 units of unitary pressure against player 23 and 4.459 units of unitary pressure against player 24), player 3 (7.520 units of unitary pressure against player 24 and 5.075 units of unitary pressure against player 23) and player 9 (4.301 units of unitary pressure against the player 20).
2. **Collective exerted unitary pressure.** The last column—blue balls—explains the collective exerted pressure of each player. In this case, the three players exerting the most collective pressure—also for the figure above—are the numbers 3 (18.366 units), 9 (16.860 units) and 2 (16.493 units).
3. **Collective received unitary pressure.** In the first row—also blue balls—we can see the total pressure received for each player—of the attacking team. Again in the figure above, the player receiving the most collective pressure are numbers 23 (22.004 units), 24 (18.120 units) and 20 (11.916 units).
4. **Unitary pressure exerted along the match.** Finally, a gray ball—on the right corner of the table—shows the total exerting pressure of the match—which coincides with the total received pressure. When team A defends is a total amount of 119.443 units—the figure above—there are 149.623 units when B is the defender team—figure below. Note that the unit pressure has been calculated with a precision of 4 decimal places, so there may be small discrepancies in the total sum of the unit pressures with respect to the teams, for example, when team A is defending the respective sums of the top row and right column give values of 119.435 and 119.438 respectively when the sum made directly through the unit pressures amounts to 119.443 (gray box), reaching an maximum absolute non-significant error of 8 units ($6,70 \cdot 10^{-5}$)

Recall that a value of 1.000 units in the unit pressure would be equivalent to a player having been in total contact with another player along 1.000 frames, i.e., 40 s.

Regarding the (unitary) collective pressure—both exerted and received—we can also present a new metric: the defensive balance. Again, we are using the unitary pressure, $p_{a,b}^r(\mathcal{F})$, as defined in (3.10). We have chosen a butterfly chart (<https://datavizproject.com/data-type/butterfly-chart/>, accessed on 24 October 2024), as it is shown in Figure 3.9. In order to do that, we assign to each player a minus sign to the total received unitary pressures—representing it with red color—and a plus sign to the corresponding total exerted unitary pressure—now with blue color. Therefore, the **defensive balance** is given by the difference between the exerted and the received unitary pressure. In Figure 3.9, it can be seen that the A team has a global negative balance of pressure in the match: a total amount of -30.180 units—see the blue box of the figure above, which is the difference between the exerted unitary pressure (119.443 units) and the received unitary pressure (149.623 units).

But much more information can be obtained from this representation. In figure above, we can see from each player of A team the total exerted unitary pressure—the right part in blue color—and also the total received unitary pressure—left part in red color. Analogously, we see the same results from each player of B team —the figure below. Of course, the exerted unitary pressure of a player (which is in the defender team) is the sum of the exerted unitary pressure between this player and all the players from the opposite team (who is suppose to have the ball). In a similar way, we have the received unitary pressure. For example, we can see that the most pressured players where players 9 from A team (−31.807 units from figure above) and 23 from B team (−22.004 units from figure below). On the other hand, player 3 from A team and player 23 from B team where the ones who exerted more pressure throughout the match (18.366 and 20.001 units of pressure, respectively)(Figure 3.9).

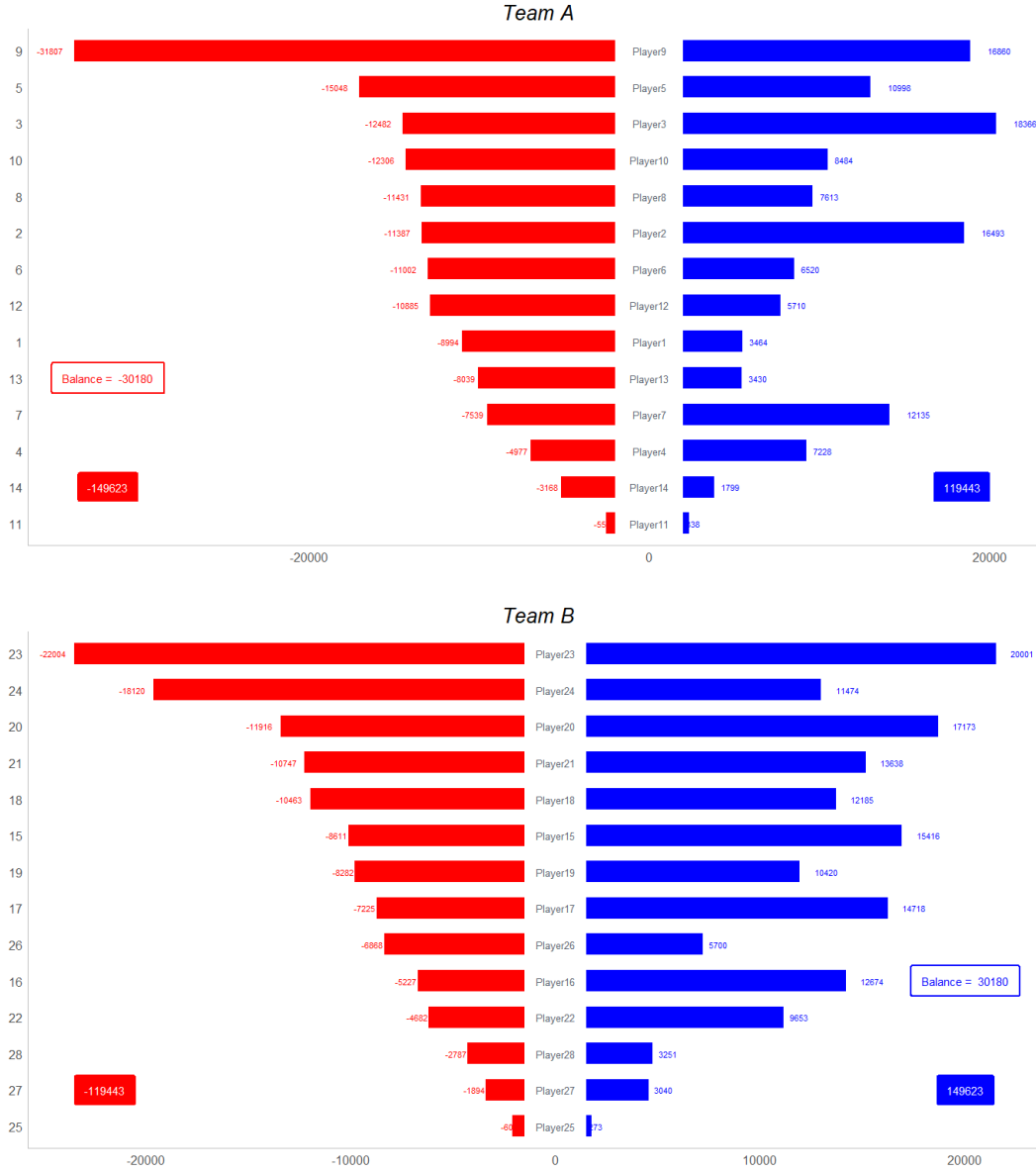


Figure 3.9: Defensive balance obtained from the unitary pressure $p_{a,b}^r(\mathcal{F})$. Above: team A players against B team. Below: team B players against A team.

We finish with the efficiency of the pressure. As we said before, not all the players play the same time. Hence, we use the mean unitary pressure notion, obtaining the **mean defensive balance**. This is shown in Figure 3.10. As we explained before, here, we use the mean unitary pressure, $\hat{p}_{a,b}^r(\mathcal{F})$, as defined in (3.11) instead of $p_{a,b}^r(\mathcal{F})$. During the match, there were the following replacements.

- team A: (1) $\leftarrow$ (12), (6) $\leftarrow$ (13), (10) $\leftarrow$ (14).
- team B: (22),(24) $\leftarrow$ (26),(27), (19) $\leftarrow$ (28).

Therefore, comparing Figures 3.9 and 3.10 we see that, for instance, player 3 from A team is now the player with more exerted pressure (with the value 0.4497), but player 9 (−0.6709) is not longer the most defended player, as this position has passed to player 1 (−0.673). With respect to team B, player 23 is still the player with the best pressure (0.4219) but is now closely followed by player 26 with (0.3507). This means that player 26 of the B team has exerted a great deal of pressing during the time he has been on the pitch.

In addition, through figures 3.9 and 3.10 we can evaluate the goodness of the substitutions made in the match from the defensive point of view, that is, we observe that the unit pressures generated by player 12 (0.2100) are lower than those of player 1 (0.2537) which he has substituted but that of player 13 (0.3410) has been much higher than that of player 6 (0.2118) and that of player 14 (0.2519) somewhat higher than those of player 10 (0.2517) so that especially the substitution (6) ← (13) has been very productive for team A in terms of pressure generation.

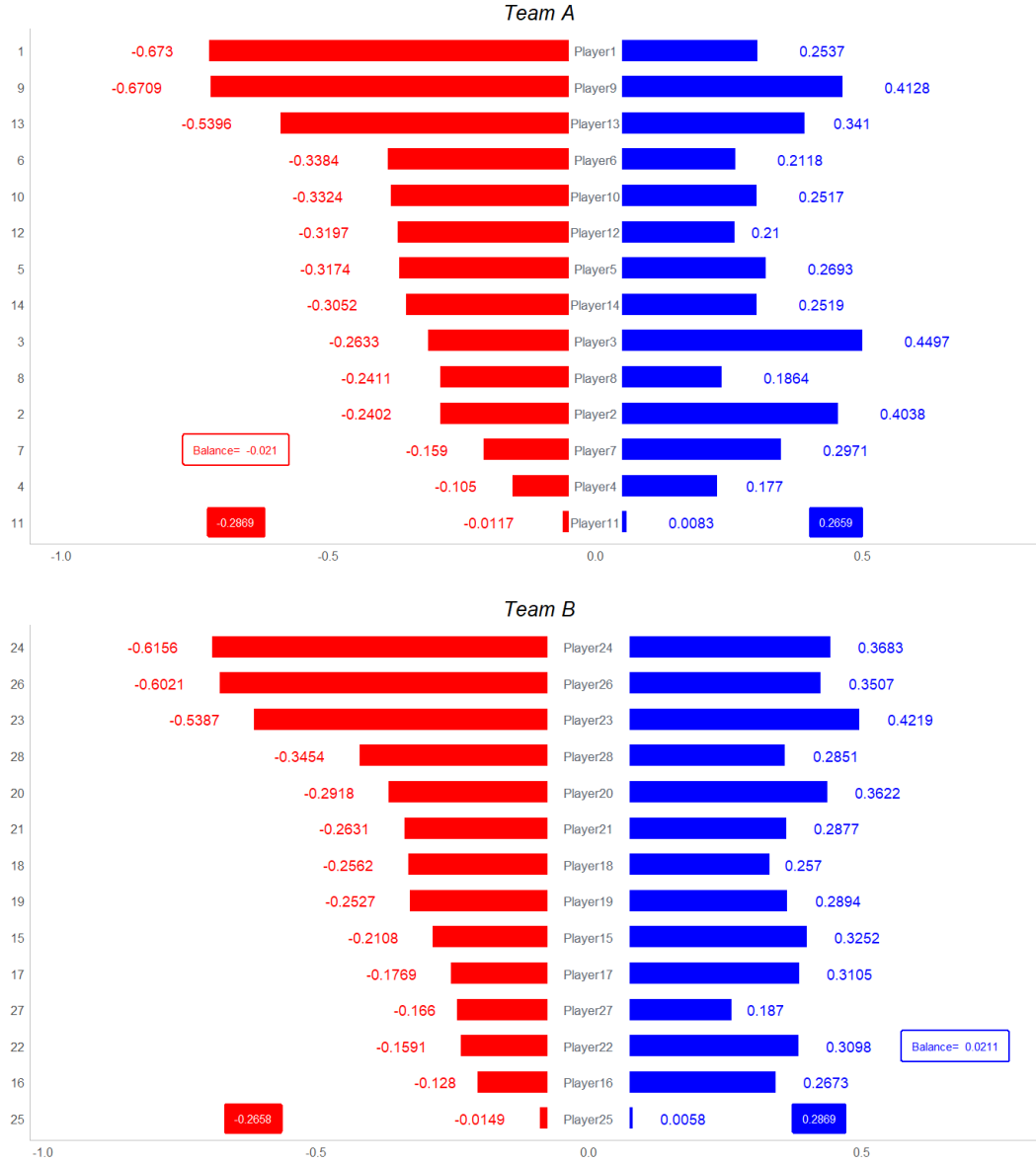


Figure 3.10: Mean defensive balance obtained from the mean unitary pressure $\hat{p}_{a,b}^r(\mathcal{F})$. Above: team A players against the B team. Below: team B players against the A team.

Of course, other ways of measuring the balance would be possible: for example, those associated with the Manhattan distance or the Euclidean distance. However, we think that our approach is better because: (1) on the one hand it is more intuitive and easier to interpret as a net balance; (2) but mainly it preserves the direction of the imbalance (positive or negative), which is crucial for the analysis of the team's behaviour; and (3) it also allows a global and direct comparison between blue and red contributions, which is computationally simple. To measure the total magnitude of the imbalance, regardless of the direction, absolute differences or Euclidean norm would be better options. However, consider the following:

1. In our approach, the sum of blue minus red bars for all players is interpreted as a net balance. This metric directly reflects whether the overall team is more inclined towards the “blue side” or the “red side”. In other words, a positive value implies a higher contribution of blue values (positive pressure), and a negative value indicates a predominance of red values (negative pressure). This facilitates a direct comparison between the offensive and defensive contributions of the team as a whole. However, by summing the absolute differences between red and blue bars for each player, we get a metric that captures the magnitude of imbalances without considering direction. This could be useful for measuring total imbalance, but you would lose key information about which side predominates (positive or negative). In a similar way, the square root of the sum of squares gives more weight to large differences (due to the quadratic effect). This might be useful if we are interested in highlighting extreme cases, but it is not as intuitive as the current metric for interpreting net balance.
2. In our metric, the sign of the balance (positive or negative) is essential, as it allows you to assess which side dominates: If the balance is negative, it means that the “red” (negative) contributions are stronger, which might reflect a defensive behaviour. If the balance is positive, the team shows an offensive behaviour or stronger positive pressure. Again, with the absolute differences, we completely lose this direction, as all differences would be positive. This does not allow to distinguish whether the team leans more towards offensive or defensive behaviour. Although the Euclidean norms retains some magnitude information, they also does not offer a direct interpretation of which side (red or blue) dominates, as it focuses only on the magnitude of the differences.
3. Finally, our approach, as a net sum, considers the team as a whole, integrating information from all players to assess their collective performance. It is a global metric that allows a clear assessment of group behaviour. The absolute differences metric would measure individual player imbalance but would not integrate the net effect on the team. The resulting value would not show whether the team, as a whole, is balanced towards one side or the other. Similarly, the Euclidean norm would measure the magnitude of the imbalance, but it does not give a clear idea of the overall tendency of the team towards blue or red.

3.4 Discussion

In football, the pressure exerted both between players and between teams is a very important tactical performance indicator, and it becomes fundamental to design game strategies, determining changes in players’ positions on the pitch, substitutions of players, spaces between lines, et cetera.

Pressure evaluation is difficult because it is a metric that involves both individual players and groups of players. Therefore, having a simple and easily computable method to evaluate it from the data available in football can be useful for coaches and technical staffs.

In this paper, we deal with a geometrical model of the definition of pressure. It starts from the elementary notion of area of influence, the area that is under the control of a player, which allows us to define the pressure exerted on a player based on the intersection of the areas of influence between the player who pressures and the player who is pressured. This distinction between the player exerting pressure and the player under pressure is made on the basis of which team is in possession of the ball at any given moment. This area of influence is established on the basis of a circumference centred on the player's position with radius r . This radius, a unique value for the complete match, is determined via a compromise between, on the one hand a significant number of interactions, N , to allow accurate estimates and, on the other hand, avoiding to consideration of very distant players as participants in the pressure. We want to note that this is the only parameter of the model whose determination from the data is straightforward as explained in Section 3.2.2.

An important advantage of our model is that, once the elementary amount of pressure between players is defined, with the same model and the same assumptions, it is very easy to obtain simultaneously pressure received by a player from the whole opponent team and also pressure exerted by a team on the whole opponent team, with everything using the same dataset information on individual players. In particular, we can evaluate the goodness of the substitutions made in the match from the defensive perspective, analyzing the pressure exerted by the player and by the substitute. It is that we call the efficiency of the pressure.

Our pressure metrics (individual and team) are defined frame by frame. Therefore, they can also be extended to the whole duration of a football match, which implies that not only can they give an static game perspective but they can also provide a time-dependent perspective on a given match. In future work, we plan to explore this time dependence via this detailed frame-by-frame analysis.

We have performed an analysis of data from a real match, showing the versatility of this metric. Is doing that, we have used tracking data with eventing that allows us to determine at each moment the position of each player on the pitch, i.e. their relative positions, as well as knowing which team is in possession of the ball at each moment. In particular, we have shown that, in a simple way, a global balance can be obtained between two teams over the course of a complete match. This analysis can be restricted to any time interval within the match, determining the balance of pressures at different instants of the game. The same analysis can be performed player by player on the same conceptual basis.

In summary, the model presented here is mathematically simple, easily computable and flexible in that it supports players and teams, as well as static and dynamic time-dependent metrics, and we believe it can be a useful tool for the tactical analysis of football matches by coaches and technical teams, as well as broadcasters and audiovisual media.

Finally, we would like to stress that this work can improve the accuracy, scope and depth of the analysis if we complement the tracking data with other types of data (eventing and scouting data).

Indeed, on the one hand, the pressure radius can vary, depending on the player's tactical role and position on the field, as well as the team's strategic characteristics. In our work, we only use tracking data, and this has inherent limitations that make it difficult to adjust r for each player accurately. Some limitations are, for instance: (1) A lack of explicit tactical context: Tracking data only provide information about players' positions and movements but do not include direct information about tactical roles or specific instructions from the coach. For example, we do not know whether a player is assigned to a zonal or individual marking task. We cannot infer whether a player is actively pressing or simply occupying space; (2) Difficulty in distinguishing functional roles: Although some patterns (such as greater mobility of midfielders) may be visible, it is not possible to assign with certainty a different radius for each position without knowing the context of the game plan. This would require information that is only available through scouting (detailed role and task reports). And (3) dynamic positions: Player roles in a modern game are very dynamic. A midfielder can act as a defender or attacker depending on the moment of the game. The tracking data does not allow transitions between roles to be identified.

To adjust r more realistically, we would need **Scouting data** to provide: (1) the identification of each player's role, information on whether a player is an attacker, defender or midfielder, and his specific responsibility (e.g., defensive pivot, winger); (2) the strategic context, knowledge of the team's strategy, such as pressing high, retreating, or holding specific positions; and (3) indications of intensity and quantitative data on each player's participation in pressing situations, based on tactical decisions.

In this study, using a uniform radius is a justifiable methodological simplification in the absence of scouting data, which is necessary to capture tactical variability accurately. We believe that a fixed r is a reasonable approximation since: (1) it is a good simplification to avoid arbitrary bias—by assuming a uniform radius for all players, we try to eliminate possible biases arising from interpreting the tracking data subjectively or incorrectly. Without scouting information, any adjustment to r based on tracking alone would be speculative. And (2), when using a constant r value, a common metric is established that allows comparisons between players and teams without introducing additional variability due to assumptions about tactical roles. In any case, we believe that using scouting data is a good approach for future work.

On the second hand, to separate the calculation of pressure in specific situations (penalties, direct and indirect free kicks or corners) we can use **eventing data**. Eventing data include detailed information about key events in the game, such as shots, passes, stoppages and restarts, providing additional context that the tracking data alone does not offer. This makes it possible to identify the following: (1) the type of events, in that a distinction can be made between penalty kick, direct free kick, indirect free kick or corner kick, something that is not directly inferred from player movement; (2) the outcome of the event information such as whether the free kick resulted in a pass, a shot or a loss of possession, which helps to understand the associated pressure dynamics; and (3) accurate event location—while tracking data provides positions, eventing data allows those positions to be associated with specific events and contextualised in the flow of the

game. One way to combine both types of data would be, for instance:

1. Penalties: **Tracking**: Only shows the movement of the goalkeeper and the penalty taker. Here, the pressure can be considered as non-existent (or fixed) for the other players, as they are restricted by regulation until the ball is touched. **Eventing**: Allows us to identify the exact moment of the penalty kick and exclude this situation from the overall pressure calculation, as it is not relevant for the usual pressure model.
2. Direct free kicks: **Tracking**: The data reflects the positions of the defensive barrier and other players. Pressure could be calculated based on the proximity of the collecting player and the barrier. **Eventing**: Provides the context to differentiate whether the free kick is direct towards goal or part of a rehearsed play. This allows the pressure calculation to be adjusted according to the nature of the event.
3. Indirect free kicks: **Tracking**: The data shows the movements of players anticipating a pass or shot, but does not explain the purpose of those movements. **Eventing**: Identifies that it is an indirect free kick and the exact timing of the pass, allowing pressure to be calculated on key receivers rather than the initial collector.
4. Corners: **Tracking**: This reflects the initial positions and movement of players (attack and defence) in the area, but does not differentiate whether the corner is played short, long or in another variant. **Eventing**: Helps to classify the type of corner (e.g., short or direct into the box) and identify the exact timing of key events (e.g., the shot or clearance). This allows for the calculation of the effective pressure of defenders on attackers in specific situations.

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4

Introduction

This chapter is situated within the broader methodological framework developed throughout this doctoral thesis. It follows the exploratory application of machine learning techniques to event data (Chapter 2) and the mathematical formalisation of defensive pressure at multiple analytical levels (Chapter 3). Within this context, the present study introduces a **dynamic and spatio-temporal modelling of defensive pressure** in football.

Traditional approaches to defensive analysis have largely relied on **static or event-based metrics**, which provide limited insight into the continuous and interactive nature of defensive behaviour. Such metrics often fail to capture how pressure emerges, evolves, and dissipates over time as a result of coordinated player movements and collective interactions. This limitation motivates the need for a framework capable of representing defensive pressure as a **dynamic process**, rather than as an isolated or discrete outcome.

In this chapter, defensive pressure is modelled using **tracking data** and continuous positional information, explicitly incorporating temporal evolution and player interactions. The proposed framework is grounded in concepts inspired by **classical physics**, where defensive pressure is treated analogously to a physical field generated by the spatial configuration and motion of players. This perspective allows the formalisation of pressure through well-defined mathematical constructs, providing a principled and interpretable foundation that departs from purely heuristic or empirical approaches commonly found in football analytics.

By embedding these physical principles in a time-dependent formulation, the model extends the concepts introduced in the previous chapter and enables the analysis of how pressure develops, propagates, and decays throughout defensive sequences and across different phases of play. In doing so, it offers a coherent link between player positioning, collective movement, and the resulting defensive effect on the opposition.

A key objective of this chapter is to move beyond static representations and towards metrics that can capture **defensive intensity, persistence, and distribution over time**. By doing so, the framework provides a more faithful representation of the tactical demands placed on players and teams during defensive phases, and enables the comparison of defensive behaviours across matches, teams, and competitive contexts.

From an applied perspective, the proposed model is designed with **operational feasibility** in mind. The computational formulation allows for the extraction of meaningful defensive indicators in **near real time**, making the framework suitable for use by **coaches, performance analysts, analytics platforms, and media outlets**. In this sense, the study bridges theoretical modelling and practical application, demonstrating how dynamic, physics-inspired representations can support performance analysis and tactical communication in professional football.

Overall, this chapter establishes a comprehensive and scalable framework for measuring defensive pressure in football, representing a decisive step from static descriptions towards a dynamic and physically grounded understanding of collective defensive behaviour.

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A Dynamic Framework for Defensive Pressure Assessment in Football

Abstract

This study introduces a novel physics-inspired framework to quantify defensive pressure in football from tracking data. We model defender–attacker interactions as a variable-mass dynamical system, translating Newtonian mechanics into operational metrics that combine spatial configuration and motion. From this formulation we derive interpretable quantities at dyad, player, and team level, including a Center of Pressure (CP), Defensive Momentum, Defensive Force, and Defensive Work. We illustrate the framework in a single-match proof-of-concept using professional optical tracking data, analysing both full-match behaviour and football-specific phases such as counter-pressing, set-pieces, and throw-ins. Results show how the proposed metrics separate persistent spatial constraint (pressure) from energetically demanding defensive actions (work), enable identification of high-cost match-ups and workload concentration, and support time-resolved descriptions of coordinated pressing sequences. The framework provides a transferable, mechanically grounded toolkit for applied defensive performance analysis and motivates future validation on larger datasets.

keywords: football analytics · tracking data · defensive pressure · physics-inspired modelling · spatio-temporal metrics

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4.1 Introduction

The advent of player-tracking technology has fundamentally transformed football analytics, shifting the focus from traditional performance statistics to sophisticated physics-inspired modeling of player and team dynamics. Modern tracking systems capture positional data at frequencies of 10–25 Hz, providing rich kinematic information including velocity, acceleration, and trajectory patterns [9]. This growing availability of data has necessitated the development of physics-informed frameworks to extract actionable insights. Such approaches draw inspiration from classical mechanics, statistical physics, and collective motion theory, where complex system behaviour emerges from simple interaction rules [23].

Teams can be modelled as many-body systems, with their collective behaviour analysed through physics-inspired metrics. Early work in this direction framed football as a dynamical system in which spatial occupation and player interactions shape tactical outcomes. Building on geometric principles, [10] used Voronoi diagrams to objectively define a player's dominant region on the pitch. Related spatial control formulations have been proposed using pitch control probabilities and motion models grounded in kinematics [20]. A more advanced, field-theoretic approach was introduced by [19], who modelled each player as generating a decaying influence field. Similar influence-based or potential-field formulations have also been used to describe how contributions of the players influence in the overall spread of the team in space[1]. Finally, [6] integrated these concepts into a dynamic decision-making framework, measuring a defender's impact by how their positioning and pressure reduced the attacking team's Expected Possession Value (EPV).

In recent years, increasing attention has been devoted to modelling collective dynamics through quantitative frameworks. Early work of [18] introduced concepts such as *dimensional compression* and *reciprocal compensation*, linking spatial organisation to team coordination. Subsequent studies [4, 15, 17] applied hypernetwork theory, synergies, and network dynamics to describe team organisation as an emergent and adaptive system. From a physics perspective, collective variables such as centroids, dispersion, and synchronisation have been shown to capture key aspects of tactical behaviour [8, 12]. More recently, [14] identified tactical situations and playing styles as key performance indicators, while [21] systematically reviewed the application of artificial intelligence to football analytics. Recently, by introducing an automatic, machine-learning-based approach to detect (potential) assists from event and tracking data, Klemmer et al.[11] highlighted the potential of objective assist definitions to improve applied football performance analysis beyond traditional rule-based metrics.

Defensive individual and team analysis are increasingly recognised as both important and relevant in football performance research. Whereas attacking actions such as passing networks, expected goals (xG), and possession value models have been widely adopted, equivalent metrics for defensive behaviour remain comparatively underdeveloped. This imbalance is striking, given that successful defensive organisation and pressure strategies can be decisive in determining match outcomes, particularly in modern football, where high pressing and rapid transitions play a central tactical role.

Addressing this gap is crucial because robust defensive metrics could support coaches and analysts in evaluating pressing strategies, identifying weaknesses in team shape, quantifying the energetic demands of counter-pressing, and benchmarking defensive intensity. Recent studies have highlighted the need for defensive indicators that explicitly integrate space, time, and motion rather than relying solely on event-based summaries [7, 16]. Moreover, as tracking data become increasingly available, the opportunity to extract reliable and interpretable defensive indicators with spatio-temporal resolution suggests an urgent area of research.

Despite these emerging approaches, existing defensive indicators remain limited in scope. Most studies have focused either on aggregate measures such as PPDA (passes allowed per defensive action) or on qualitative assessments of pressing intensity, these provide useful summaries but lack the spatial and temporal resolution needed to capture the dynamics of collective pressure. From a physical perspective, such static or aggregate measures cannot fully characterise how defensive interactions evolve, propagate, and dissipate across the pitch. Current methods are therefore unable to quantify in detail how defensive pressure evolves, how it is coordinated across players, or how it translates into energetic costs such as force and work.

In recent years, the emergence of dynamical metrics based on defensive actions has begun to influence tactical decision-making. These measures aim to capture the static positioning of players, the spatio-temporal dynamics of pressing, recovery, and workload distribution [5, 13]. Despite these advances, defensive analytics has not yet received the same level of systematic attention from football science as offensive analytics, leaving an important gap in the literature.

In previous work [3], we introduced the concept of the area of influence between two players as the basis for defining Defensive Pressure through. This defensive metric was designed to be aggregative and scalable, enabling the calculation of collective pressures between players and teams. However, this framework did not explicitly address the spatial arrangement or dynamic (in-motion) nature of these pressures on the pitch.

The present study extends the concept of defensive pressure from a static to a dynamic domain by adopting a completely new point of view. Drawing on analogies with variable-mass dynamical systems in classical mechanics, we introduce formulations that capture the magnitude of pressure as well as its temporal evolution through momentum, force, and work. This extension allows us to treat collective defensive activity as a dynamic system in which pressure is continuously generated, displaced, and dissipated across the pitch.

4.2 Methods

In this section, we will describe, first of all, the original framework we have designed for the new metrics. This framework is based on Newtonian physics of variable mass systems, making an analogy that, in our opinion, is both easily understandable and clearly interpretable.

4.2.1 Theoretical Framework

The model aims to establish a physical and mathematical equivalence between a dynamic system of variable-mass particles and the defensive pressure interactions observed in football. In classical mechanics, a particle system with variable mass is characterized by continuous changes in its constituent masses, which, in turn affect the system's momentum, force and work over time. By analogy, in football, the defensive pressure exerted by players can be considered as a “variable-mass” entity: the magnitude and location of pressures fluctuate dynamically as players approach or move away from each other, enter or leave the pressing zone, or change their tactical role.

Each player–opponent interaction is modelled as an individual “**particle of pressure**” with a variable magnitude, defined through its area of influence and density. The set of all simultaneous interactions between players of opposing teams therefore constitutes a Dynamic System of Variable Pressures, structurally comparable to a variable-mass particle system. In this framework, the Center of Pressure (CP) plays an analogous role to the Center of Mass: it provides a spatial reference point for the collective distribution of pressures across the pitch.

This equivalence allows for the direct application of Newton’s Laws of Motion to the football context. The first law supports the assumption of inertial reference frames on the pitch; the second law provides a rigorous definition of Defensive Force as the time derivative of Defensive Momentum; and the third law reflects the reciprocity of player–opponent interactions. In this way, classical physical principles become the foundation for deriving new dynamic defensive metrics—Defensive Momentum, Defensive Force, and Defensive Work—capturing the static presence of pressure and its temporal evolution and collective coordination.

Defensive pressure

If we consider the equation for the mass of a planar body, given by:

$$M(t)[Kg] = \rho \left[\frac{Kg}{m^2} \right] \cdot A(t)[m^2], \quad (4.1)$$

we can establish a formal equivalence with classical mechanics by defining the **absolute defensive pressure between the players $a \in A$ and $b \in B$** as follows:

$$\mathcal{P}_{a,b}^r(t)[up] = \rho_{a,b}(t) \left[\frac{up}{m^2} \right] \cdot \mathcal{A}_{a,b}^r(t)[m^2], \quad (4.2)$$

where up denotes an arbitrary unit of defensive pressure, $\rho_{a,b}(t)$ is the pressure density (up/m^2)—which can be interpreted as the intensity per unit influence area assigned to the interaction—, and $\mathcal{A}_{a,b}^r$ is the **influence area between the players $a \in A$ and $b \in B$** , defined as the truncated area of $B(a, r) \cap B(b, r)$ for a given $r > 0$ by:

$$\mathcal{A}_{a,b}^r(t) := \begin{cases} 2r^2 \arccos\left(\frac{d_{a,b}}{2r}\right) - \frac{d_{a,b}}{2} \sqrt{4r^2 - d_{a,b}^2}, & 0 \leq d_{a,b} \leq r, \\ 0, & d_{a,b} > r, \end{cases} \quad (4.3)$$

Here $d_{a,b}$ is the euclidean distance, $d_{a,b} = \|a - b\|_2$, and a and b stands for the coordinates of each player—in $\mathbb{R}^2$ —. Note that the influence area is maximum when the players are extremely closed ($d_{a,b} = 0$) and equals zero when $d_{a,b} > r$ to take into account that when two players are far away enough this area cannot be considered. In this definition the dependence of the influence area with time is *implicit* because the coordinates of each player, necessary to calculate $d_{a,b}$, are depending on time for players in movement.

For computational efficiency and improved interpretability, we can also introduce a simple linear approximation, which closely matches the exact model over the relevant distance range—see [3]—and is used in the subsequent derivations to enable fast evaluation:

$$\mathcal{A}_{a,b}^r(t) := \begin{cases} \pi r^2 - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right) r d_{a,b}, & 0 \leq d_{a,b} \leq r, \\ 0, & d_{a,b} > r. \end{cases} \quad (4.4)$$

With this formulation (4.2) mirrors the classical relation given in (4.1), with football pressure understood as the product of a density term and an influence area and playing the role of a “mass”. In the present study, **we assume a constant pressure density** $\rho_{a,b}(t) = \rho = 1$ (up/m^2) for all player interactions and a dependence in time of $\mathcal{A}_{a,b}^r(t)$. Under this assumption, the weighting of pressure exerted by any player a on an opponent b is identical, regardless of pitch location, ball position, velocity profiles, or any other contextual variables, thus yielding a tractable baseline representation of the system. A further development of the model can consider the density to incorporate situational dependencies, making it a powerful placeholder for more complex formulations. In principle, $\rho_{a,b}$ may vary as a function of player–ball distance, distance to goal, pitch zones, or instantaneous velocities, among others giving rise in some cases a non-local definitions of density.

It is preferable to use the **area-normalised pressure density**, $p_{a,b}^r(t)$, rather than absolute values. To achieve this, we normalise the system with respect to the maximum area of influence— $\mathcal{A}_{max}^r = \pi r^2$ —. This normalisation, analogous to the concept of relative density in physics, ensures that the values are bounded by $\rho = 1$ up/m^2 and directly comparable across players, contexts, and teams, thereby enhancing interpretability and extending the versatility of the model. The area-normalised pressure density is:

$$p_{a,b}^r(t) \left[\frac{up}{m^2} \right] = \frac{1}{\pi r^2} \left[\frac{1}{m^2} \right] \cdot \mathcal{A}_{a,b}^r(t) [up]. \quad (4.5)$$

Therefore, the dynamics and the temporal “shapes” of the signals do not change; only the overall scale does.

We can count the number of players of a team exerting pressure—in the sense that de pressure is different to zero—. Let $b \in B$. We will write $n_{A,b}^r$ for the number of players on team A who exert (non null) pressure over the player $b \in B$:

$$n_{A,b}^r = \text{card}(\{a \in A : \mathcal{P}_{a,b}^r \neq 0\}) = \text{card}(\{a \in A : p_{a,b}^r \neq 0\}), \quad (4.6)$$

and we define the **total number of pressure interactions** on the pitch as

$$N^r = \sum_{b \in B} n_{A,b}^r. \quad (4.7)$$

Observe that, the total number of interactions computed from each side—the pressure that all the members of B receive from the players of A , and the pressure that the members of A exert over the team B —coincide.

Finally, we define our inertial reference system, ϕ , with origin $O(0,0)$ located at the geometric Center of the pitch, which serves as the coordinate system for all subsequent computations. For notational simplicity, we omit the explicit dependence on r in the remainder of the paper.

Center of Pressure

We define the **Center of Pressure (CP)** between players $a \in A$ and $b \in B$ at time t , denoted by $\mathbf{r}_{a,b}(t)$, as the average value of the position vectors of these players, $\mathbf{r}_a(t)$ and $\mathbf{r}_b(t)$,

$$\mathbf{r}_{a,b}(t) = \frac{1}{2} (\mathbf{r}_a(t) + \mathbf{r}_b(t)). \quad (4.8)$$

For readability, the explicit time dependence t is omitted when clear from context.

The **CP** coincides with the geometric center of the pressure area, $p_{a,b}$, between players $a \in A$ and $b \in B$. This is because the **CP** coincides with the midpoint of the segment joining the centers of the circles of radius r belonging to players $a \in A$ and $b \in B$. Moreover, **CP** is equidistant from the trajectories of players a and b , generating its own trajectory (Fig.4.2). This allows the defensive pressure $p_{a,b}$ to be formalised in analogy with classical mechanics, where mass is a scalar property assigned to a particle at a given position. Accordingly, defensive pressure $p_{a,b}$ is defined as a scalar magnitude endowed with a spatial location through its position vector $\mathbf{r}_{a,b}$, which enables the computation of kinematic quantities such as velocity and acceleration. The spatial location of this defensive pressure, $\mathbf{r}_{a,b}$, is determined by the associated Center of Pressure (CP).

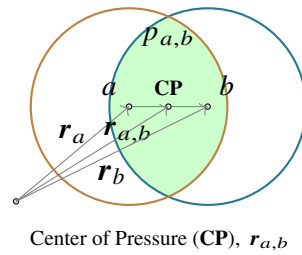


Figure 4.1: Geometric representation of the Center of Pressure (CP) generated between players $a \in A$ and $b \in B$. The **CP** corresponds to the midpoint of the segment connecting the Centers of the influence areas (circles of radius r) associated with each player. This location, $\mathbf{r}_{a,b}$, is consistent with the formal definition in Equation 4.8, where the **CP** is given by the mean of the players' position vectors, $\mathbf{r}_a$ and $\mathbf{r}_b$. The shaded region illustrates the overlap of the individual influence areas, which defines the effective pressure $p_{a,b}$.

The velocity $\dot{\mathbf{r}}_{a,b}$ and acceleration $\ddot{\mathbf{r}}_{a,b}$ of the pressure generated by players $a \in A$ and

$b \in B$, located at their **CP**, can be defined as

$$\dot{\mathbf{r}}_{a,b} = \frac{d}{dt}[\mathbf{r}_{a,b}] = \frac{1}{2}(\dot{\mathbf{r}}_a + \dot{\mathbf{r}}_b), \quad (4.9)$$

$$\ddot{\mathbf{r}}_{a,b} = \frac{d^2}{dt^2}[\mathbf{r}_{a,b}] = \frac{1}{2}(\ddot{\mathbf{r}}_a + \ddot{\mathbf{r}}_b). \quad (4.10)$$

Figure 4.2 illustrates the magnitude of pressure $\mathcal{P}_{a,b}(t)$ and the **CP** trajectory $\mathbf{r}_{a,b}$ at two consecutive time instants.

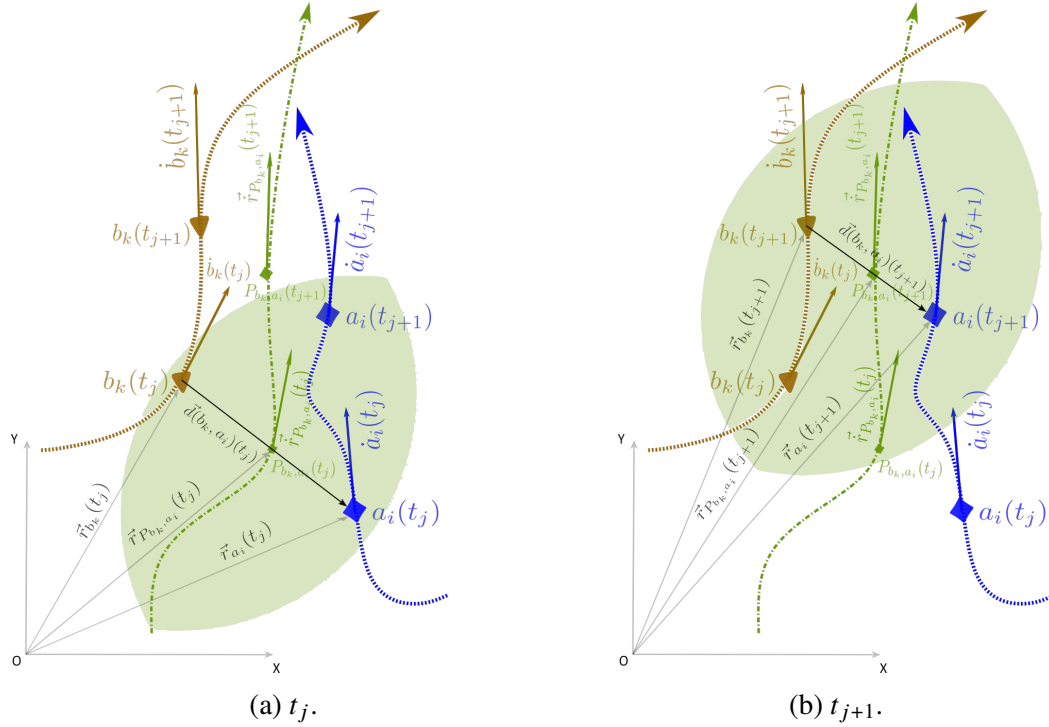
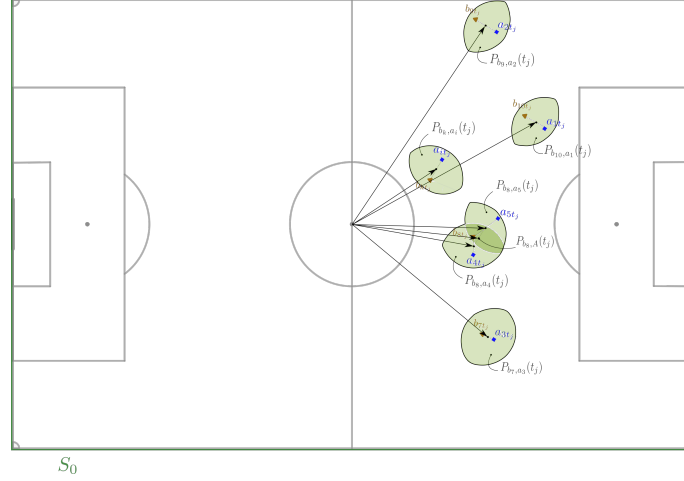


Figure 4.2: Generation of defensive pressure $p_{a,b}(t)$ at two consecutive instants, t_j (4.2a) and t_{j+1} (4.2b). In brown we show the trajectory and velocity of player $b_k \in B$, and in blue those of player $a_i \in A$. The midpoint of the distance between their trajectories represents the **CP**, whose position vector $\mathbf{r}_{a,b}(t_j)$ defines the trajectory of the defensive pressure in the model. The **CP** thereby generates its own fictitious trajectory with associated velocity $\dot{\mathbf{r}}_{a,b}(t_j)$ and acceleration $\ddot{\mathbf{r}}_{a,b}(t_j)$. The light green region illustrates the instantaneous magnitude of defensive pressure $p_{a,b}(t_j)$ (4.2).

These **CP** trajectories define a **Dynamic System of Variable Pressures**, whose positions, velocities, and accelerations are given by $\mathbf{r}_{a,b}$, $\dot{\mathbf{r}}_{a,b}$, and $\ddot{\mathbf{r}}_{a,b}$ respectively. This system is composed of $N(t)$ instantaneous interactions (4.7) exerted by players from teams A and B , which vary throughout the match. The value and location of pressures $p_{a,b}$ fluctuate not only in magnitude ($p_{a,b}$ is bounded by $\rho = 1 \text{ up}/\text{m}^2$) but also in velocity ($\dot{\mathbf{r}}_{a,b}$) and acceleration ($\ddot{\mathbf{r}}_{a,b}$) as time progresses. Figure 4.3 illustrates the dynamic system of pressures at a given instant t , showing the generation of six simultaneous defensive pressures ($N = 6$) arising from the interactions between five players from team B and six from team A .

Figure 4.3: Dynamical Pressure System at t_j .

Although dyad-level metrics can be summarised compactly in matrix form, their interpretation benefits from spatial anchoring on the pitch. We therefore associate each dyad (a, b) with a CP location (the mean position of the defender and attacker) and use this location to visualise where high-cost interactions occur. Figure 4.4 provides an example of how a matrix heatmap of Defensive Work can be mapped back to the field via dyad CP positions. As usual, a dyad here refers to a specific defender–attacker pair (a, b) .

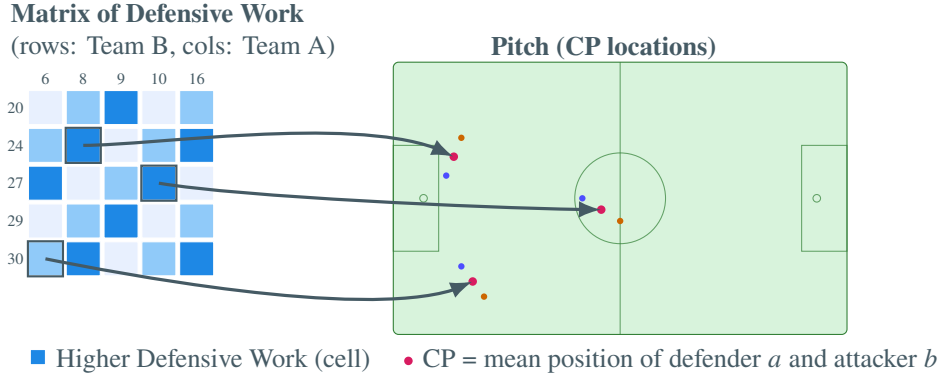


Figure 4.4: Linking a *matrix heatmap* of Defensive Work (left) to spatial locations on the pitch (right). Each selected cell corresponds to a dyadic interaction (a, b) and is mapped to its Center of Pressure (**CP**), defined as the midpoint between the positions of the two players. This preserves the spatial context of the workload while retaining a compact matrix summary.

Dynamic defensive metrics: Momentum, Force and Work

In a dynamic system of particles with variable mass, the linear momentum $\mathcal{L}_k(t)$ of a particle k of mass $m_k(t)$ in a reference frame ϕ is a time-dependent vector quantity defined as the product of the instantaneous mass $m_k(t)$ of the particle and its velocity $\dot{\mathbf{r}}_k$ in ϕ ,

$$\mathcal{L}_k(t) = m_k(t) \dot{\mathbf{r}}_k. \quad (4.11)$$

By analogy, and as introduced at the beginning of this section, the equivalence between a variable-mass particle system and our dynamic pressure system leads naturally to the

following formulation. We define the **Defensive Momentum between player $a \in A$ and player $b \in B$ at the moment t** , as

$$\mathcal{L}_{a,b}(t) = p_{a,b}(t) \dot{\mathbf{r}}_{a,b}. \quad (4.12)$$

The Defensive Momentum $\mathcal{L}_{a,b}(t)$ is therefore directly proportional to both the magnitude of the pressure exerted between two players and to the velocity of its **CP**. Hence, defensive activity increases either when the pressure itself intensifies (i.e., smaller inter-player distance) or when the CP moves more rapidly across the pitch. By construction, $\mathcal{L}_{a,b}(t)$ has the same direction as the **CP** velocity $\dot{\mathbf{r}}_{a,b}$, reflecting the spatial orientation of the defensive action.

It is important to note that this magnitude explicitly captures *pressure in motion*. A player $a \in A$ may exert static pressure on an opponent $b \in B$, but such pressure only contributes to the Defensive Momentum if the velocity of the CP is non-zero, $\dot{\mathbf{r}}_{a,b} \neq 0$. In other words, *Defensive Momentum quantifies the dynamic component of pressing actions, rather than static proximity alone*. This property is particularly relevant when considered alongside *Defensive Force* and *Defensive Work*—both of which are detailed in the following sections—as it provides a comprehensive description of the intensity and temporal evolution of defensive behaviour.

In classical dynamics, according to the second law, the total force $\mathcal{F}_k(t)$ exerted on a particle k in a reference frame ϕ is equal to the time derivative of its linear momentum ([22]). In the case of a system with variable mass, this leads to

$$\mathcal{F}_k(t) = \frac{d}{dt} [\mathcal{L}_k(t)] = \frac{d}{dt} [m_k(t) \dot{\mathbf{r}}_k] = \frac{d}{dt} [m_k(t)] \dot{\mathbf{r}}_k + m_k(t) \ddot{\mathbf{r}}_k. \quad (4.13)$$

We define the **Defensive Force between player $a \in A$ and player $b \in B$ at time t and located at their Center of Pressure**, as the time derivative of their defensive momentum

$$\mathcal{F}_{a,b}(t) = \frac{d}{dt} [\mathcal{L}_{a,b}(t)] = \frac{d}{dt} [p_{a,b}(t) \dot{\mathbf{r}}_{a,b}] = \frac{d}{dt} [p_{a,b}(t)] \dot{\mathbf{r}}_{a,b} + p_{a,b}(t) \ddot{\mathbf{r}}_{a,b}. \quad (4.14)$$

Conceptually, the Defensive Force $\mathcal{F}_{a,b}(t)$ depends on two additive components. The first term is proportional to the rate of variation in the inter-player distance $d_{a,b}$ and to the velocity of their **CP**. It represents the instantaneous reaction to relative displacement and therefore reflects how quickly the pressure exerted by a defender changes as the attacker moves. The second term is proportional to both the magnitude of the pressure and the acceleration of the **CP**, capturing the effect of sudden increases in defensive intensity, such as explosive pressing actions or rapid adjustments to ball progression.

Thus, Defensive Force not only quantifies the strength of the pressing action but also its dynamic responsiveness. The direction of the force vector aligns with the velocity and acceleration of the **CP** trajectory, linking the measure to the actual flow of defensive actions on the pitch. By analogy with mechanics, one can interpret $\mathcal{F}_{a,b}(t)$ as the “effective push” exerted by a defender on an opponent, expressed through their ability to accelerate and sustain pressure within the spatio-temporal context of the game.

Once the *Defensive Force* has been established, it becomes natural to define the *Defensive Work* carried out along a trajectory Γ . This quantity allows to quantify the effective defensive effort (dynamic work) exerted by an individual player, or by a whole team, upon an opposing player or team.

In classical mechanics, the work $\mathcal{W}_k$ of particle k performed by a force $\mathcal{F}_k(t)$ acting along a trajectory Γ with instantaneous velocity $\dot{\mathbf{r}}(t)$ is defined as

$$\mathcal{W}_k = \int_{\Gamma} \mathcal{F}_k(t) \cdot d\mathbf{s}_k(t), \quad (4.15)$$

where $d\mathbf{s}_k(t) = \dot{\mathbf{r}}_k(t)dt$ denotes the infinitesimal displacement vector along the trajectory.

This formulation expresses work as the cumulative effect of the force projected onto the actual displacement of the system over time. If the force is constant and collinear with the displacement, the expression reduces to the familiar scalar product $W = \mathbf{F} \cdot \mathbf{s}$, where $\mathbf{s}$ is the total displacement. In the context of football analytics, this general principle is reformulated to quantify the *Defensive Work*, where $\mathcal{F}$ corresponds to the Defensive Force generated by pressure interactions, and the trajectory is described by the motion of the associated Center of Pressure (**CP**). From this perspective, Defensive Work is conceptually equivalent to the mechanical energy expended by the defensive system in applying and maintaining pressure against the opponent's attacking dynamics.

We define the **Defensive Work generated by a defensive force, $\mathcal{F}_{a,b}$, exerted on a discrete trajectory, Γ , created between a defender $a \in A$ and an attacker $b \in B$** , as the sum of the scalar product between the instantaneous force and the corresponding displacement vectors of the **CP**:

$$\mathcal{W}_{a,b} = \sum_{f \in \mathbb{F}_{a,b}} \mathcal{F}_{a,b}(t_f) \cdot \Delta \mathbf{s}_{a,b}(t_f), \text{ where } \Delta \mathbf{s}_{a,b}(t_f) = \dot{\mathbf{r}}_{a,b}(t_f) \frac{1}{h}. \quad (4.16)$$

Here, h denotes the sampling frequency (in Hz) of the tracking data (25 Hz in this study). This discrete formulation enables the calculation of Defensive Work on specific sets of frames $f \in \mathbb{F}_{a,b}$, thereby facilitating temporal segmentation of pressing actions.

Collective variables: the CP-system (short summary)

To characterise team-level defensive behaviour, we define a **global Center of Pressure (CP) for the system** as the pressure-weighted average of all instantaneous dyadic CP locations. Let

$$p_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} p_{a,b}(t), \quad \mathbf{r}_{A,B}^{CP}(t) = \frac{1}{p_{A,B}(t)} \sum_{b \in B} \sum_{a \in A} p_{a,b}(t) \mathbf{r}_{a,b}(t).$$

Using $p_{a,b}(t)$ instead of $\mathcal{P}_{a,b}(t)$ yields the same CP location because the constant scaling cancels in the weighted average; absolute totals can be recovered via

$$\mathcal{P}_{A,B}(t) = \pi r^2 \sum_{b \in B} \sum_{a \in A} p_{a,b}(t).$$

This global CP provides a single, spatially anchored reference to visualise and compare collective defensive actions. The associated velocity, momentum, force, and work can be defined analogously; importantly, these CP-based quantities are not, in general, equal to the simple sums of the corresponding dyadic quantities because they incorporate the creation and dissolution of pressure interactions over time. Full derivations and the remaining CP-system expressions are provided in the Supplementary Material. Table 4.1 shows a summary of the defensive metrics defined in this article for the different aggregations as well as those belonging to the system’s CP.

Metrics	SINGLE		SYSTEM				CP-SYSTEM			
	a, b		a, B		A, B		CP_{aB}		CP_{AB}	
	Sym	Source	Sym	Source	Sym	Source	Sym	Source	Sym	Source
Position	$\mathbf{r}_{a,b}$	Main					$\mathbf{r}_{a,B}^{CP}$	Supp	$\mathbf{r}_{A,B}^{CP}$	Main
Velocity	$\dot{\mathbf{r}}_{a,b}$	Main					$\dot{\mathbf{r}}_{a,B}^{CP}$	Supp	$\dot{\mathbf{r}}_{A,B}^{CP}$	Supp
Acceleration	$\ddot{\mathbf{r}}_{a,b}$	Main					$\ddot{\mathbf{r}}_{a,B}^{CP}$	Supp	$\ddot{\mathbf{r}}_{A,B}^{CP}$	Supp
Pressure	$\mathcal{P}_{a,b}$	Main	$\mathcal{P}_{a,B}$	Main	$\mathcal{P}_{A,B}$	Main	$\mathcal{P}_{a,B}^{CP}$	Supp	$\mathcal{P}_{A,B}^{CP}$	Supp
Defensive Momentum	$\mathcal{L}_{a,b}$	Main	$\mathcal{L}_{a,B}$	Main	$\mathcal{L}_{A,B}$	Main	$\mathcal{L}_{a,B}^{CP}$	Supp	$\mathcal{L}_{A,B}^{CP}$	Supp
Defensive force	$\mathcal{F}_{a,b}$	Main	$\mathcal{F}_{a,B}$	Main	$\mathcal{F}_{A,B}$	Main	$\mathcal{F}_{a,B}^{CP}$	Supp	$\mathcal{F}_{A,B}^{CP}$	Supp
Defensive work	$\mathcal{W}_{a,b}$	Main	$\mathcal{W}_{a,B}$	Main	$\mathcal{W}_{A,B}$	Main	$\mathcal{W}_{a,B}^{CP}$	Supp	$\mathcal{W}_{A,B}^{CP}$	Supp

Table 4.1: Summary of the proposed metrics / KPIs and their relationships. “Main” indicates definitions provided in the main text; “Supp” indicates definitions and derivations provided in the Supplementary Material (CP-system).

4.2.2 Data and preprocessing

To illustrate and evaluate the proposed dynamic defensive framework in a single-match proof-of-concept, we used publicly available open-source tracking and event data provided by Metrica Sports BV [2]. Specifically, the dataset `Sample_Game_3` was analysed, containing synchronised optical tracking data and event annotations from a professional football match.

Tracking coordinates are provided in normalised pitch units and were converted to metric units using the official field dimensions (105 m $\times$ 68 m). Player and ball positions were sampled at a frequency of 25 Hz, with all data anonymised and fully synchronised across tracking and events. Metadata include team formations and attacking directions, allowing consistent spatial alignment across halves.

Dataset integrity and player participation were verified by cross-checking total frame counts, ball-in-play segments, and individual player involvement. Figure 4.5 summarises player participation and possession phases, confirming internal consistency and the absence of systematic data loss.

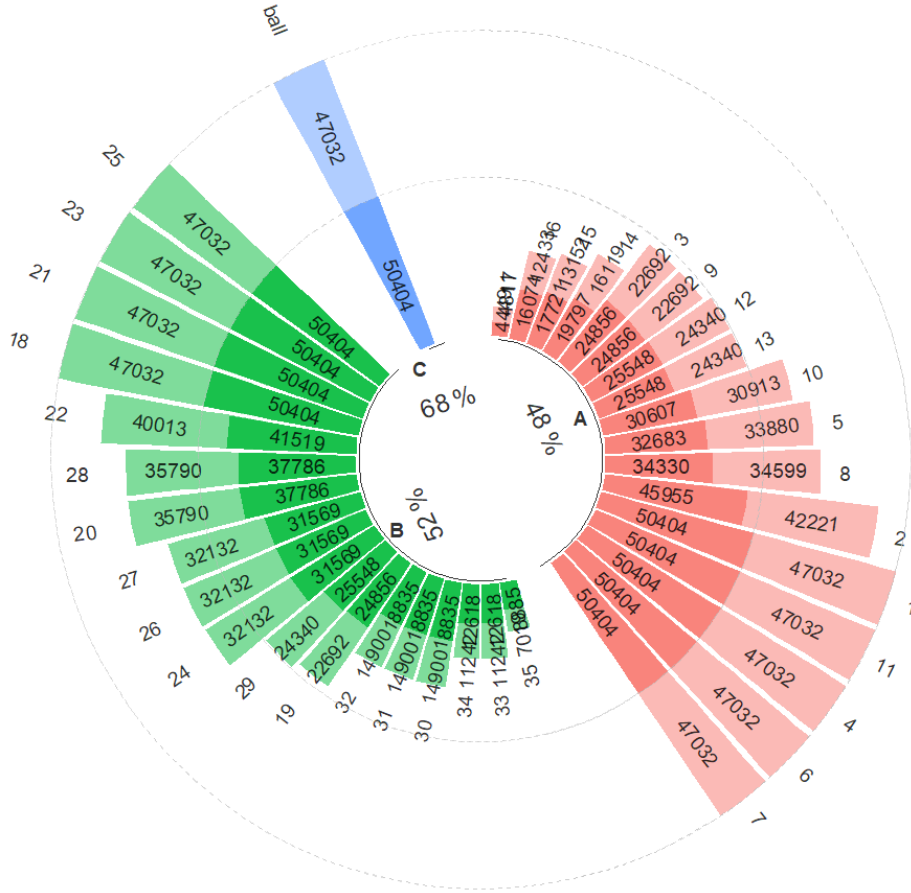


Figure 4.5: Player participation and possession timeline for both teams. Each bar corresponds to one player and its length indicates time on pitch (frames). Shading indicates team possession, and the blue band marks ball-in-play frames used for the analysis.

We then applied the following pre-processing steps to obtain consistent kinematic variables and ensure data quality.

An inertial reference frame was defined with the origin at the pitch Center, requiring a transformation of the original dataset. The second half of the match was reoriented so that each team maintained a consistent attacking orientation for analysis.

A plausibility check on the player velocities was performed. As a reference, the maximum sprinting speeds achievable by humans without assistance can be approximated from Usain Bolt's 100-meter world record at the 2009 World Championships in Berlin, where he achieved an average speed of 37.58 km/h (10.44 m/s) and a peak instantaneous speed of 44.72 km/h (12.42 m/s) between 60 and 80 meters. While short, abrupt changes in movement may yield slightly higher instantaneous values, sustained velocities above these thresholds are physiologically implausible. From a total of 2143592 observations, only 552 exceeded 12 m/s. No additional smoothing or filtering was applied to preserve the spatio-temporal properties of the pressure dynamics under study. The formulation of

defensive pressure metrics inherently reduces the impact of isolated velocity spikes.

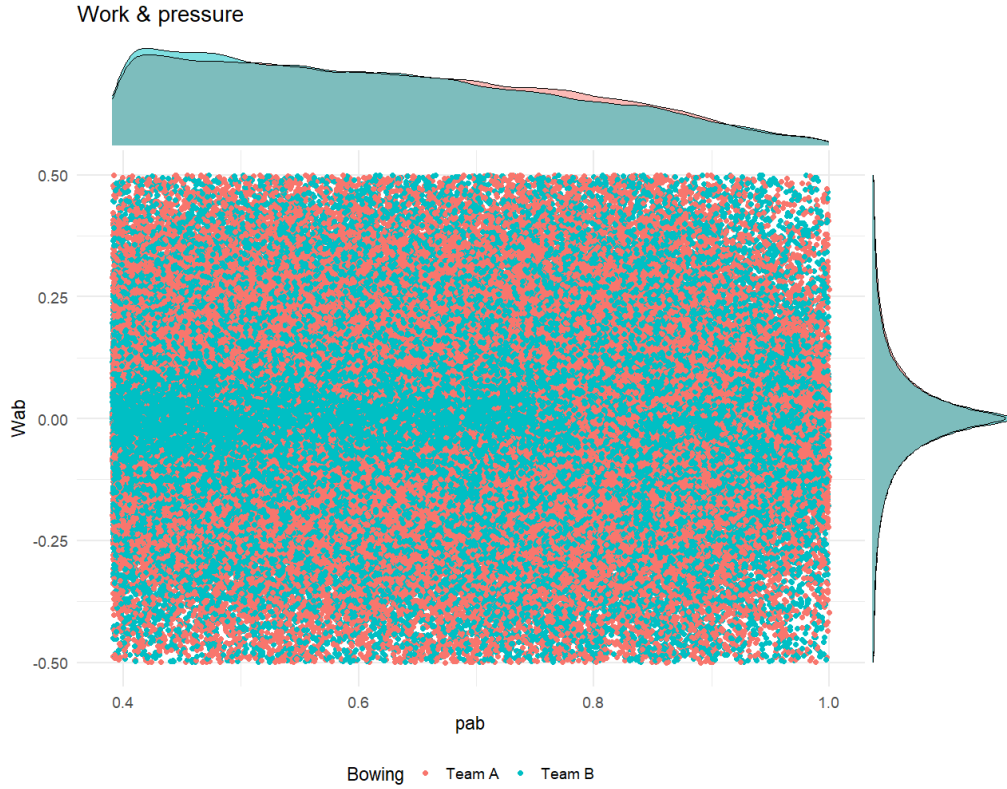


Figure 4.6: Joint distribution of area-normalised pressure density $p_{a,b}$ (horizontal axis) and Defensive Work $W_{a,b}$ (vertical axis), segmented by ball possession for Teams A and B. Pressure is broadly distributed, whereas $W_{a,b}$ is centered near zero with heavy tails, reflecting intermittent high-demand interactions.

4.2.3 Statistical analysis

We have performed a statistical analysis focused on characterising the empirical distributions of defensive Pressure and Defensive Work, both at player-player and team levels, and on examining how these quantities vary as a function of ball possession. Frame-wise values were computed for all valid interactions and segmented according to possession state (Team A or Team B in possession).

To explore global properties of the model, marginal density distributions of area-normalised pressure density and defensive work were analysed (Fig. 4.6). Pressure exhibits a stable distribution skewed towards moderate-to-high values, reflecting persistent spatial constraint, whereas Defensive Work is centered near zero with heavy tails, capturing intermittent bursts of energetic defensive effort.

In addition, team-level Defensive Work distributions were compared across possession phases using violin plots (Fig. 4.7). These representations display central tendency and dispersion while preserving the full distributional structure. Percentile thresholds (Q_{75} and Q_{99}) were subsequently used to identify high-intensity defensive episodes, enabling

the analysis of both frequent demanding situations and extreme peaks of defensive workload.

Dyad matrices can be mapped back to pitch locations via the dyad CP, preserving spatial context (Fig 4.4).

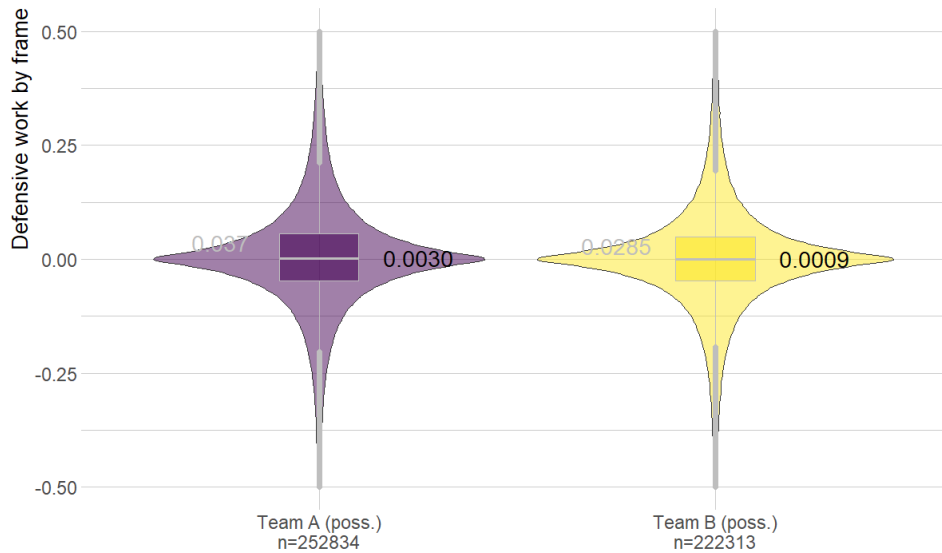


Figure 4.7: Team-level distributions of Defensive Work by possession phase (Team A defending vs. Team B defending). Violin plots summarise central tendency and dispersion while preserving the full distributional shape.

4.2.4 Reliability and methodological evaluation

No additional reliability procedures (e.g., test–retest or inter-rater analyses) were conducted, as all proposed metrics are computed directly from optical tracking data whose positional accuracy and temporal resolution have been extensively tested in previous studies. In the present study, which is intended as a single-match methodological demonstration, we therefore focus on technical and construct-oriented evaluation rather than population-level validation.

Specifically, we examine whether the derived quantities—Pressure, Defensive Momentum, Defensive Force, and Defensive Work—exhibit coherent behaviour across distinct phases within the match (open play, transitions, and set-pieces) and whether the observed spatial and temporal patterns are consistent with established tactical principles (e.g., compactness, pressing waves, and possession-dependent defensive demands).

Finally, the internal balance relationships implied by the formulation (e.g., near-balance of work between teams) are used as an additional internal consistency check. Future work will extend the evaluation to larger datasets and will benchmark the metrics against independent annotations and existing baselines.

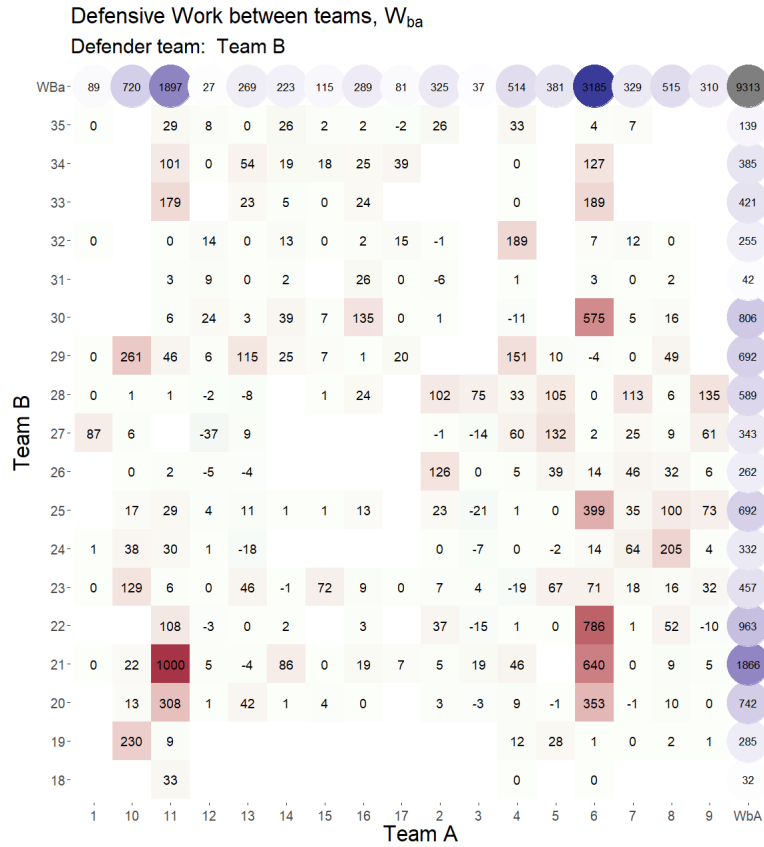
4.3 Results

This section presents results from a single full-match dataset used as a proof-of-concept demonstration of the proposed framework. Results are organised to progressively illustrate spatial, distributional, individual, and temporal properties of the derived metrics, with emphasis on their interpretability and practical relevance for applied performance analysis.

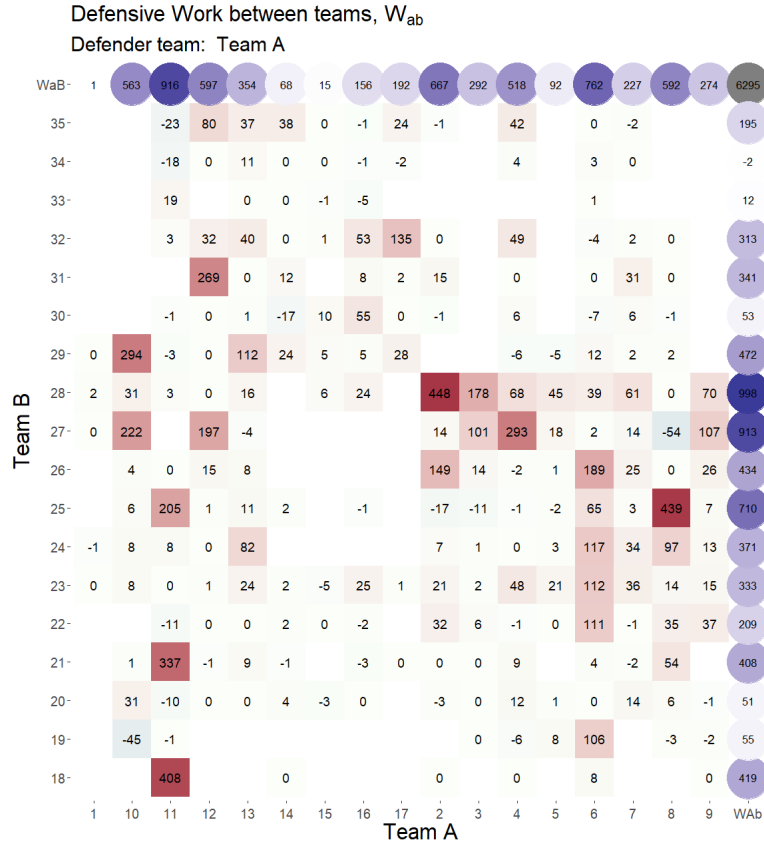
4.3.1 Interaction-level defensive workload

The matrix heatmaps in Figures 4.8a and 4.8b show the distribution of Defensive Work ($\mathcal{W}$) accumulated in interactions between players of opposing teams. Each cell reports the match-accumulated work associated with a specific defender–attacker dyad, while the marginal sums provide the aggregated contribution of each individual player across all their interactions.

When Team A defends, the workload is concentrated in a limited number of dyads, indicating targeted defensive match-ups and a more selective distribution of effort. In contrast, Team B exhibits a more homogeneous distribution of Defensive Work across players and interactions, suggesting a collective defensive organisation with shared responsibilities.



(a) Team B is defending



(b) Team A is defending

Figure 4.8: Dyad-level Defensive Work matrices for the match. Each cell reports the match-accumulated signed Defensive Work $\mathcal{W}_{a,b}$ for a defender–attacker dyad (a, b) —red indicates higher positive work; negative values are shown within cells—. Row and column marginals show per-player aggregates, and the outer circles summarise team-level totals.

(a) Team B defending against Team A. (b) Team A defending against Team B.

To quantify workload concentration from the visualisations, we computed two complementary summaries from the dyad-level work matrices: (i) the Top- k share of total absolute work, $\sum |\mathcal{W}_{a,b}|$, which captures how much of the workload is carried by a small set of highest-cost match-ups; and (ii) the Gini coefficient, which summarises overall concentration across all dyads. Because $\mathcal{W}_{a,b}$ is signed, concentration is quantified using $|\mathcal{W}_{a,b}|$ to avoid cancellations.

Table 4.2 shows that when Team A is defending (Fig. 4.8b), the workload is more concentrated in a small number of dyads: Top-5: 34.9%, Top-10: 48.8%, than when Team B is defending (Fig. 4.8a): Top-5: 27.8%, Top-10: 44.9%, while overall concentration remains similar (Gini ≈ 0.78 in both cases).

Defender team (heatmap)	#dyads	Top-5	Top-10	Gini
Fig. 4.8a (Team B defends)	214	27.8%	44.9%	0.782
Fig. 4.8b (Team A defends)	217	34.9%	48.8%	0.785

Table 4.2: Dyad-level defensive workload concentration (match-level). Workload is computed using absolute Defensive Work, $|\mathcal{W}_{a,b}|$, to avoid cancellations due to sign. Top- k quantify the contribution of the highest-work dyads to the total, while the Gini coefficient summarises overall concentration.

From an applied perspective, these results enable rapid identification of high-cost match-ups and assessment of whether defensive workload is concentrated or evenly distributed across the defensive unit.

4.3.2 Player-level defensive workload profiles

Figures 4.9a and 4.9b present the average Defensive Work per player, normalised by minutes played, using butterfly plots. Workload exerted against opponents and workload received are shown separately, allowing direct comparison at individual level.

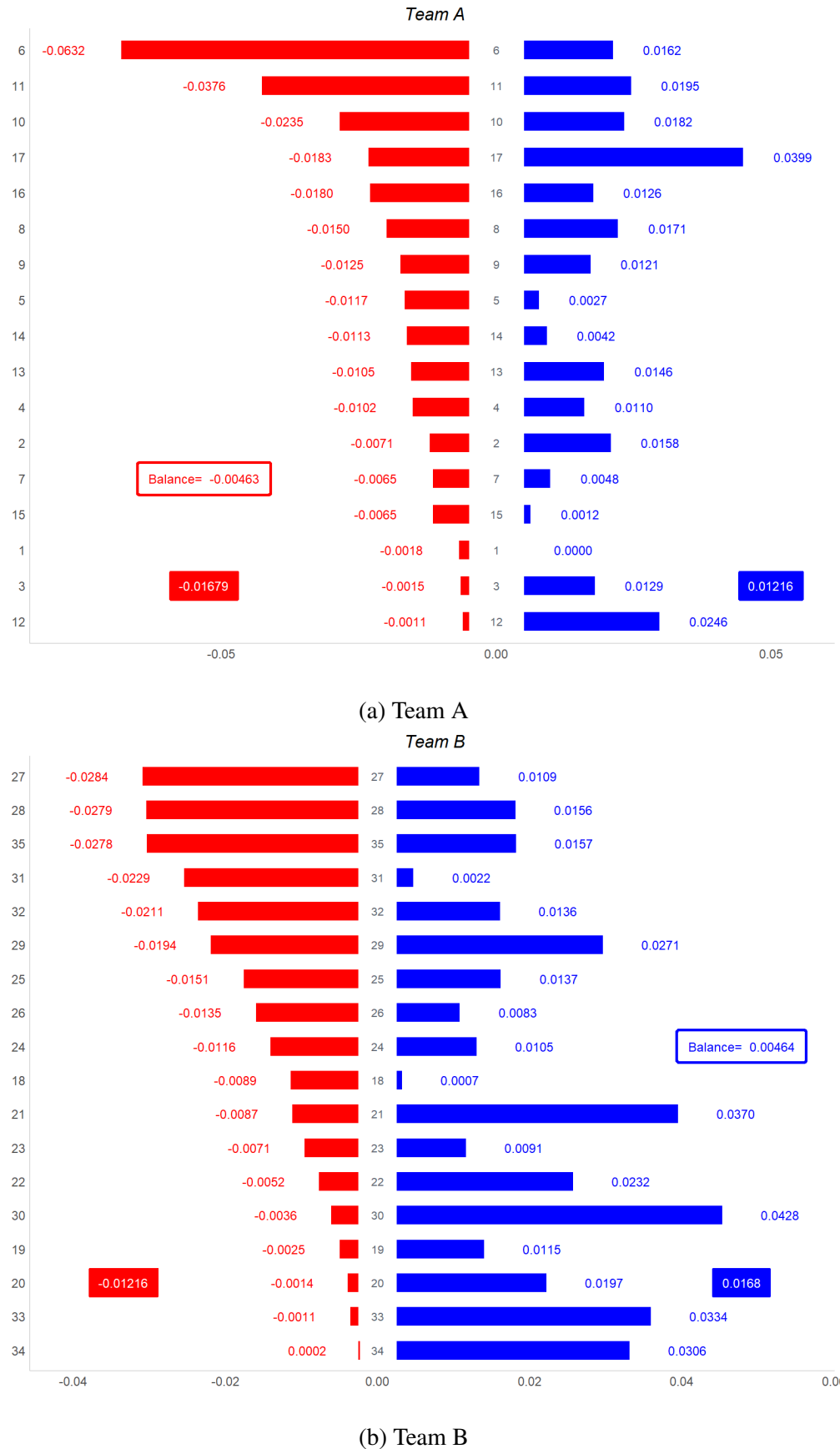


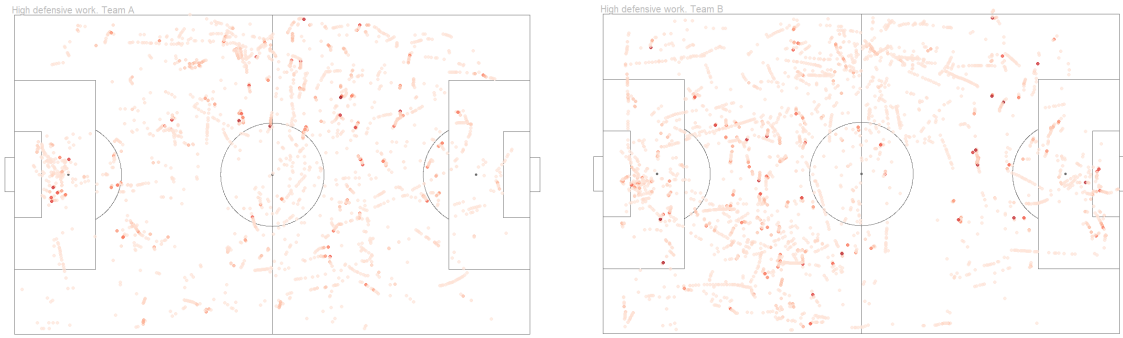
Figure 4.9: Butterfly plots of average Defensive Work per player. Red bars indicate workload received and blue bars workload exerted. The two panels correspond to Team A and Team B, enabling a direct comparison of how defensive workload is distributed across roles.

The differences between teams reflect distinct tactical roles and defensive responsibilities. Certain players accumulate frequent moderate defensive actions, while others experience fewer but more extreme workload peaks. The balance of total Defensive Work between teams is preserved (the small discrepancy observed, -0.00463 for Team A in front of $+0.00464$ for Team B, arises only from rounding in numerical calculations), confirming internal consistency of the metric.

These profiles provide a compact tool for role-based workload assessment and support individual monitoring in applied performance settings.

4.3.3 High-intensity defensive actions

Extreme defensive actions were identified using percentile-based thresholds of Defensive Work, focusing on events above Q_{99} , the 99th percentile, $\mathcal{W} > Q_{99}$. The spatial distribution of these events is shown in Fig. 4.10.



(a) High defensive work developed by Team A. (b) High defensive work developed by Team B.

Figure 4.10: Spatial distribution of extreme Defensive Work events ($\mathcal{W} > Q_{99}$). Team A (attack direction: right to left) shows a concentration of high-work actions in advanced zones near the opponent's penalty area, whereas Team B (attack direction: left to right) displays a wider distribution of such events across different areas of the pitch.

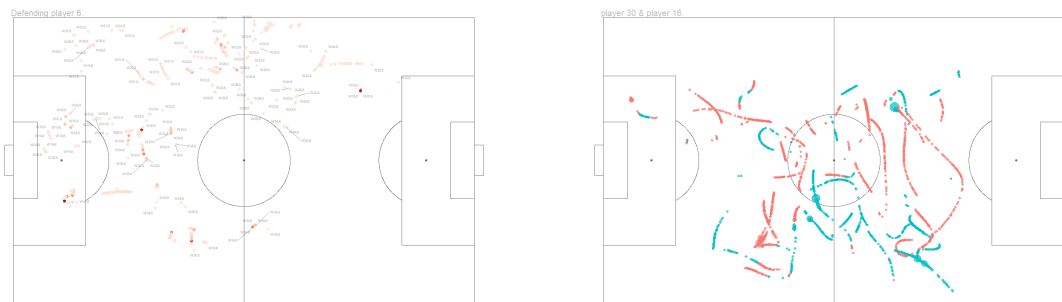


Figure 4.11: Examples of high-demand dyadic interactions identified from Defensive Work thresholds. Left: a peak event above the Q_{99} threshold. Right: spatial footprint of repeated high-demand events above Q_{75} for a representative defender-attacker pair; Team A is shown in red and Team B in blue.

Team A displays a higher concentration of extreme defensive actions in advanced zones, consistent with sustained high pressing. Team B shows a more dispersed spatial pattern, including deeper areas of the pitch.

This analysis isolates the most demanding defensive efforts and provides actionable information for managing high-intensity load and evaluating pressing strategies.

4.3.4 Temporal dynamics during counter-pressing

Figures 4.12a and 4.12b analyse a 6-second counter-press. Team-level metrics show Pressure declining as the attack develops, while a sharp rise and convergence in Momentum indicates the onset of coordinated pressing in motion. Peaks in Defensive Work reflect intense duels. Player contributions reveal the reaction was initiated by the nearest defender (Player 20), with tracking back from others (Players 21, 27) aligning the press, before wider players supported containment.

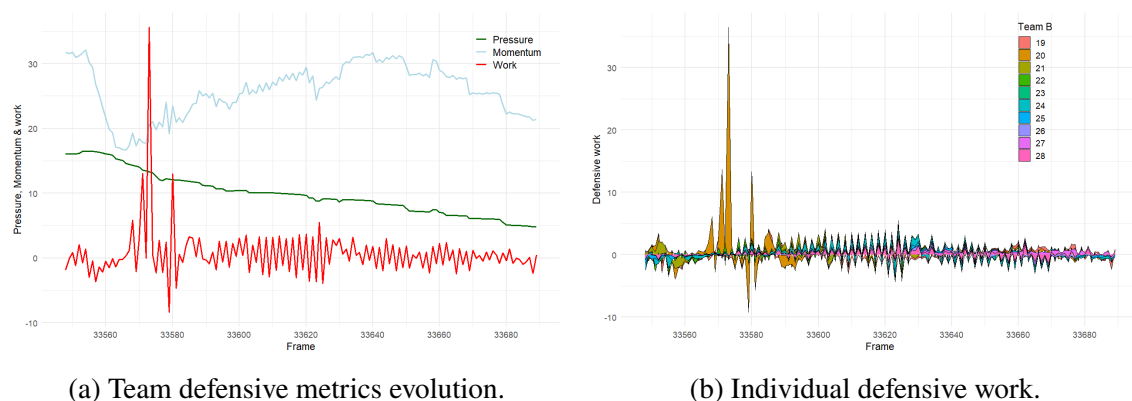


Figure 4.12: Counter-pressing sequence. (a) Pressure drops, Momentum rises with convergence, and Work shows peaks. (b) Player contributions shift from initiator (Player 20) to supporting players.

Figures 4.13a and 4.13b depict a longer 11-second counter-attack. Pressure is stable, while a pronounced rise in Momentum curve shows coherent pressing in motion. Sustained Defensive Work oscillations indicate prolonged effort. Player contributions shift from advanced players (10, 7) initially to midfield and defensive reinforcements (13, 14, 15) later.

4.3.5 Defending a Corner and a Throw-in

Figure 4.14 shows defensive metrics across eight corners. Typical profiles feature very high initial Pressure due to player proximity, which then decreases. Tactical variations are evident: a short corner (subfigure 4) led to lower initial metrics, rising only as attackers entered the box, while a low-commitment final play (subfigure 8) showed lower initial pressure and a workload shift to defending the rebound.

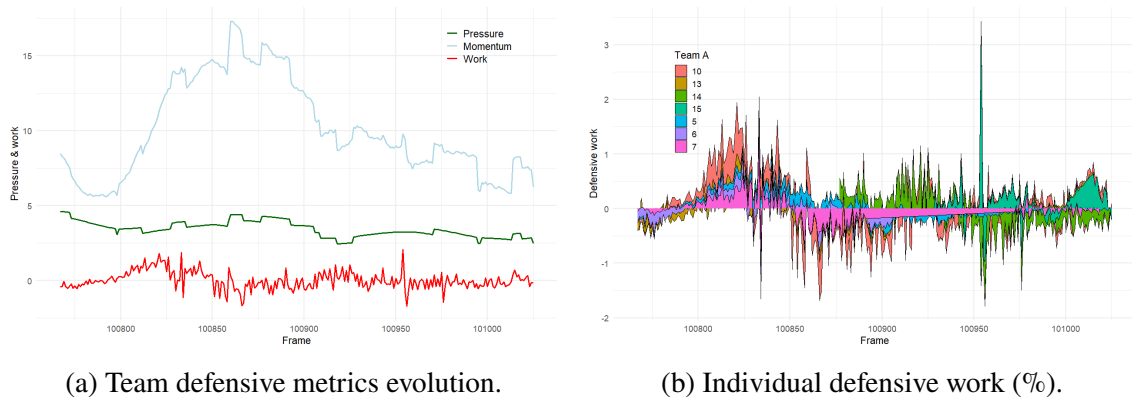


Figure 4.13: Extended counter-attack defense. (a) Stable pressure, rising coherent Momentum, and sustained Work. (b) Defensive workload redistributes from forwards to midfielders and defenders.

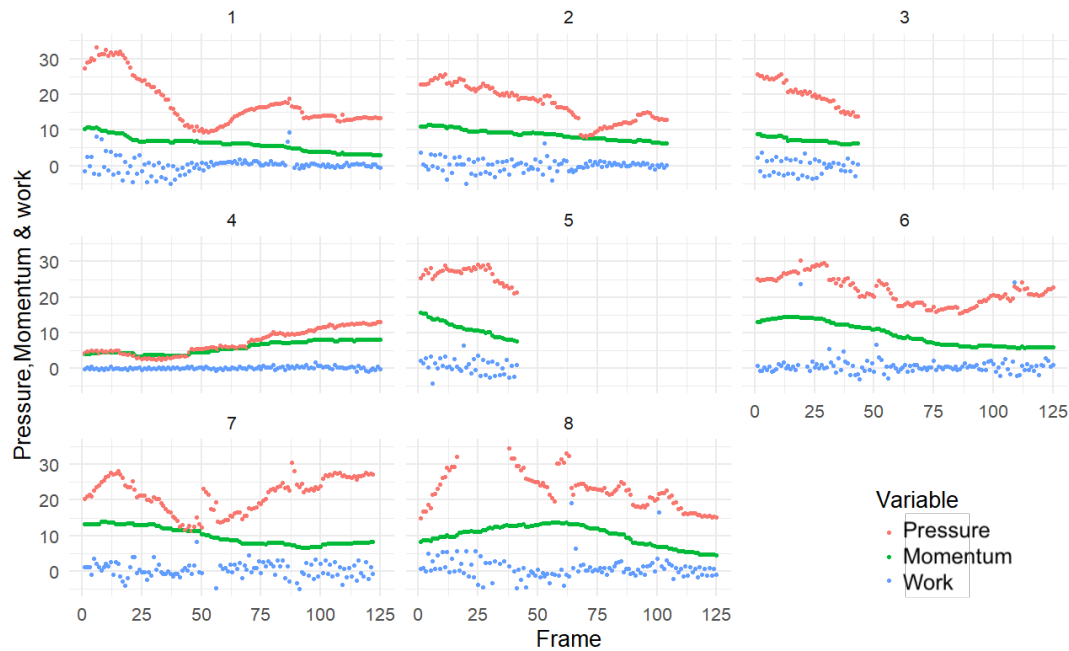


Figure 4.14: Defensive metrics for all match corners. General pattern shows high initial pressure decreasing post-delivery. Subfigures 4 and 8 exemplify how tactics and context alter the profile.

Figure 4.15 presents the aggregated profile of 50 throw-ins. Pressure rises gradually from a moderate baseline as play resumes. Momentum increases and converges, showing progressive defensive alignment. Defensive Work oscillates with small peaks, reflecting the low-to-moderate, repeated efforts characteristic of throw-in duels, contrasting with the sharper spikes seen in corners or counter-attacks.

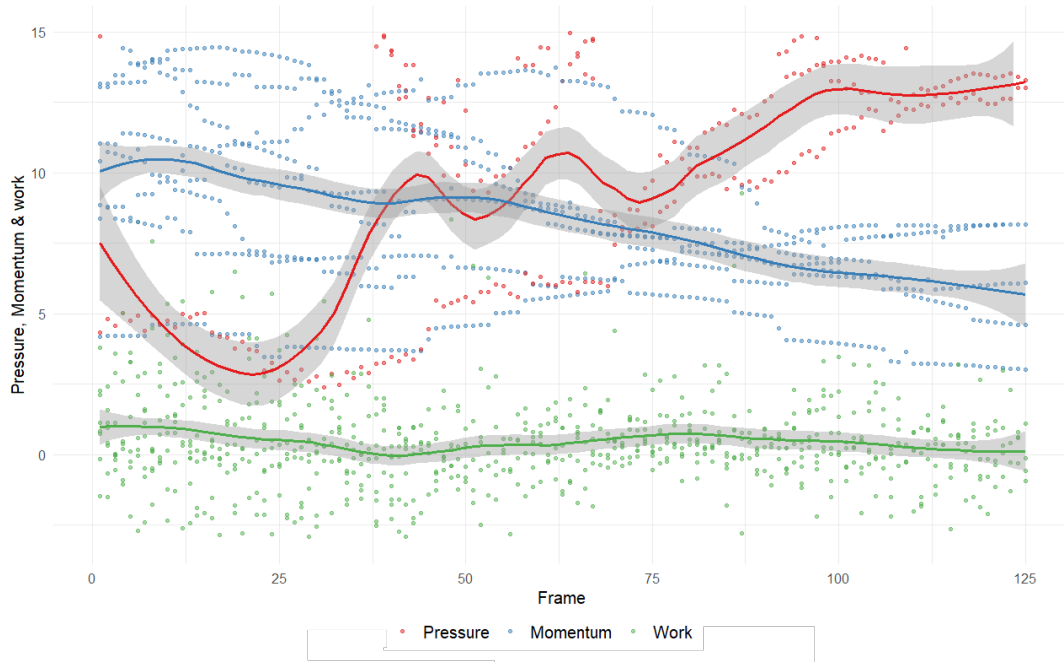


Figure 4.15: Aggregated defensive metrics for throw-ins. Gradual increase in Pressure and Momentum alignment. Work shows sustained, low-intensity oscillations.

4.4 Discussion

Using real match tracking data as a single-match proof-of-concept, the proposed dynamic defensive framework provides a coherent and operational description of defensive behaviour in football by integrating spatial configuration, motion, and coordination into a unified set of metrics. The results illustrate how the framework can support post-match review of pressure evolution, and motivate future work towards near-real-time implementations. In this section, we discuss the main implications of these findings for applied performance analysis and the limitations of the current approach.

4.4.1 Interpretation of defensive workload and coordination

A central contribution of the framework is the explicit separation between spatial constraint (Pressure) and energetically demanding defensive actions (Defensive Work). The results show that these two dimensions exhibit markedly different distributions and temporal behaviours, supporting the idea that defensive effectiveness and defensive cost should not be conflated. Because Defensive Work is signed, workload summaries are more robust when computed on $|\mathcal{W}|$ (to avoid cancellations), while signed $\mathcal{W}$ remains informative for interpreting directionality in specific interactions.

From an applied perspective, this distinction allows practitioners to differentiate between defensive phases characterised by positional discipline and those requiring repeated high-cost interventions. This addresses a common limitation of traditional physical metrics, which often fail to capture the collective and contextual nature of defensive actions.

4.4.2 Interaction and player-level insights

The interaction-level analysis highlights how defensive workload is distributed across specific match-ups of players rather than uniformly across the pitch. Concentration of Defensive Work in a limited number of dyads may reflect targeted defensive match-ups, whereas more homogeneous distributions are consistent with a more collective distribution of defensive responsibility.

At player level, the observed variability in Defensive Work profiles illustrates how tactical roles translate into distinct defensive demands. These results indicate that the proposed metrics can support role-specific evaluation, enabling comparisons between players occupying similar positions and informing individualised conditioning or recovery strategies.

For Team A (Fig. 4.9a), the distribution of defensive work shows strong asymmetries. Certain players—particularly players 6, 11 and 10 of Team A—are heavily targeted, receiving the highest average workload values, whereas the exertion of work is more dispersed across multiple players (notably players 17 and 12). This indicates that while Team A generated collective defensive activity, their own key playmakers were consistently pressured by the opposition.

By contrast, Team B (Fig. 4.9b) presents a different pattern. The highest received workloads fall on players 27, 28 and 35, corresponding to their forwards and left-back, suggesting that Team A's pressing strategy concentrated on closing down attacking outlets. However, Team B's exerted work is more prominent through several midfielders and wide players (21, 22, 33, and 34), producing some of the largest positive values in the dataset. This distribution reflects a more decentralised defensive response, where several individuals contributed significantly to collective pressing.

4.4.3 High-intensity defensive actions

The identification of extreme defensive events using percentile-based thresholds (e.g., Q_{99}) provides a practical mechanism for isolating the most demanding phases of play. The spatial patterns of high-intensity Defensive Work observed in the results are consistent with different pressing strategies, such as sustained high press versus deeper, more reactive defending. In this study, percentile thresholds were computed per defending team over ball-in-play frames, and extreme events were defined as $\mathcal{W} > Q_{99}$.

In practical terms, both the heatmap and butterfly analyses consistently reveal concentrated defensive work in Team A and a more distributed response in Team B, offering coaches and analysts a complementary tool to diagnose pressing structures and player workloads while preserving spatial context through the Centers of Pressure framework.

In applied environments, this information is directly relevant for managing peak defensive load, designing training drills that reproduce competitive demands, and assessing the sustainability of tactical approaches over the course of a match or season.

4.4.4 Temporal dynamics and applied monitoring

The analysis of short counter-pressing sequences demonstrates the ability of the framework to capture rapid transitions from spatial constraint to coordinated motion. Peaks in Defensive Work coincide with phases of synchronised defensive effort, highlighting the importance of coordination in determining defensive cost.

This temporal resolution supports potential applications in near-real-time analysis (subject to implementation constraints), where identifying demanding defensive passages can inform substitution strategies or in-game tactical adjustments. In addition, aggregated temporal profiles provide reference patterns that can be used to calibrate training intensity against match demands.

4.4.5 Methodological considerations and limitations

Several limitations should be acknowledged. First, the analysis is based on a single match and is intended primarily as a methodological illustration rather than a generalised performance study. Future work should evaluate the robustness of the proposed metrics across larger datasets, different competition levels, and varying tactical contexts.

Second, the framework relies on tracking data accuracy and does not explicitly account for contextual factors such as opponent quality, match status, or fatigue. Integrating contextual variables and extending the model to longer temporal horizons represent promising directions for future research.

In addition, the interaction radius r controls the spatial horizon of the model and may influence absolute magnitudes; a sensitivity analysis and data-driven calibration of r are left for future work.

Finally, while the physical analogy underlying the framework provides internal consistency, the interpretation of absolute metric values should be approached cautiously. The primary strength of the approach lies in comparative and relative analyses rather than absolute thresholds.

The proposed framework provides new tools that translate complex collective behaviour into interpretable, quantifiable metrics grounded in physical principles. By bridging spatial, temporal, and energetic aspects of defensive play, the approach offers a complementary perspective to existing physical and tactical indicators and supports applied decision-making in professional football environments.

Overall, the framework offers a physically grounded and interpretable set of metrics that complements existing physical and tactical indicators, and provides a basis for future validation on larger datasets.

4.5 Conclusions

This work introduced a physics-inspired, tracking-based framework to characterise defensive behaviour in football through a set of dynamic metrics that integrate spatial configuration, motion, and coordination. By modelling defensive interactions as a continuous, time-varying process, the approach enables a time-resolved description of collective defending that complements traditional event-based summaries.

The first main contribution of the study is the explicit separation between spatial constraint (Pressure) and energetically demanding defensive actions (Defensive Work). This distinction supports analyses that differentiate structured compactness from high-cost interventions, addressing a common limitation of aggregate physical indicators.

Second, Defensive Work is defined at defender–attacker level and can be aggregated consistently across players, time windows, and teams. This yields interpretable summaries of match-ups, defensive responsibilities, and role-specific demands, and enables concise measures of workload concentration (e.g., Top- k and Gini) from dyad matrices.

Third, the proposed metrics capture short, high-intensity defensive sequences (e.g., counter-pressing) and longer passages of sustained effort, supporting the identification of demanding phases and the attribution of contributions at team and player level. These properties motivate applied use cases in tactical review and workload monitoring, and provide a basis for future near-real-time implementations.

As this study is based on a single match and is intended as a methodological illustration, broader conclusions about performance or tactical effectiveness require validation across larger datasets, competition levels, and match contexts. Future work will also examine sensitivity to modelling parameters (e.g., the interaction radius r) and benchmark the proposed indicators against existing defensive baselines and independent annotations.

Overall, the framework provides a principled and operational mapping from tracking data to interpretable defensive metrics grounded in classical mechanics. By bridging spatial, temporal, and energetic aspects of defensive play within a unified formulation, it offers a transferable toolkit for football performance analysis and a foundation for future large-scale validation and applied deployment.

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4.6 Appendix: Aggregated metrics

This appendix presents the formulas for metrics aggregated at higher organizational levels: player-team and team-team.

4.6.1 Collective pressures

The **pressure exerted by player $a \in A$ over the team B** and the **pressure exerted by team A over team B** are defined as,

$$\mathcal{P}_{a,B}(t) = \sum_{b \in B} \mathcal{P}_{a,b}(t) \quad , \quad \mathcal{P}_{A,B}(t) = \sum_{a \in A} \mathcal{P}_{a,B}(t) = \sum_{a \in A} \sum_{b \in B} \mathcal{P}_{a,b}(t) . \quad (4.17)$$

4.6.2 Defensive Momentum

We extend the definition of Defensive Momentum to aggregated or collective variables of the system. We define the **Defensive Resultant Momentum generated in the game by a single player $a \in A$ against all opponents B , at time t** as

$$\mathcal{L}_{a,B}(t) = \sum_{b \in B} \mathcal{L}_{a,b}(t) . \quad (4.18)$$

Also, the **Defensive Resultant Momentum generated in the game between the two teams A and B , at time t** is defined by:

$$\mathcal{L}_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} \mathcal{L}_{a,b}(t) \quad (4.19)$$

4.6.3 Total Momentum

We define the **Total Momentum** generated in the game by a single player $a \in A$ against all opponents B , at time t as

$$\mathcal{L}T_{a,B}(t) = \sum_{b \in B} \|\mathcal{L}_{a,b}(t)\|_2 . \quad (4.20)$$

Note that $\|\mathcal{L}_{a,B}(t)\|_2 \leq \mathcal{L}T_{a,B}(t)$.

Also, we define the **Total Momentum** generated in the game between the two teams A and B , at time t as

$$\mathcal{L}T_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} \|\mathcal{L}_{a,b}(t)\|_2 . \quad (4.21)$$

Again, $\|\mathcal{L}_{A,B}(t)\|_2 \leq \mathcal{L}T_{A,B}(t)$.

The Total Momentum is an scalar quantity $\mathcal{L}T_{A,B}(t)$ and represents the theoretical maximum **Defensive Momentum** of the system, which would be reached if all individual pressures were exerted in perfectly aligned directions. For this reason, the ratio between

the vectorial and scalar forms provides a useful indicator of the spatial coherence of defensive activity:

$$0 \leq \frac{\|\mathcal{L}_{a,B}(t)\|_2}{\mathcal{L}\mathcal{T}_{a,B}(t)} \leq 1, \quad 0 \leq \frac{\|\mathcal{L}_{A,B}(t)\|_2}{\mathcal{L}\mathcal{T}_{A,B}(t)} \leq 1, \quad (4.22)$$

This normalised index takes values between 0 and 1 and reflects the extent to which defensive momentum is spatially coordinated across the pitch. Values approaching 1 indicate that individual player contributions are directionally aligned and mutually reinforcing, generating a coherent collective pressing action that adapts effectively to the attacking structure of the opposition. Conversely, values closer to 0 suggest that individual momenta are more scattered or oriented in divergent directions, often as a consequence of the opponent's attacking patterns, thereby reducing the overall efficiency of the collective pressure.

Accordingly, this ratio provides an indicator of defensive synchronisation, both at the micro level (a single player against multiple opponents) and at the macro level (the interaction between two teams), while also accounting for the tactical demands imposed by the opposition's style of play. Tracking its temporal evolution makes it possible to identify specific phases of play: periods of highly coordinated defensive activity, such as during organised high pressing against structured build-up play, and phases of greater disorganisation, such as transitional moments or when facing unpredictable attacking movements that disrupt defensive shape.

Temporal dynamics during counter-pressing

Figures 4.16a and 4.16b show relation between **Resultant** and **Total Momentum** in the Counter-pressing sequence that we have showed in the main article (Figures 4.12a and 4.13a).

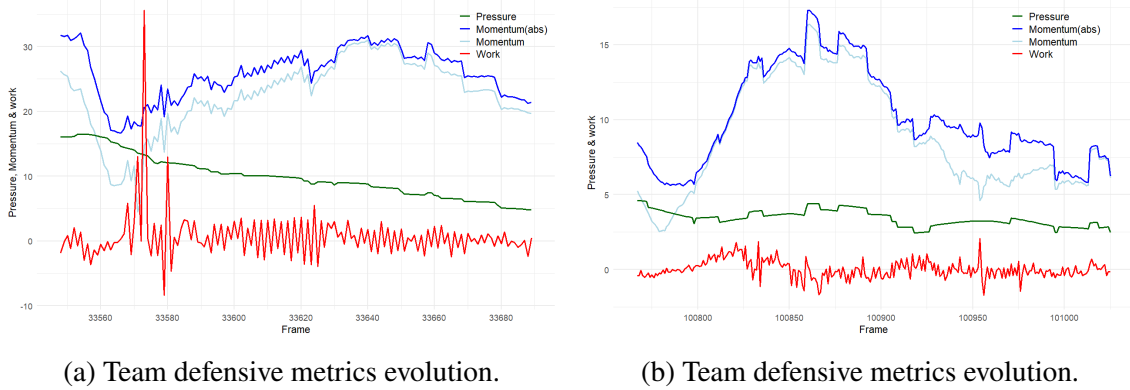


Figure 4.16: Counter-pressing sequence evolution including the Total Momentum.

Defending a Corner

Figure 4.17 shows the **Resultant** and **Total Momentum** evolution across the eight corners we have showed in the main article (Figure 4.14).

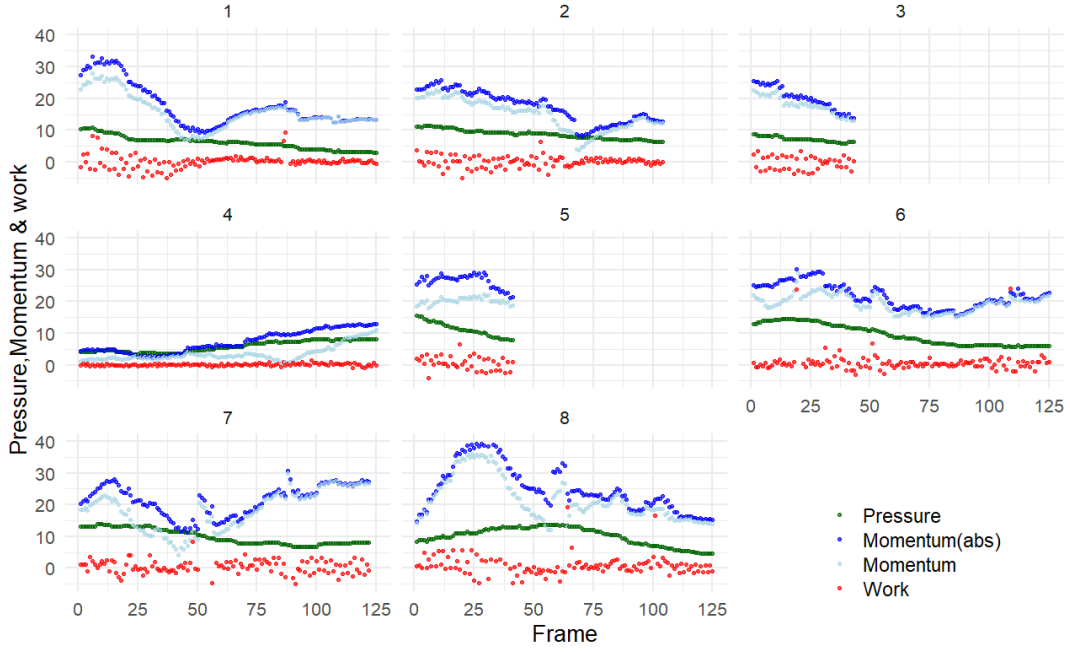


Figure 4.17: Defensive metrics for all match corners. General pattern shows high initial pressure decreasing post-delivery. Subfigures 4 and 8 exemplify how tactics and context alter the profile.

4.6.4 Defensive Force

We define the **Defensive Force** both generated by player $a \in A$ against the whole team B and the generated collectively by team A against team B as

$$\mathcal{F}_{a,B}(t) = \sum_{b \in B} \mathcal{F}_{a,b}(t), \quad (4.23)$$

$$\mathcal{F}_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} \mathcal{F}_{a,b}(t). \quad (4.24)$$

In analogy with physics, these aggregated measures may be viewed as the “net force” exerted by a player on an entire opposing team, or by a whole team on its opponent. Their temporal evolution provides insight into the capacity of a defensive system to adapt collectively to the attacking dynamics imposed by the rival team, such as ball circulation, dribbling actions or structured build-up play.

4.6.5 Defensive Work

We define the **Defensive Work** generated by player $a \in A$ against the whole team B as the sum of the scalar products between all defensive forces $\mathcal{F}_{a,b}(t)$ and their associated displacement vectors, across the discrete trajectory Γ generated by the frames, $f \in \mathbb{F}_A$ as

$$\mathcal{W}_{a,B} = \sum_{f \in \mathbb{F}_A} \sum_{b \in B} \mathcal{F}_{a,b}(t_f) \cdot \Delta s_{a,b}(t_f). \quad (4.25)$$

This formulation reflects the effective dynamic contribution of a single defender to the suppression of an entire opponent team across time.

We define the **Defensive Work generated collectively by team A against team B** as the total sum of the scalar products between all defensive forces exerted and their respective displacements over the discrete trajectory Γ generated by the frames, $f \in \mathbb{F}_A$ as

$$\mathcal{W}_{A,B} = \sum_{f \in \mathbb{F}_A} \sum_{b \in B} \sum_{a \in A} \mathcal{F}_{a,b}(t_f) \cdot \Delta \mathbf{s}_{a,b}(t_f). \quad (4.26)$$

4.7 Appendix: CP-System

In analogy with the calculation of the Center of Mass in variable-mass dynamical systems, it is conceptually useful to consider the total defensive pressure

$$\mathcal{P}_{A,B}^{CP}(t) = \mathcal{P}_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}(t). \quad (4.27)$$

This construct allows us to spatially locate, graph and visualise $\mathcal{P}_{A,B}(t)$, $\mathcal{L}_{A,B}(t)$, $\mathcal{F}_{A,B}(t)$, and others directly on the pitch, providing a single point of reference for collective defensive behaviour. It is straightforward to define

$$\mathbf{r}_{A,B}^{CP}(t) = \frac{1}{\mathcal{P}_{A,B}(t)} \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}(t) \mathbf{r}_{a,b}(t). \quad (4.28)$$

The velocity is defined as

$$\begin{aligned} \dot{\mathbf{r}}_{A,B}^{CP}(t) &= \frac{d}{dt} [\mathbf{r}_{A,B}^{CP}(t)] \\ &= \frac{1}{\mathcal{P}_{A,B}(t)} \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}(t) \dot{\mathbf{r}}_{a,b}(t) \\ &\quad + \frac{1}{\mathcal{P}_{A,B}(t)} \left[\sum_{b \in B} \sum_{a \in A} \dot{\mathcal{P}}_{a,b}(t) \mathbf{r}_{a,b}(t) - \frac{\dot{\mathcal{P}}_{A,B}(t)}{\mathcal{P}_{A,B}(t)} \mathbf{r}_{A,B}^{CP}(t) \right], \end{aligned} \quad (4.29)$$

and the acceleration

$$\ddot{\mathbf{r}}_{A,B}^{CP}(t) = \frac{d}{dt} [\dot{\mathbf{r}}_{A,B}^{CP}(t)]. \quad (4.30)$$

Then

$$\mathcal{L}_{A,B}^{CP}(t) = \mathcal{L}_{A,B}(t) + \sum_{b \in B} \sum_{a \in A} \dot{\mathcal{P}}_{a,b}(t) \mathbf{r}_{a,b}(t) - \frac{\dot{\mathcal{P}}_{A,B}(t)}{\mathcal{P}_{A,B}(t)} \mathbf{r}_{A,B}^{CP}(t). \quad (4.31)$$

Here, $\mathcal{L}_{A,B}(t)$ represents the *Defensive Momentum* of the System:

$$\mathcal{L}_{A,B}(t) = \sum_{b \in B} \sum_{a \in A} \mathcal{P}_{a,b}(t) \dot{\mathbf{r}}_{a,b}(t). \quad (4.32)$$

Observe that $\mathcal{L}_{A,B}(t) \neq \mathcal{L}_{A,B}^{CP}(t)$. The discrepancy arises because the latter incorporates the time variation of pressures themselves, analogous to variable-mass systems in physics. In football terms, this difference captures not only the motion of pressure zones but also their **creation and dissolution as player interactions** appear and disappear. This $\mathcal{L}_{A,B}(t)$ is computationally straightforward and describes the aggregate magnitude of defensive momentum, but without spatial anchoring. In contrast, $\mathcal{L}_{A,B}^{CP}(t)$ attaches this momentum to a concrete location on the pitch, enabling intuitive visualisation of team-level defensive dynamics. This makes the **CP**-based approach especially useful for analysing and comparing collective defensive actions during specific match contexts, such as immediately after ball loss, following a goal kick or throw-in, or in transitional phases.

We define the **Defensive Force of the CP** and **Defensive Work of the CP**

$$\begin{aligned}
 \mathcal{F}_{A,B}^{CP}(t) &= \frac{d}{dt} [\mathcal{L}_{A,B}^{CP}(t)] \\
 &= \mathcal{F}_{A,B}(t) + \frac{d}{dt} \left[\sum_{b \in B} \sum_{a \in A} \dot{\mathcal{P}}_{a,b}(t) \mathbf{r}_{a,b}(t) - \frac{\dot{\mathcal{P}}_{A,B}(t)}{\mathcal{P}_{A,B}(t)} \mathbf{r}_{A,B}^{CP}(t) \right] \\
 &= \mathcal{F}_{A,B}(t) + \sum_{b \in B} \sum_{a \in A} \ddot{\mathcal{P}}_{a,b}(t) \mathbf{r}_{a,b}(t) + \sum_{b \in B} \sum_{a \in A} \dot{\mathcal{P}}_{a,b}(t) \dot{\mathbf{r}}_{a,b}(t) \\
 &\quad - \left(\frac{\ddot{\mathcal{P}}_{A,B}(t)}{\mathcal{P}_{A,B}(t)} - \frac{\dot{\mathcal{P}}_{A,B}^2(t)}{\mathcal{P}_{A,B}^2(t)} \right) \mathbf{r}_{A,B}^{CP}(t) - \frac{\dot{\mathcal{P}}_{A,B}(t)}{\mathcal{P}_{A,B}(t)} \dot{\mathbf{r}}_{A,B}^{CP}(t)
 \end{aligned} \tag{4.33}$$

$$\mathcal{W}_{A,B}^{CP} = \sum_{f \in \mathbb{F}_A} \mathcal{F}_{A,B}^{CP}(t_f) \cdot \Delta \mathbf{s}_{A,B}^{CP}(t_f), \quad \text{where} \quad \Delta \mathbf{s}_{A,B}^{CP}(t) = \dot{\mathbf{r}}_{A,B}^{CP}(t) h^{-1}. \tag{4.34}$$

The present article prioritises the explicit formulation of player–player and team–team dynamics, leaving the player–team variants as directly inferable extensions for practitioners and applied researchers.

In an analogous manner, it is also possible to compute the **(CP) between a single player and the opposing team**, thereby allowing the visualisation of the interventions of an individual player relative to the rival collective. This naturally gives rise to the following **CP**-based metrics:

$$\mathbf{r}_{a,B}^{CP}(t), \quad \dot{\mathbf{r}}_{a,B}^{CP}(t), \quad \ddot{\mathbf{r}}_{a,B}^{CP}(t), \quad \mathcal{P}_{a,B}^{CP}(t), \quad \mathcal{L}_{a,B}^{CP}(t), \quad \mathcal{W}_{a,B}^{CP}. \tag{4.35}$$

These quantities are not formally computed here, since their deduction follows directly and unambiguously from the player–player and team–team formulations presented in the body text.

5

Introduction

This chapter constitutes the last study included in the present doctoral thesis and is incorporated as a non-published manuscript at the time of submission. It builds directly upon the defensive pressure metric introduced and formally defined in Chapter 3, where unitary defensive pressure was modelled using tracking data to quantify proximity-based defensive interactions between players.

While Chapter 3 focuses on the mathematical formulation and validation of the metric itself, the present article shifts the emphasis towards its *interpretation at the individual player level*. In particular, this study explores how the empirical distributions of unitary pressure values can be used to characterise individual defensive behaviour, moving beyond aggregate or volume-based indicators that are commonly employed in performance analysis.

The central objective is to investigate whether the defensive pressure metric can be used to characterise and discriminate defensive player behaviours by analysing the *shape* of their pressure distributions and assessing whether classical descriptive statistics can reveal meaningful and interpretable defensive typologies.

To this end, this chapter introduces a framework based on distributional shape parameters—specifically Fisher’s skewness coefficient and the kurtosis coefficient—to describe two fundamental dimensions of defensive behaviour: aggressiveness in approaching opponents and consistency in the execution of defensive pressure. By combining these descriptors with unsupervised clustering techniques, the article proposes the concept of a player-specific *defensive signature*, offering a compact, quantitative, and interpretable representation of individual defensive styles.

Within the broader structure of the thesis, this chapter serves as a conceptual and methodological bridge between the formal definition of defensive pressure metrics and

their applied interpretation in tactical analysis and player profiling. It illustrates how mathematically defined pressure measures can be translated into actionable insights at the player level, providing a foundation for subsequent work on defensive intensity, workload, and the temporal evolution of pressure in football.

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The Defensive Signature of the Player: Characterising Aggressiveness and Consistency in Defensive Pressure

Abstract

This study investigates individual defensive behavior in football through the analysis of unitary defensive pressure derived from player tracking data. Building on a previously defined pressure metric, the work focuses on the shape of pressure distributions at the player level rather than on aggregate or volume-based indicators. The aim is to characterise the *shape* of the distributions of these **unitary pressures** for each player throughout the match. For doing that, classical descriptive statistics are used, with particular attention to Fisher's skewness coefficient and the Kurtosis coefficient, to characterize two key dimensions of defensive behavior: aggressiveness to approach opponents and consistency in the execution of defensive pressure.

The analysis is conducted on tracking data from a complete professional football match, considering only active phases of play and interactions that involve defensive pressure. Individual player distributions are examined using histograms, density functions, and box-whisker diagrams. The resulting shape descriptors are subsequently used as input variables for unsupervised clustering, allowing the identification of distinct defensive pressure profiles.

The results show that the players exhibit clearly differentiated defensive behaviors that cannot be captured by mean pressure values alone. The proposed framework provides an interpretable and quantitative basis for defining player-specific defensive signatures, with potential applications in tactical analysis, player profiling, and scouting.

keywords: football analytics · tracking data · defensive pressure · scouting · spatio-temporal metrics

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5.1 Introduction

Defensive pressure has become one of the most influential tactical elements in modern football, particularly in settings where collective structures rely heavily on the ability of individual defenders to restrict the time and space available to opponents. In a previous article ([1]), a formal metric of defensive pressure was introduced using exclusively tracking data, enabling the identification and quantification of individual pressure episodes based on the effective proximity between attackers and defenders.

In this work it was defined the quantitative descriptive metric termed **unitary pressure**, $p_{a,b}(f)$ that depends exclusively on the distance between each player a from team A ($a \in A$) and each player b from team B ($b \in B$), and is defined for each of the 135,000 frames that make up the match, sampled at $h = 25$ fps ($f \in \mathcal{F}$). The value of the **unitary pressure** is a real number between 0 and 1 ($0 \leq p_{a,b}(f) \leq 1$). This value reaches its maximum when player a is located spatially in the same region as player b , representing the situation of maximum pressure. As the distance between the two players increases, the value of $p_{a,b}(f)$ decreases continuously until it reaches 0.391, beyond which the function becomes identically zero. This formulation provides a continuous, interpretable, and normalised metric of defensive proximity.

Building on that methodological foundation, the present article focuses on the *typology of players from a defensive perspective*, analysing the characteristic ways in which each player performs their defensive pressure throughout a match. The aim of this work is to *evaluate and compare* two defensive qualities—aggressiveness in approaching opponents and regularity in defensive execution—in order to deepen our understanding of individual defensive pressure behaviours in football. This approach shows how studying the shape of defensive pressure distributions can provide new quantitative tools for describing player roles and defensive styles within and across teams.

The central aim of this study is to characterise the *shape* of the distributions of these **unitary pressures** for each player throughout the match. To this end, we analyse individual empirical distributions using histograms, density functions, and box–whisker diagrams, with particular attention to **shape parameters**: the **Fisher’s skewness coefficient** (CA) and the **Kurtosis coefficient** (CC).

Fisher’s **skewness coefficient** (CA) allows us to determine the displacement of the mean of the distribution relative to the median or to the normal distribution. A decreasing value of CA_a indicates that the distribution shifts to the right, producing a longer left-hand tail, which suggests *higher defensive quality*: player a has approached players from team B more frequently. Thus, low or negative CA values enable the quantification of a typically qualitative characteristic associated with aggressive or proactive defensive behaviour.

The **Kurtosis coefficient** (CC) describes the degree to which a distribution is more peaked or flattened than the normal distribution. A positive value ($CC_a > 0$) indicates a leptokurtic distribution, associated with a *more regular and disciplined* defensive behaviour: the player applies pressure following stable and repeatable patterns, regardless

of whether the distances achieved are optimal or not from a defensive standpoint. This measure allows us to distinguish between players who press aggressively but irregularly and those who maintain consistent defensive patterns. Further theoretical details on descriptive and graphical analytics can be found in [3, 4].

Consequently, a player who exhibits both a **low or negative** skewness coefficient (indicating frequent reductions in defensive distance) and a **high** Kurtosis coefficient (indicating stronger behavioural consistency) may be characterised as a defender who “bites” opponents in a *disciplined and recurrent* manner. The joint analysis of CA and CC provides a robust quantitative basis for describing individual defensive styles. This study explores player typologies along two complementary axes. First, we consider players who “bite” more aggressively, that is, those whose defensive pressure is defined by frequent and consistent approaches towards their opponents. Secondly, we examine players who adopt a more stable and regular pattern of approach, tending to position themselves at similar distances from opponents regardless of whether those distances are relatively short or long. The joint analysis of these two behavioural axes enables a richer description of defensive profiles and contributes to tactical interpretation, training design, and player recruitment.

Consequently, a player who exhibits both a **low or negative** skewness coefficient (indicating frequent reductions in defensive distance) and a **high** Kurtosis coefficient (indicating stronger behavioural consistency) may be characterised as a defender who “bites” opponents in a *disciplined and recurrent* manner. The joint analysis of CA and CC provides a robust quantitative basis for describing individual defensive styles.

Therefore, the final objective of this work now is to know if the pressure metrics developed in ([1]) can be used to **discriminate defensive qualities** in football players **evaluating and comparing** —*aggressiveness in approaching opponents* and *consistency in the pressure pattern*— both between players and between teams, thereby contributing to a deeper understanding of defensive typologies in professional football.

5.2 Materials and Methods

The analysis was performed on tracking data from a complete professional football match. After removing all frames corresponding to stopped play and all player pairs for which no defensive pressure was exerted, a total of 165,735 observations of the unitary pressure variable $p_{a,b}(f)$ were retained for team A, and 232,768 observations for team B.

Fifteen players from team A and sixteen players from team B participated in the match, with team B acting as the away team. For each player, the empirical distribution of unitary pressure values was constructed over the full duration of the match and subsequently analysed through descriptive statistics and shape parameters, forming the basis for the comparative and clustering analyses presented in the following sections.

Finally, the resulting shape descriptors were used as input features for unsupervised clustering in order to identify defensive pressure typologies among players. In particular,

the K-means clustering algorithm was selected due to its suitability for low-dimensional continuous feature spaces and its widespread use in exploratory data analysis. Since the feature space is defined by the shape parameters CA and CC, which are continuous and comparable across players, K-means provides an efficient and interpretable framework to partition players into groups with similar defensive pressure profiles. Moreover, the centroid-based nature of the algorithm facilitates the tactical interpretation of each cluster in terms of aggressiveness and consistency of defensive behaviour. The number of clusters was set to $K = 3$, allowing for a parsimonious yet expressive categorisation of defensive pressure styles, capturing distinct profiles ranging from aggressive and consistent behaviours to more conservative or irregular patterns, while avoiding over-segmentation of the player set.

The computation of the **Fisher's skewness coefficient** (CA) and the **Kurtosis coefficient** (CC) was carried out using the functions `skewness()` and `kurtosis()` available in RStudio (Posit.co, [2]). In both cases, missing values were excluded from the calculations (`na.rm = TRUE`), and the estimators corresponding to `type = 1` were employed, ensuring consistency with the classical definitions of these shape parameters.

```
skewness(pab, na.rm = TRUE, type = 1)
kurtosis(pab, na.rm = TRUE, type = 1)
```

5.2.1 Theoretical framework

The **total unitary pressure** exerted by player a on all players of team B during a single frame f is defined as

$$p_{a,B}(f) = \sum_{b \in B} p_{a,b}(f), \quad (5.1)$$

where $p_{a,b}(f)$ denotes the unitary pressure exerted by player a on player b at frame f .

Accordingly, the **total unitary pressure** exerted by player a on team B over the entire match is given by

$$p_{a,B}(\mathcal{F}) = \sum_{f \in \mathcal{F}_B} p_{a,B}(f) = \sum_{b \in B} \sum_{f \in \mathcal{F}_B} p_{a,b}(f) = \sum_{f \in \mathcal{F}_B} \sum_{b \in B} p_{a,b}(f), \quad (5.2)$$

where $\mathcal{F}_B \subset \mathcal{F}$ denotes the set of frames in which defensive pressure between teams A and B is present.

This quantitative descriptive metric represents the cumulative defensive pressure exerted by player a on team B throughout the match. Its simplicity of computation, together with its direct interpretability, makes it particularly suitable for applied performance analysis.

Let $n_{a,B}(\mathcal{F})$ denote the **number of defensive interactions** between player a and players of team B . This quantity corresponds to the total number of observations available

for player a . The total population size is then defined as

$$N_{A,B}(\mathcal{F}) = \sum_{a \in A} n_{a,B}(\mathcal{F}), \quad (5.3)$$

representing the total number of unitary pressure observations generated by team A .

At team level, the **total unitary pressure** exerted by team A on team B over the entire match is given by

$$p_{A,B}(\mathcal{F}) = \sum_{a \in A} p_{a,B}(\mathcal{F}) = \sum_{a \in A} \sum_{b \in B} \sum_{f \in \mathcal{F}_B} p_{a,b}(f) = \sum_{f \in \mathcal{F}_B} \sum_{a \in A} \sum_{b \in B} p_{a,b}(f). \quad (5.4)$$

The fundamental variable analysed in this study is the **unitary pressure** $p_{a,b}(f)$ exerted by player a on each player $b \in B$ at every frame $f \in \mathcal{F}_B$. For each player a , the empirical distribution of these values over all interactions constitutes the basis for the computation of the corresponding shape parameters.

The **Fisher's skewness coefficient** and the **Kurtosis coefficient** for player a are therefore defined as

$$CAp_{a,B} = \frac{\sum_{b \in B} \sum_{f \in \mathcal{F}_B} (p_{a,b}(f) - \bar{p}_{a,B})^3}{n_{a,B}(\mathcal{F}) \cdot \sigma_{p_{a,B}}^3}, \quad (5.5)$$

$$CCp_{a,B} = \frac{\sum_{b \in B} \sum_{f \in \mathcal{F}_B} (p_{a,b}(f) - \bar{p}_{a,B})^4}{n_{a,B}(\mathcal{F}) \cdot \sigma_{p_{a,B}}^4} - 3, \quad (5.6)$$

where

$$\bar{p}_{a,B} = \frac{p_{a,B}(\mathcal{F})}{n_{a,B}(\mathcal{F})}, \quad \sigma_{p_{a,B}}^2 = \frac{1}{n_{a,B}(\mathcal{F})} \sum_{b \in B} \sum_{f \in \mathcal{F}_B} (p_{a,b}(f) - \bar{p}_{a,B})^2, \quad (5.7)$$

and where the coefficient of variation is defined as

$$CV_{p_{a,B}} = \frac{\sigma_{p_{a,B}}}{\bar{p}_{a,B}}. \quad (5.8)$$

It should be noted that, in this study, the analysis is not based on samples drawn from a larger population, but rather on the *complete population* of defensive interactions observed during the match, segmented by player. For this reason, the normalisation in the definitions of CA and CC is performed using $n_{a,B}(\mathcal{F})$ instead of $n_{a,B}(\mathcal{F}) - 1$. In any case, given the large magnitude of $n_{a,B}(\mathcal{F})$, the numerical difference between both normalisations is negligible.

To calculate the aggregate metrics for teams A and B , the aggregate shape coefficients have been defined as

$$CA_{p_{AB}} = \frac{\sum_{a \in A} \sum_{b \in B} \sum_{f \in \mathcal{F}_B} (p_{a,b}(f) - \bar{p}_{A,B})^3}{N_{A,B}(\mathcal{F}) \cdot \sigma_{p_{A,B}}^3}, \quad (5.9)$$

$$CC_{p_{AB}} = \frac{\sum_{a \in A} \sum_{b \in B} \sum_{f \in \mathcal{F}_B} (p_{a,b}(f) - \bar{p}_{A,B})^4}{N_{A,B}(\mathcal{F}) \cdot \sigma_{p_{A,B}}^4} - 3, \quad (5.10)$$

where

$$\bar{p}_{A,B} = \frac{p_{A,B}(\mathcal{F})}{N_{A,B}(\mathcal{F})}, \quad \sigma_{p_{A,B}}^2 = \sum_{a \in A} \sum_{b \in B} \sum_{f \in \mathcal{F}_B} \frac{1}{n_{a,b}(f)} (p_{a,b}(f) - \bar{p}_{A,B})^2, \quad (5.11)$$

and

$$CV_{p_{A,B}} = \frac{\sigma_{p_{A,B}}}{\bar{p}_{A,B}}. \quad (5.12)$$

5.3 Results

5.3.1 Group-level team analysis

To initiate the comparative analysis between teams A and B , a descriptive statistical study of the unitary pressure distributions was first conducted at team level. This preliminary step is essential in order to characterise the global behaviour of defensive pressure and to provide a reference framework for the subsequent player-level analysis.

Table 5.1 reports the main statistical descriptors computed for the two populations, corresponding to the unitary pressures exerted by team A on team B (p_{AB}) and vice versa (p_{BA}). These descriptors include measures of central tendency, dispersion, and distribution shape.

	n_{AB}	p_{AB}	$\bar{p}_{AB}$	$Medp_{AB}$	$\sigma_{p_{AB}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{AB}}$	$CC_{p_{AB}}$
A	165,735	103,047	0.6218	0.6096	0.1475	0.2419	0.4936	0.6096	0.7387	0.2829	-0.9711
	n_{BA}	p_{BA}	$\bar{p}_{BA}$	$Medp_{BA}$	$\sigma_{p_{BA}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{BA}}$	$CC_{p_{BA}}$
B	232,768	144,210	0.6195	0.6038	0.1489	0.2465	0.4916	0.6038	0.7366	0.3252	-0.9469

Table 5.1: Statistical descriptors of unitary pressures exerted between teams A and B

Both teams exhibit remarkably similar values in terms of mean and median unitary pressure, indicating comparable overall levels of defensive proximity during the match. Dispersion measures, including the standard deviation and coefficient of variation, are also closely aligned, suggesting that the variability of defensive pressure is similar across both teams.

The most relevant differences emerge when analysing the *shape parameters*. Team A presents a slightly lower skewness coefficient ($CA_{p_{AB}}$) than team B , indicating a marginally higher tendency to generate pressure values closer to the upper range of the metric. Conversely, both distributions are characterised by negative Kurtosis values, revealing relatively flat (platykurtic) distributions at team level. This indicates a broad range of defensive pressure values and the absence of highly concentrated pressure patterns when all players

are aggregated.

Figure 5.1 displays the histograms of p_{AB} and p_{BA} , allowing for a visual comparison of the empirical distributions. The overlap between both histograms confirms the numerical findings, showing very similar global pressure behaviours for the two teams, with subtle differences in asymmetry that anticipate the presence of more pronounced individual-level variability. The histograms for both teams exhibit clearly differentiated peaks. Team A reaches a maximum frequency of approximately 2500 observations, whereas team B approaches 4000. This difference can be mainly attributed to two factors. First, team A holds a higher share of ball possession (around 60%), which naturally reduces the time spent in defensive phases. Second, the teams differ in the defensive intensity displayed when they are out of possession, measured here as the number of defensive pressure interactions generated per frame of the match. In particular, during the defensive phases of team B, a total of 232768 pressure interactions are recorded, compared to 165735 for team A.

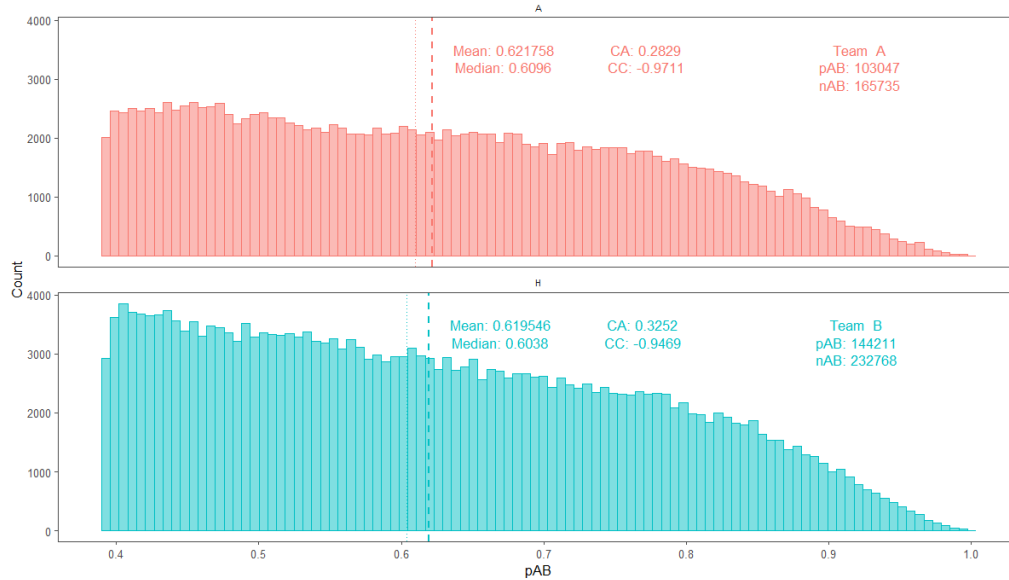


Figure 5.1: Histograms of p_{AB} and p_{BA} : unitary pressure distributions for teams A and B.

5.3.2 Player-level analysis

To compare defensive behaviours at individual level, Box-Whisker diagrams and histograms were computed for the unitary pressure values generated by each player throughout the match. Figures 5.2 and 5.3 show that players exhibit clearly distinct distributions in terms of central tendency (mean and median) and dispersion (interquartile range), indicating that the unitary pressure metric is able to discriminate between individual defensive profiles.

A player exhibiting a higher mean value of unitary pressure against the opposing team tends, on average, to remain closer to opponents during defensive phases. In practical terms, this suggests a more proactive or aggressive defensive behaviour, characterised by frequent reductions in the spatial distance to rival players. Conversely, lower mean values may indicate players who defend more conservatively, maintaining greater distances or operating within more positional defensive roles.

The dispersion of the distributions also provides relevant tactical information. Players with wider interquartile ranges or longer whiskers tend to apply pressure at more variable distances, which may reflect different defensive responsibilities, contextual tactical adjustments, or involvement in multiple defensive scenarios across the pitch. In contrast, narrower distributions suggest more consistent defensive distances and potentially more stable positional roles.

It is important to note that the distributions shown in Figure 5.2 are aggregated with respect to all opposing players. While this provides a global view of each player's defensive behaviour, the same methodology could be applied to partial subsets of opponents. For instance, unitary pressure distributions could be computed separately against forwards, midfielders, or defenders of the opposing team. Such segmented analyses would allow a more detailed characterisation of defensive roles, revealing how players adapt their pressure behaviour depending on the positional context of the opponent.

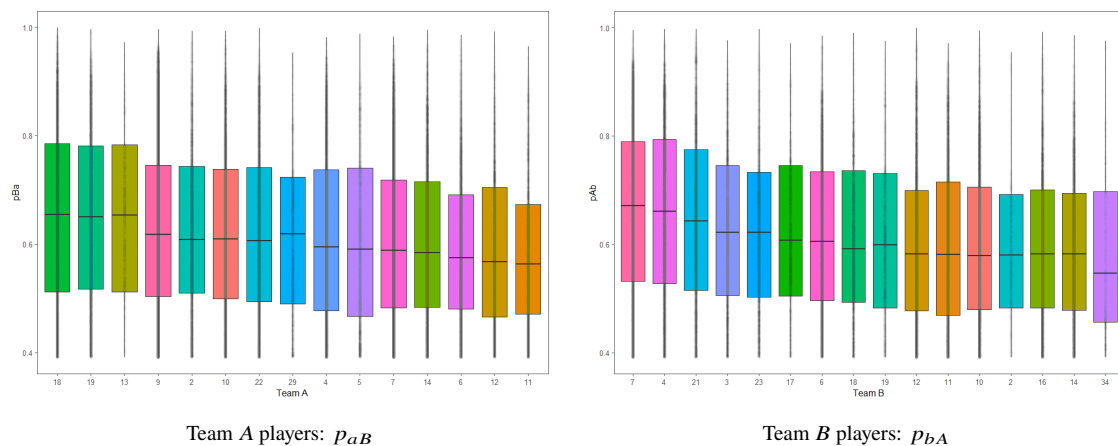


Figure 5.2: Box–Whisker diagrams of player-level unitary pressure distributions for teams A and B, ordered by mean value.

Figure 5.3 presents the histograms of unitary pressure for team B, ordered by increasing values of the Fisher's skewness coefficient (CA). Under the interpretation adopted in this study, lower values of CA correspond to distributions shifted towards higher pressure values, indicating a greater frequency of close defensive approaches. Accordingly, the player with the smallest CA value (top-left, shirt number 19) can be interpreted as the one who “bites” opponents most frequently.

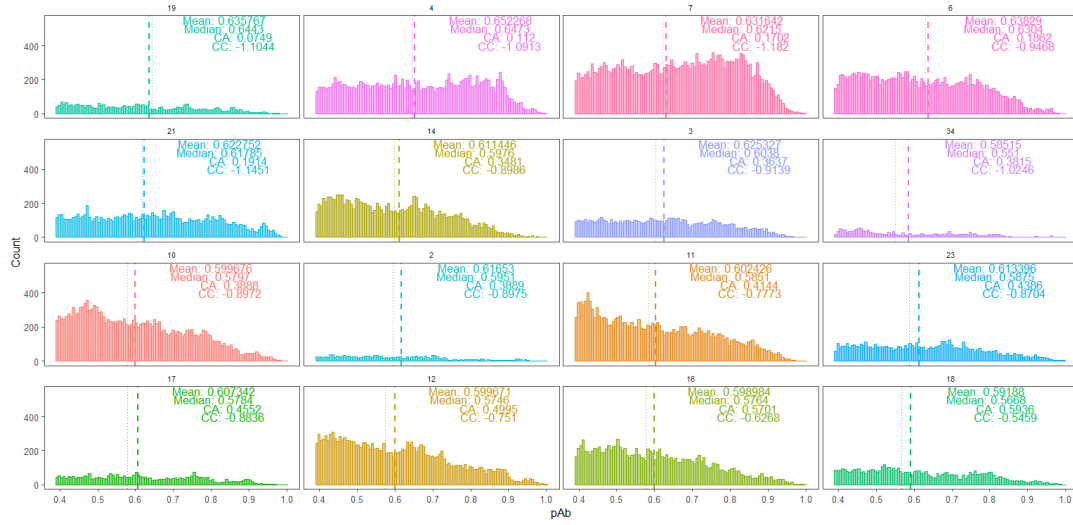


Figure 5.3: Team B players: histograms of p_{bA} ordered by increasing CA .

It should be emphasised that the focus of the analysis is placed on the *shape and quality* of the pressure distributions rather than on their magnitude. Differences in histogram height mainly reflect unequal playing time, such as substitutions during the second half, rather than differences in defensive behaviour. For instance, player 19 entered the pitch around minute 70, as did players 2, 17, 18 and 21, resulting in smaller sample sizes. In contrast, the very low number of observations for player 34 corresponds to the goalkeeper, whose pressure values are mostly associated with set-piece situations.

For visual comparison of distribution shape, density functions provide a clearer representation than raw histograms, as they are not directly influenced by sample size. Figure 5.4 displays the estimated density functions $f(p_{bA})$ for team B , also ordered by increasing CA , highlighting differences in tail behaviour across players.

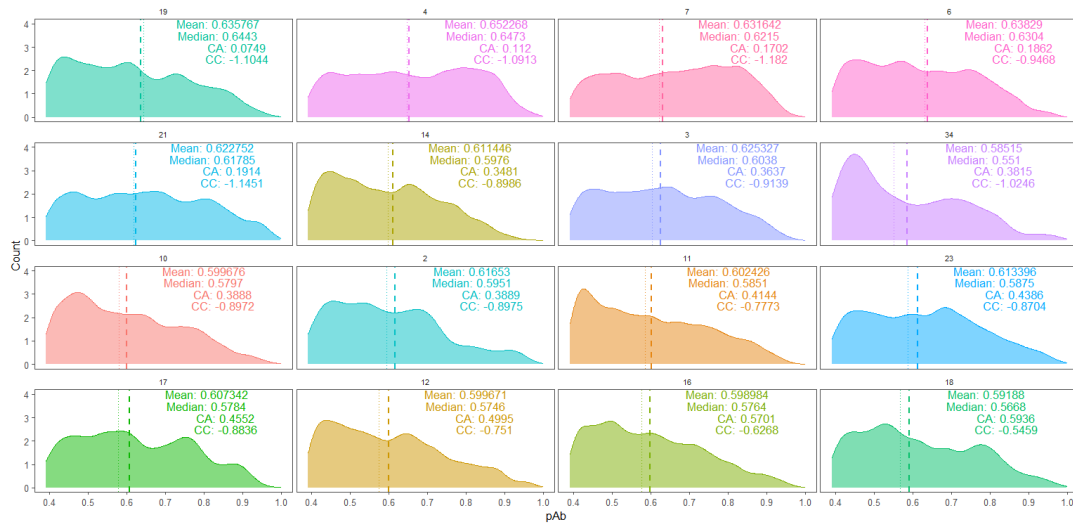


Figure 5.4: Team B players: density functions $f(p_{bA})$ ordered by increasing CA .

A complementary perspective is obtained by ordering players according to the Kurtosis coefficient (CC), which reflects the degree of concentration of the distribution around its central values. Figure 5.5 shows the density functions for team B ordered by decreasing CC . Although all distributions remain platykurtic ($CC_{A,b} < 0$), meaning that the pressure values are relatively spread with respect to a normal distribution, noticeable differences appear across players. In particular, the player with the highest kurtosis value is player 18 ($CC_{A,18} = -0.5459$), while the lowest value corresponds to player 7 ($CC_{A,7} = -1.182$). These differences indicate varying degrees of concentration of the defensive pressure values around their central tendency.

Players with higher (less negative) kurtosis exhibit density curves that are more concentrated around their typical pressure values, suggesting a more consistent reproduction of similar defensive distances over time. In contrast, players with more negative kurtosis values display flatter and more dispersed distributions, indicating a greater variability in the distances at which pressure is applied. From a tactical perspective, this variability may reflect differences in defensive roles, positional responsibilities, or contextual adaptations during the match. Players with more concentrated distributions tend to reproduce a more stable defensive pattern, whereas players with flatter distributions appear to adjust their pressure distances more flexibly depending on the evolving situations of play.

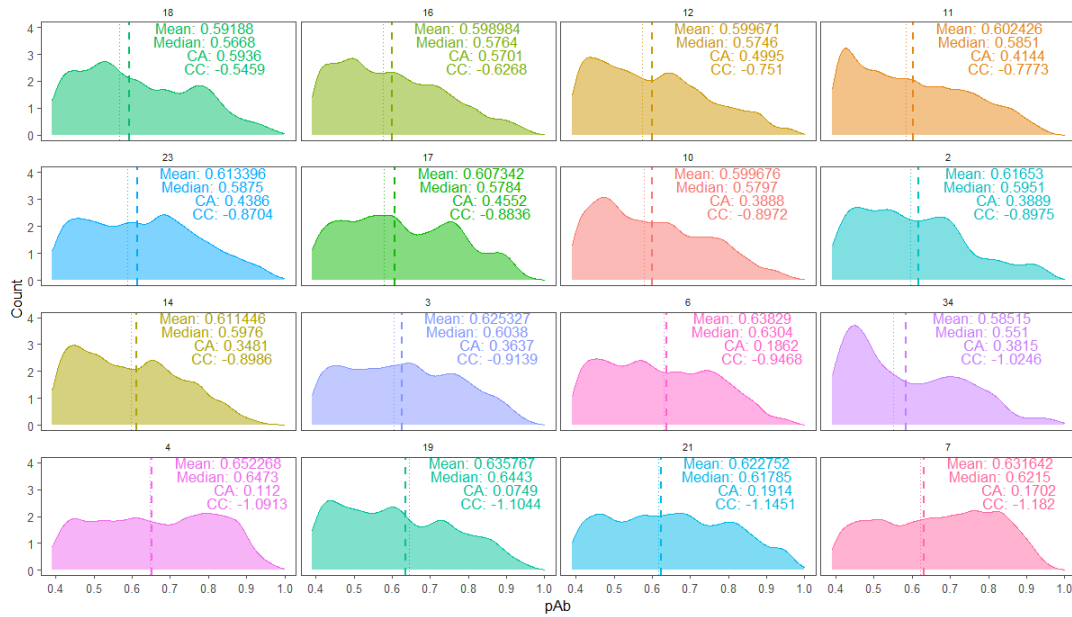


Figure 5.5: Team B players: density functions $f(p_{bA})$ ordered by decreasing CC .

Tables 5.2 and 5.3 report the full set of descriptive statistics computed for the unitary pressure distributions of each player in teams A and B , respectively. In both tables, players are sorted in ascending order of Fisher's skewness coefficient (CA), so that the

most aggressive “biting” profiles appear at the top. Shaded rows highlight players whose CA values are below the corresponding team-average skewness (i.e., $CA_{p_{aB}} \leq CA_{p_{AB}}$ for team A and $CA_{p_{bA}} \leq CA_{p_{BA}}$ for team B). Under the interpretation proposed in this work, these players tend to generate distributions shifted towards higher unitary pressure values and can therefore be regarded as applying above-average aggressive pressure within their team context.

Tables 5.2 and 5.3 complement the graphical analysis by providing the full set of descriptive statistics for each player. While the previous figures allow a visual comparison of the distributions of unitary pressure, these tables incorporate additional quantitative information such as the number of observations (n_{aB} or n_{bA}), the total accumulated pressure (p_{aB} or p_{bA}), and the mean pressure value ($\bar{p}_{aB}$ or $\bar{p}_{bA}$). These metrics allow a more precise comparison of the defensive behaviour of players within each team.

Players in table 5.2 and 5.3 are sorted according to the skewness coefficient CA , which in this study is interpreted as a proxy for defensive aggressiveness or “biting” behaviour. Shaded rows correspond to players whose skewness values are below the team average ($CA_{p_{aB}} \leq CA_{p_{AB}}$ for team A , and $CA_{p_{bA}} \leq CA_{p_{BA}}$ for team B). Under this criterion, eight players from team A and five players from team B can be considered to exhibit above-average aggressive defensive behaviour relative to their teammates.

An important observation derived from these tables is that the mean unitary pressure value alone does not fully characterise defensive behaviour. For instance, player 2 in team A presents a mean pressure value of $\bar{p}_{2,B} = 0.63$, while the goalkeeper (player 13) reaches the highest mean value $\bar{p}_{13,B} = 0.67$. However, this does not imply that player 13 performs the most aggressive or consistent defensive pressure. In fact, the goalkeeper accumulates very few interactions ($n_{13,B} = 764$), and the distribution of his pressure values is mainly influenced by specific situations such as set pieces.

Player	n_{aB}	p_{aB}	$\bar{p}_{aB}$	$Medp_{aB}$	$\sigma_{p_{aB}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{aB}}$	$CC_{p_{aB}}$
19	7486	4931.58	0.66	0.66	0.15	0.23	0.53	0.66	0.79	-0.04	-1.22
18	22206	14592.18	0.66	0.66	0.15	0.23	0.53	0.66	0.78	0.02	-1.06
13	764	510.05	0.67	0.67	0.14	0.21	0.57	0.67	0.76	0.09	-0.64
2	8260	5236.73	0.63	0.63	0.15	0.24	0.50	0.63	0.76	0.16	-1.07
10	9996	6357.18	0.64	0.63	0.15	0.24	0.50	0.63	0.76	0.19	-1.06
29	5727	3690.33	0.64	0.63	0.16	0.25	0.50	0.63	0.77	0.24	-1.03
12	8654	5371.08	0.62	0.61	0.14	0.23	0.50	0.61	0.73	0.25	-0.87
22	18634	11609.41	0.62	0.61	0.15	0.24	0.50	0.61	0.74	0.26	-1.02
5	4858	3061.16	0.63	0.61	0.16	0.25	0.50	0.61	0.76	0.29	-1.00
7	17069	10324.36	0.60	0.60	0.14	0.22	0.49	0.60	0.71	0.31	-0.85
4	12338	7632.21	0.62	0.61	0.14	0.23	0.49	0.61	0.73	0.31	-0.87
14	12356	7580.30	0.61	0.60	0.14	0.24	0.49	0.60	0.73	0.32	-0.91
9	21434	12846.90	0.60	0.58	0.14	0.24	0.47	0.58	0.71	0.45	-0.89
6	11426	6607.05	0.58	0.56	0.13	0.23	0.47	0.56	0.67	0.59	-0.50
11	4527	2696.48	0.60	0.56	0.14	0.24	0.48	0.56	0.70	0.64	-0.54
Team	n_{AB}	p_{AB}	$\bar{p}_{AB}$	$Medp_{AB}$	$\sigma_{p_{AB}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{AB}}$	$CC_{p_{AB}}$
A	165.735	103.047	0.6218	0.6096	0.1475	0.2419	0.4936	0.6096	0.7387	0.28288	-0.97112

Table 5.2: Team A player statistics. Players are sorted in ascending order of $CA_{p_{aB}}$. Shaded rows indicate $CA_{p_{aB}} \leq CA_{p_{AB}}$.

Player	n_{bA}	p_{bA}	$\bar{p}_{bA}$	$Medp_{bA}$	$\sigma_{p_{bA}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{bA}}$	$CC_{p_{bA}}$
19	3453	2195.30	0.64	0.64	0.15	0.23	0.50	0.64	0.76	0.07	-1.10
4	29213	19054.69	0.65	0.65	0.15	0.24	0.52	0.65	0.78	0.11	-1.09
7	16562	10461.26	0.63	0.62	0.15	0.25	0.49	0.62	0.77	0.17	-1.18
6	33687	21502.09	0.64	0.63	0.14	0.23	0.52	0.63	0.75	0.19	-0.95
21	8436	5253.53	0.62	0.62	0.15	0.24	0.48	0.62	0.75	0.19	-1.15
14	27805	17001.26	0.61	0.60	0.14	0.23	0.49	0.60	0.72	0.35	-0.90
3	13555	8476.31	0.63	0.60	0.15	0.24	0.50	0.60	0.74	0.36	-0.91
34	1526	892.94	0.59	0.55	0.13	0.23	0.48	0.55	0.71	0.38	-1.02
10	19187	11505.99	0.60	0.58	0.14	0.24	0.48	0.58	0.71	0.39	-0.90
2	3707	2285.48	0.62	0.60	0.15	0.25	0.48	0.60	0.74	0.39	-0.90
11	16098	9697.86	0.60	0.59	0.14	0.24	0.48	0.59	0.71	0.41	-0.78
23	14553	8926.76	0.61	0.59	0.15	0.24	0.49	0.59	0.73	0.44	-0.87
17	4324	2626.14	0.61	0.58	0.15	0.24	0.48	0.58	0.72	0.46	-0.88
12	21623	12966.69	0.60	0.57	0.15	0.24	0.47	0.57	0.71	0.50	-0.75
16	13432	8045.56	0.60	0.58	0.15	0.25	0.47	0.58	0.70	0.57	-0.63
18	5607	3318.67	0.59	0.57	0.14	0.23	0.48	0.57	0.68	0.59	-0.55
Team	n_{BA}	p_{BA}	$\bar{p}_{BA}$	$Medp_{BA}$	$\sigma_{p_{BA}}$	CV	$Q1$	$Q2$	$Q3$	$CA_{p_{BA}}$	$CC_{p_{BA}}$
<i>B</i>	232.768	144.210	0.6195	0.6038	0.1489	0.2465	0.4916	0.6038	0.7366	0.32521	-0.94691

Table 5.3: Team *B* player statistics. Players are sorted in ascending order of $CA_{p_{bA}}$. Shaded rows indicate $CA_{p_{bA}} \leq CA_{p_{BA}}$.

To analyse the joint behaviour of the two shape descriptors, Figure 5.6 presents the scatter plot of the standardised variables (i.e., z -scores) for CA and CC . This representation allows the identification of atypical players and facilitates comparison across teams within a common reference frame.

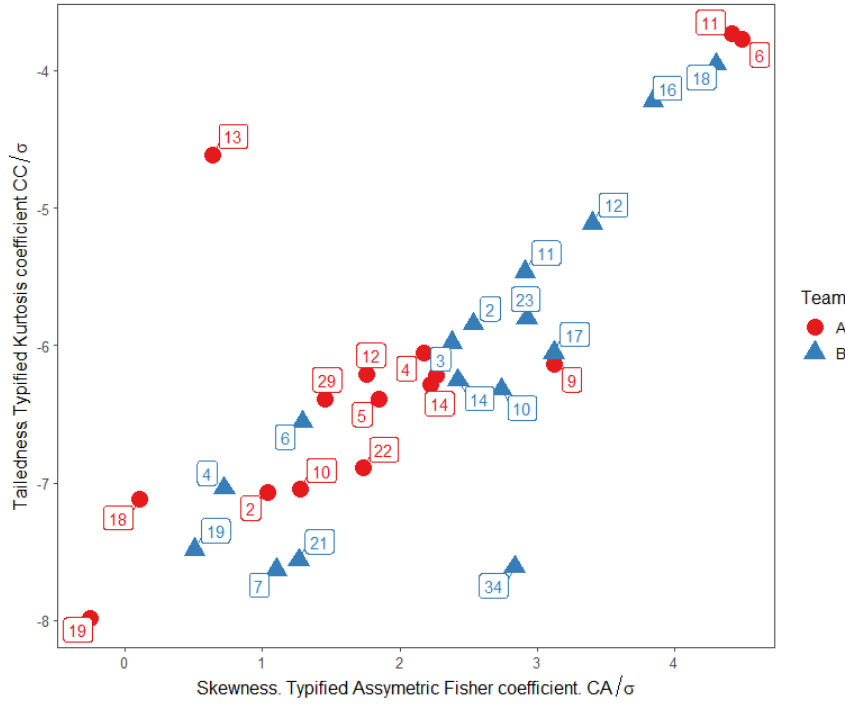


Figure 5.6: Teams A and B: scatter plot of standardised shape parameters (z-scores) for CA and CC.

As a rule of thumb, standardised values within the interval ± 2 are considered compatible with non-extreme behaviour. Figure 5.6 shows that the most evident outliers correspond to the two goalkeepers (player 13 on team A and player 34 on team B). This behaviour is expected, as goalkeepers are not primarily involved in systematic defensive pressure actions and as we have just commented their unitary pressure values are mainly driven by set-piece situations ($n_{13,B} = 764$ and $n_{34,A} = 1526$), consequently both goalkeepers were excluded from the clustering analysis.

Subsequently, K-means clustering with $K = 3$ was applied to the remaining outfield players using the standardised shape parameters $\frac{CA}{\sigma_{CA}}$ and $\frac{CC}{\sigma_{CC}}$ as input features. The resulting partition is shown in Figure 5.7 for the combined set of players and in Figure 5.8 separated by team.

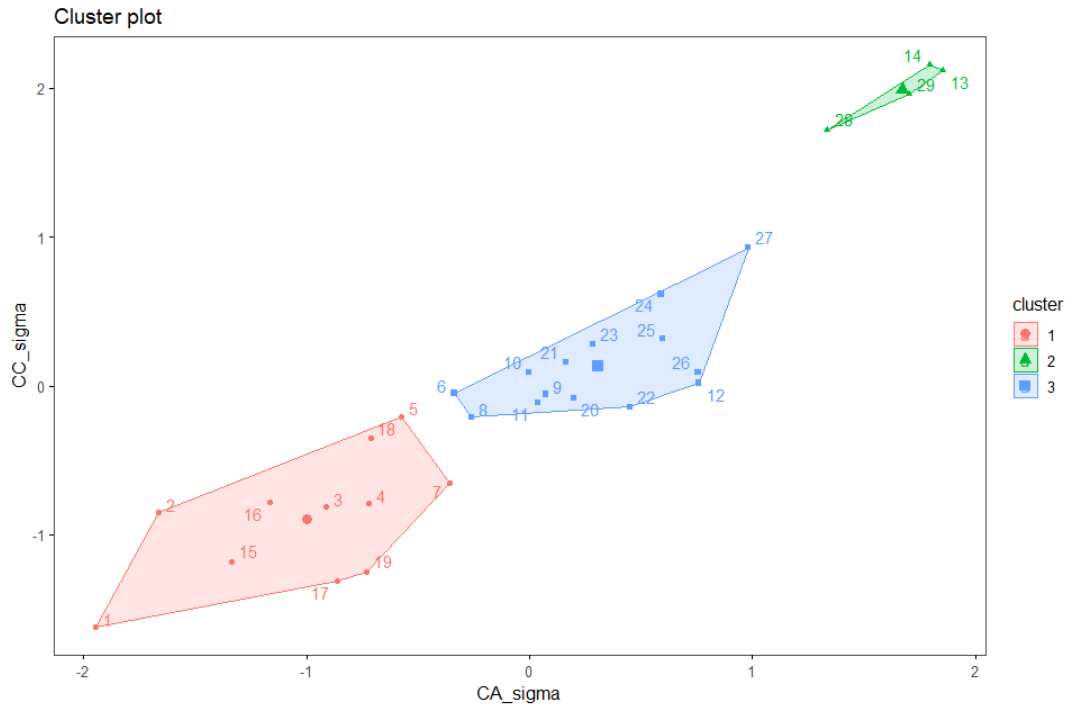


Figure 5.7: K-means clustering results for outfield players ($K = 3$) using standardised CA and CC .

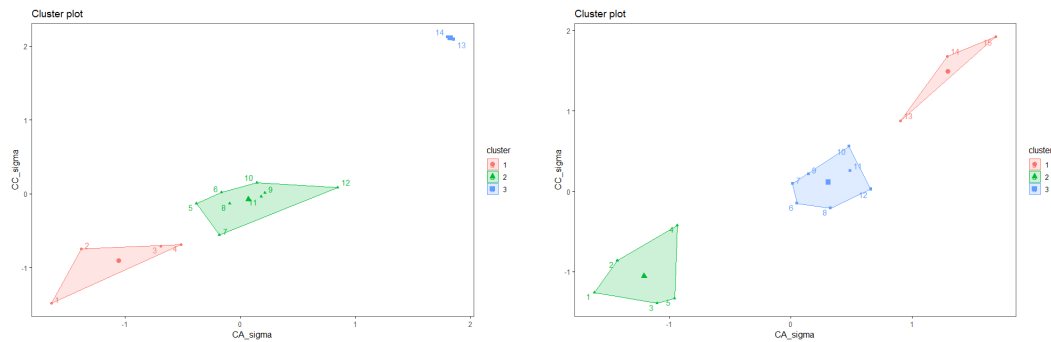


Figure 5.8: Team-specific K-means clustering results for outfield players ($K = 3$).

Table 5.4 summarises the final classification obtained through the K-means clustering procedure ($K = 3$), applied to the standardised shape descriptors of defensive pressure. The three resulting groups should not be interpreted merely as numerical partitions of the data, but as distinct defensive pressure profiles that emerge from the joint behaviour of aggressiveness and consistency. These groups represent distinct combinations of aggressiveness and consistency, providing an interpretable typology of individual defensive behaviours and reinforcing the concept of a player-specific *defensive signature*.

Group	Defensive pressure characteristics	Team	Players
I	High aggressiveness, low consistency	A	19, 18, 2, 10, 22, 29
		B	19, 4, 7, 21, 6
II	Moderate aggressiveness, moderate consistency	A	12, 5, 4, 14, 7, 9
		B	3, 14, 2, 10, 23, 11, 17, 12
III	Low aggressiveness, high consistency	A	11, 6
		B	16, 18

Table 5.4: Classification of players from teams *A* and *B* according to their defensive pressure profiles obtained through K-means clustering ($K = 3$).

Group I comprises players whose pressure distributions are characterised by lower values of the skewness coefficient and relatively low values of Kurtosis. From a tactical perspective, these are players who tend to reduce the distance to opponents more frequently, thereby exhibiting a more aggressive defensive behaviour. However, the lower concentration of their pressure distributions indicates that such aggressiveness is not usually reproduced at stable distances, but rather in a more irregular or context-dependent way. This group therefore corresponds to players who *bite* more often, although without maintaining a highly consistent positional pattern in the execution of pressure.

Group II contains players with intermediate values in both shape descriptors, representing a more balanced defensive profile. These players neither exhibit the strongest aggressive tendency of Group I nor the highest degree of consistency of Group III. Instead, they combine moderate proximity-based pressure with a certain degree of regularity, suggesting defensive behaviours that are adaptable and context-sensitive. In practical terms, this group may be interpreted as comprising players whose defensive actions are neither highly disruptive nor purely positional, but rather lie in an intermediate tactical zone between intensity and order.

Finally, Group III includes players whose pressure behaviour is characterised by lower aggressiveness but greater consistency. These players tend to maintain larger defensive distances on average, which places them farther from the most aggressive pressure profiles. Nevertheless, their distributions are comparatively more concentrated, indicating that they reproduce similar pressure distances more regularly throughout the match. From a tactical

point of view, this group may be associated with players whose defensive contribution is based more on positional discipline, spatial control, and stable defensive habits than on frequent close-range pressure.

Taken together, these three groups provide an interpretable typology of defensive behaviour, ranging from highly aggressive but less regular pressure profiles to less intense but more positionally consistent ones. In this way, the clustering analysis supports the idea that the proposed shape descriptors are capable of capturing meaningful defensive signatures at player level.

5.4 Conclusions

In this work, the unitary defensive pressure metric $p_{a,b}(f)$, is used to characterize outfield players according to their defensive profiles during a football match. The aim of the study has been to deepen the understanding of this metric through numerical and graphical descriptive techniques, in order to characterise not only in magnitude but also in **shape and quality** of defensive actions.

At team level, a global comparison between teams *A* and *B* shows that both teams exhibit very similar average unitary pressure values ($\bar{p}_{AB} = 0.6218$ and $\bar{p}_{BA} = 0.6195$, respectively), indicating comparable overall levels of defensive proximity. However, differences emerge when analysing the shape of the pressure distributions. In particular, team *A* presents a lower skewness coefficient ($CA_{p_{AB}} = 0.28288$) than team *B* ($CA_{p_{BA}} = 0.32521$), which is interpreted as a slightly higher frequency of close defensive approaches and, consequently, a marginally higher defensive pressure quality for team *A*.

The analysis of the Kurtosis coefficient (CC) reveals negative values for both teams ($CC_{p_{AB}} = -0.97112$ and $CC_{p_{BA}} = -0.94691$), indicating platykurtic distributions at team level. This result suggests that, when aggregating all players, defensive pressure is applied over a broad range of distances, without a strong concentration around specific values. In practical terms, team-level defensive behaviour appears heterogeneous and not dominated by a single, rigid proximity pattern.

At player level, an inverse relationship between the Fisher's skewness coefficient (CA) and the Kurtosis coefficient (CC) is observed. Players with lower CA values—those who tend to reduce the distance to opponents more frequently—also tend to present lower CC values, reflecting greater variability in the distances at which pressure is applied. This finding suggests that higher defensive aggressiveness is often associated with less regular or more opportunistic pressure behaviour, whereas players exhibiting greater consistency tend to apply pressure at more stable distances but with lower aggressiveness.

By segmenting the teams into individual players and analysing their histograms, density functions and descriptive statistics, this study confirms that the shape parameters CA and CC are effective quantitative descriptors for discriminating individual defensive behaviours. These parameters allow qualitative concepts such as aggressiveness, regularity and defensive habits to be translated into measurable quantities derived directly from

tracking data.

The scatter plot of the standardised shape parameters identifies both goalkeepers (player 13 from team *A* and player 34 from team *B*) as clear outliers. This behaviour is consistent with their tactical role, as goalkeepers are not primarily involved in systematic defensive pressure actions and their unitary pressure values are mainly influenced by set-piece situations. Consequently, they were excluded from the clustering analysis in order to focus on meaningful outfield defensive behaviours.

Among outfield players, the most aggressive pressure profiles correspond to player 19 from team *A* ($CA_{p_{aB}} = -0.04$) and player 19 from team *B* ($CA_{p_{bA}} = 0.07$). Although both players participated for a limited number of minutes after entering the match in the second half, their pressure distributions indicate a high frequency of close defensive approaches. At the same time, their low Kurtosis values ($CC_{p_{aB}} = -1.22$ and $CC_{p_{bA}} = -1.10$) suggest that this aggressiveness is expressed in a more irregular manner, rather than following stable or highly structured defensive patterns.

Finally, the application of K-means clustering with $K = 3$ enabled the classification of players into three distinct defensive pressure profiles. These groups range from players who apply highly aggressive but irregular pressure, through players with moderate aggressiveness and partial regularity, to players who exert lower-intensity pressure while maintaining greater consistency in their defensive distances. This classification provides a compact and interpretable typology of defensive behaviours and supports the concept of a player-specific *defensive signature*, defined by the joint interaction between aggressiveness and consistency in defensive pressure.

From an applied perspective, the proposed framework has direct implications for both player scouting and real-time tactical analysis. By characterising each player through a pair of shape descriptors (CA, CC), defensive behaviour can be summarised into a compact and interpretable *defensive signature* that goes beyond traditional volume-based metrics. In a scouting context, this allows the identification of players whose defensive profiles match specific tactical requirements, such as highly aggressive pressers, disciplined positional defenders, or balanced profiles capable of adapting to different game models.

Moreover, the fact that these descriptors can be computed over arbitrary time windows enables their use in near real-time match analysis. Variations in a player's CA and CC during a match may reflect tactical adjustments, fatigue effects, or contextual changes in team strategy. Consequently, this approach opens the door to monitoring defensive behaviour dynamically, supporting coaching staff in evaluating pressing efficiency, detecting deviations from the intended defensive model, and informing in-game decision-making.

In summary, the combined analysis of CA and CC , supported by classical statistical methods and unsupervised clustering algorithms, allows the classification and comparison of these defensive traits across teams and among players. This analysis provides a characterization, from the point of view of defensive pressure of both, teams/groups of players and individual players.

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6

Discussion

This thesis has investigated defensive performance in football through a mathematical and computational framework, progressively integrating different data sources, modelling approaches, and analytical scales. Throughout the chapters of this thesis, defensive behaviour has been conceptualised not as a collection of isolated actions or outcomes, but as a continuous spatial–temporal process emerging from interactions between players and from collective team organisation. This perspective has guided both the methodological choices and the design of the proposed metrics, with the overarching aim of advancing towards a more objective, interpretable, and operational characterisation of defence in football.

The starting point of this thesis can be traced to a practical question arising from professional football analysis. In Chapter 2, the research begins with a real tactical problem posed by a professional football club concerning the success of a specific set-play situation. From a tactical perspective, it became evident that the outcome of such situations does not depend solely on the attacking team, but also on how the opposing team organises and executes its defensive behaviour. This observation highlighted that the success of a play is inherently influenced by the defensive response, raising the broader question of how defensive behaviour can be characterised in a systematic and objective manner. A review of the existing literature showed that this problem had not been fully addressed from a quantitative perspective. In order to explore this question, the study relied on event-based data and machine learning techniques to identify patterns in how teams defended these situations. The analysis revealed the presence of distinct defensive profiles adopted by opposing teams, which could be broadly characterised as aggressive, technical, or mixed defensive approaches. While this initial study provided valuable insights into the relationship between defensive behaviour and tactical outcomes, it also exposed the limitations of event-based data for capturing the continuous spatial and temporal nature of defensive interactions, particularly those related to off-ball positioning, anticipation, and collective spatial control. These limitations motivated the transition towards tracking data and more

explicit mathematical modelling, which forms the basis of the subsequent chapters of this thesis.

Chapter 3 marked a methodological shift by introducing a mathematical formulation of defensive pressure based on player tracking data. By defining a unitary pressure metric between an attacker and a defender, the proposed model established an interpretable representation of defensive interaction grounded in spatial proximity and relative positioning. This formulation directly addressed the objective of developing interpretable defensive metrics based on spatial-temporal relationships, and demonstrated how individual-level interactions can be coherently aggregated to describe collective and team-level defensive behaviour. The scalability of the metric constitutes a key contribution, as it ensures conceptual consistency across levels of analysis while preserving interpretability, a property often lacking in more complex black-box models.

Building upon this foundation, Chapter 4 incorporated the temporal dimension of defensive pressure, enabling the analysis of its evolution over time and across different phases of play. This dynamic perspective allowed the derivation of metrics related to defensive workload and intensity, offering insights into how defensive effort is distributed throughout the match and in response to specific game contexts. In doing so, this study addressed the objective of investigating defensive behaviour in specific situations and short-term dynamics, such as transitions and sequences following discrete match events. The results emphasise that defensive performance cannot be fully understood through static summaries alone, but requires consideration of timing, persistence, and fluctuation, reinforcing the view of defence as a dynamic process.

Chapter 5 extended the analysis to the individual player level by examining whether the proposed defensive pressure metric allows players to be systematically characterised and grouped according to their defensive behaviour. To address this question, statistical descriptors and visual representations of pressure-related metrics were employed in order to analyse the distribution of defensive actions across players. Through this approach, the metric was shown to capture behavioural traits such as aggressiveness and consistency in the way defensive pressure is applied. This analysis contributed to the objective of developing metrics applicable at multiple levels of granularity, demonstrating that the proposed framework can provide meaningful player-level characterisations beyond simple aggregate counts. Importantly, the results indicated that similar aggregate defensive outputs may correspond to markedly different behavioural patterns, highlighting the value of interpretable, distribution-based analyses for scouting, player comparison, and contextual performance evaluation.

Across all studies, a recurring theme is the practical feasibility of the proposed methods. Computational performance analyses showed that the metrics derived from tracking data can be computed within time frames compatible with real-world applications. This directly addresses the objective of evaluating the robustness and operational viability of the proposed framework under realistic data and computational constraints. The feasibility of near-real-time computation opens opportunities for deployment in applied settings, including tactical decision-support systems and broadcast visualisations, thereby extending

the relevance of the proposed models beyond post-match analysis.

Taken together, the discussion of Chapters 1 to 5 demonstrates that the objectives established at the outset of the thesis have been systematically addressed through a coherent and progressive research design.

7

Conclusions

This thesis has investigated how defensive pressure in football can be described, quantified, and analysed both at the level of individual players and at aggregated levels involving groups of players, including entire teams. The aim has been to characterise defensive pressure and study its spatial evolution across the pitch and its temporal dynamics throughout the course of a match. To achieve this, several mathematical modelling approaches have been developed, combining topological–geometrical representations with a dynamic formulation inspired by variable-mass systems. These models have been constructed from real match data and supported by analytical tools drawn from statistics and machine learning.

The research presented in this thesis follows a progressive conceptual development across the different chapters. In Chapter 2, defence is first examined from a tactical perspective, considering the defensive behaviour of the opposing team as a key factor influencing the success of specific attacking plays. In this context, the study analysed how different defensive styles affect the outcome of goal-kick strategies, such as playing long or building from short distribution.

Chapter 3 then introduces a topological–geometrical representation of defensive pressure. In this framework, each player is associated with a spatial neighbourhood that defines an area of influence on the pitch. Although the initial model considers neighbourhoods of equal radius for all players, the formulation naturally allows extensions in which these areas may vary according to the characteristics of each player, their tactical role, their position on the pitch, or even the match context at a given moment in time.

In Chapter 4, defensive pressure is further developed as a dynamic quantity. As players move across the pitch and interact with one another in their attempt to control space and contest possession, their respective areas of influence intersect and evolve over time. The intensity of defensive pressure can therefore be interpreted through the intersection of

these domains of influence, reflecting the continuous spatial contest between players. The temporal evolution of these intersections is modelled as a dynamic system inspired by the behaviour of a body with variable mass, providing a novel framework for analysing the evolution of defensive pressure throughout the match. The dynamic modelling of defensive pressure introduced in this thesis establishes a novel analogy between player interactions on the pitch and the dynamics of variable-mass systems. By interpreting the evolution of overlapping areas of influence between players as a dynamic process whose effective “mass” varies over time, the proposed framework provides an original physically inspired perspective for analysing defensive pressure. To the best of our knowledge, this approach has not previously been explored in the scientific literature on football performance analysis, and it offers a mathematically grounded way to study the temporal evolution of defensive interactions.

Finally, Chapter 5 explores defensive pressure at the individual player level. In this context, pressure-related metrics are used to identify defensive signatures that characterise the behavioural tendencies of players. These signatures provide a meaningful metric for player profiling, offering potential applications in tactical analysis, player comparison, and scouting processes.

Collectively, the analyses presented across the different chapters provide a coherent framework for understanding and quantifying defensive pressure in football, leading to the following main conclusions:

1. Defensive behaviour in football can be effectively modelled through continuous, mathematically defined metrics grounded in spatial–temporal tracking data. This approach overcomes fundamental limitations of traditional event-based analyses, particularly in the representation of off-ball actions, spatial control, and collective organisation.
2. A unitary defensive pressure metric constitutes a robust and interpretable. By formalising individual attacker–defender interactions, this metric enables coherent aggregation across analytical scales, from local match-ups to collective and team-level defensive behaviour, thereby fulfilling the objective of developing interpretable and mathematically grounded defensive metrics.
3. The incorporation of temporal dynamics is essential for a comprehensive understanding of defensive performance. Analysing the evolution of pressure over time allows meaningful aspects such as defensive workload, intensity, persistence, and contextual variation to be quantified, supporting the study of defence as a dynamic process rather than a static outcome.
4. The defensive pressure metric introduced in this thesis was examined to determine whether it can meaningfully group players according to their defensive behaviour. Using statistical descriptors of the distributions of pressure-related variables, such as kurtosis and Fisher coefficients, the analysis explored whether distinct patterns emerge across players. The results provide preliminary evidence that the metric is capable of distinguishing groups of players that exhibit similar defensive behaviours, effectively defining equivalence classes in terms of how defensive pressure is applied.

Although further validation with larger datasets would be required, this finding suggests that the proposed metric has the potential to support player profiling, comparison, and scouting applications.

5. The proposed analytical framework is computationally feasible under realistic conditions. The reported execution times demonstrate that the models and metrics developed in this thesis can be integrated into operational workflows, including real-time or near-real-time analytical pipelines for tactical decision support and media communication.

Taken together, these conclusions demonstrate that the objectives defined at the outset of the thesis have been successfully addressed through a coherent and progressive research design. By combining mathematical rigour, interpretability, and computational efficiency, the thesis tries to contribute to a unified framework for the quantitative analysis of defensive behaviour in football, bridging the gap between theoretical modelling and applied performance analysis.

Importantly, the proposed metrics are inherently scalable, allowing consistent comparisons not only at the level of individual players and matches, but also across teams, full seasons, and competitive contexts. This scalability enables comparative analyses between teams within the same competition, such as *LaLiga*, as well as across different European leagues and international competitions, supporting broader benchmarking and cross-contextual performance evaluation.

The methodologies and results presented open several avenues for future research, including the integration of defensive and offensive models within a common analytical framework, the extension of the proposed approaches to other team sports, and the deployment of these metrics in live analytical and broadcasting environments. In this sense, the thesis not only contributes to the scientific understanding of defensive performance in football, but also provides practical tools with potential impact in professional and applied contexts.

